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ADITYA RANJAN SIR
(EXCISE INSPECTOR)

INDEX

ADVANCE MATHS

ARITHMETIC MATHS

यथा शिखा मयूराणा, नागानां मणयो यथा।
तद्वद् वेदाङ्गशास्त्राणां गणितं मूर्धनि संस्थितम्॥

(महर्षि लगध कृत 'वेदांग ज्योतिष')

जैसे मोरों में शिखा और नागों में मणि का स्थान सबसे ऊपर है,
वैसे ही सभी वेदांग और शास्त्रों में गणित का स्थान सबसे ऊपर है।

Just as the crest on the heads of peacocks and the gems on the heads of serpents is in the highest position, in the same way the place of mathematics in the Vedangashastras is at the top.

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CHAPTER 01

GEOMETRY

GEOMETRY
- Lines & Angle
- Triangle
- Quadrilateral
- Polygon
- Circle

LINES & ANGLE

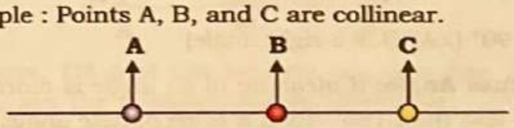
Point

A point is a circle of zero radius which determines a location. It is usually denoted by a capital letter.

Types of Points
- Collinear Points
- Non-Collinear Points
- Intersecting Points

(a) Collinear Points: If three or more points situated on a straight line, these points are called collinear points.

Example : Points A, B, and C are collinear.



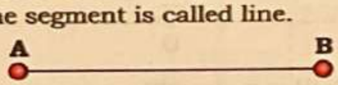
(b) Non-collinear Points: If three or more points are not situated on a straight line, these points are called non-collinear points.

(c) Intersecting Points: When two or more lines cross or meet at a common point, that point is known as intersecting point.

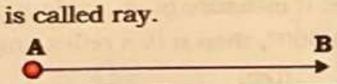
Line

A line is made up of an infinite number of points and it has only length i.e., it does not have any thickness (or width). A line is endless so, it can be extended in both directions.

Line Segment: A line segment has two end points; generally line segment is called line.



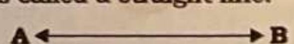
Ray: A ray extends indefinitely in one direction from any given point is called ray.



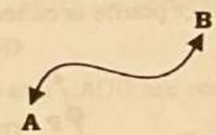
Name	Line	Line segment	Ray
Symbol	$\overleftrightarrow{AB}$	$\overline{AB}$	$\overrightarrow{AB}$
End Points	No end points	Two end points	One end points

Types of Lines

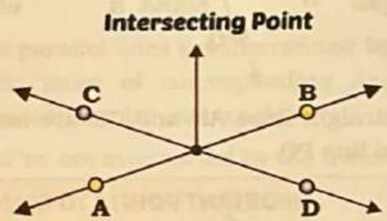
(a) Straight line: A line which does not change its direction at any point is called a straight line.



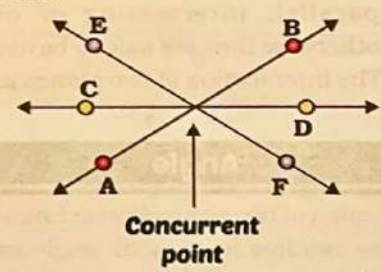
(b) Curve line: A line which changes its direction is called a curved line.



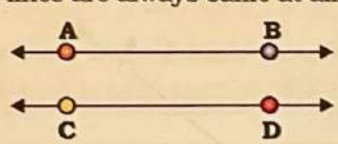
(c) Intersecting Lines: If two or more lines intersect each other, then they are called intersecting lines. In the figure AB and CD are intersecting lines. Two lines can intersect maximum at one point.



(d) Concurrent Lines : If three or more lines pass through a point, then they are called concurrent lines and the point through which these all lines pass is called point of concurrent.



(e) Parallel Lines : Two straight lines are parallel if they lie in the same plane and do not intersect even if they produced infinitely. Perpendicular distances between two parallel lines are always same at all places.

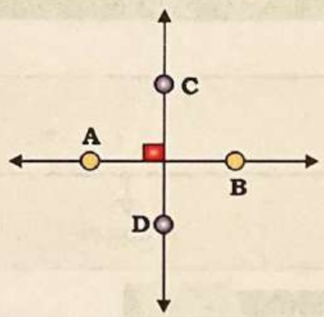


In the figure AB and CD are parallel lines.

Symbol for parallel lines is ||.

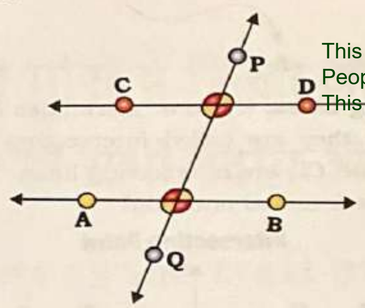
Hence, AB || CD.

(f) Perpendicular Lines : If two lines intersect each other at right angles, then two lines are called perpendicular lines. In the following figure AB and CD are perpendicular lines.



Symbolically it is represented as AB ⊥ CD or we can also say that CD ⊥ AB.

(g) Transversal Lines : A line which intersects two or more given lines at distinct points is called a transversal of the given lines.



In figure straight lines AB and CD are intersected by a transversal line PQ.

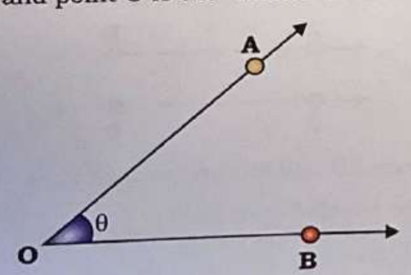
IMPORTANT POINTS TO REMEMBER

- There is one and only one line passing through two distinct points.
- Two or more lines are said to be coplanar if they lie in the same plane and can be parallel, intersecting or overlapping, otherwise they are said to be non-coplanar.
- The intersection of two planes is a line.

Angle

An angle is the union of two non-collinear line with a common initial point. The two line forming an angle are called arms of the angle and the common initial point is called the vertex of the angle.

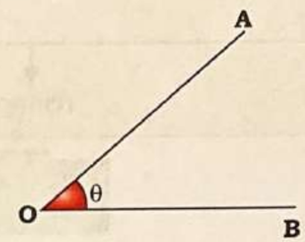
- The angle AOB is denoted by ∠AOB, is formed by line OA and OB and point O is the "vertex" of the angle.



(I) **Types of Angle**

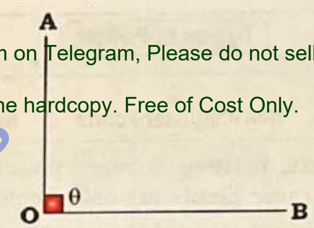
- Acute angle
- Right angle
- Obtuse angle
- Straight angle
- Reflex angle
- Complete angle

Acute Angle: If the measure of an angle is less than 90°, it is an acute angle.



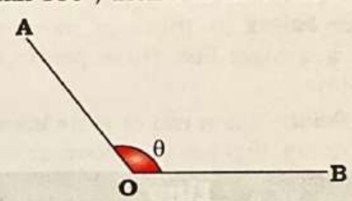
$0° < \theta < 90°$ (∠AOB is an acute angle)

Right Angle : If measure of an angle is equal to 90°, then it is a right angle.



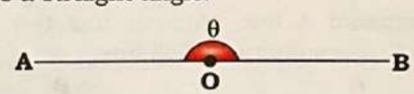
$\theta = 90°$ (∠AOB is a right angle)

Obtuse Angle: If measure of an angle is more than 90° but less than 180°, then it is an obtuse angle.



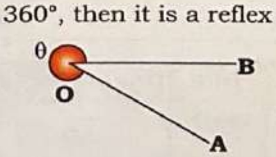
$90° < \theta < 180°$ (∠AOB is an obtuse angle)

Straight Angle: If measure of an angle is equal to 180°, then it is a straight angle.



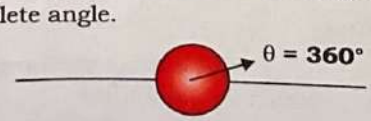
$\theta = 180°$ (∠AOB is a straight angle)

Reflex Angle: If measure of an angle is more than 180° but less than 360°, then it is a reflex angle.



$180° < \theta < 360°$ (∠AOB is a reflex angle)

Complete angle: If measure of an angle is 360° then it is a complete angle.

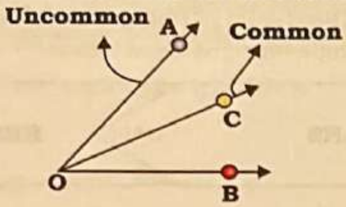


(II) **Pair of Angles**

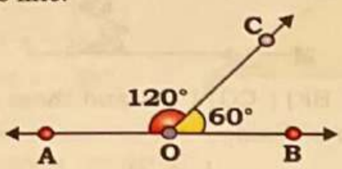
Adjacent Angle | **Linear Angle** | **Complementary Angle** | **Supplementary Angle**

Adjacent Angles: Two angles are called adjacent angles if:

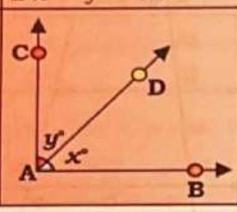
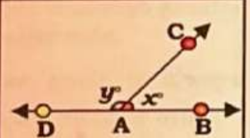
- They have the same vertex,
- They have a common arm,
- Uncommon arms are on either side of the common arm.



- In the figure, ∠AOC and ∠BOC have a common vertex O. Also, they have a common arm OC and the distinct arms OA and OB, lie on the either side of common arm OC.
- **Linear Pair of Angles:** Two adjacent angles are said to form a linear pair of angles, if their uncommon arms are two opposite line.



- In figure, OA and OB are two opposite line ∠AOC & ∠BOC are the adjacent angles. Therefore, ∠AOC and ∠BOC form a linear pair.
- If a line stand on another line, the sum of the adjacent angles so formed is 180°.

Angle	Complementary	Supplementary
Definition	If sum of two angles is equal to 90°, then the two angles are called complementary angles. • ∠BAD and ∠DAC are complementary angles, if $x^o + y^o = 90°$	If sum of two angles is equal to 180°, then the two angles are called supplementary angle. • ∠BAC and ∠DAC are supplementary angles, if $x^o + y^o = 180°$
Representation		

Ex. The measure of an angle for which the measure of the supplement is four times the measure of the complement is:

HINTS Let Angle = x.

So, Complementary angle = $90° - x$

Supplementary angle = $180° - x$

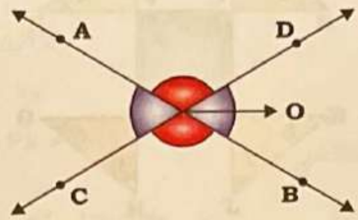
ATQ,

4 × Complementary angle = Supplementary angle

$\Rightarrow 4 \times (90° - x) = 180° - x \Rightarrow x = 60°$

(III) Vertically Opposite Angles

If arm of two angles form two pairs of opposite rays, then the two angles are called as vertically opposite angles.



- In other words, when two lines intersect, two pairs of vertically opposite angles are formed. Each pair of vertically opposite angle is equal.
- In the figure, two lines AB and CD intersect at O. We find that ∠AOC and ∠BOD are vertically opposite angles

$\Rightarrow$ ∠AOC = ∠BOD

Similarly, ∠BOC and ∠AOD are vertically opposite angles.

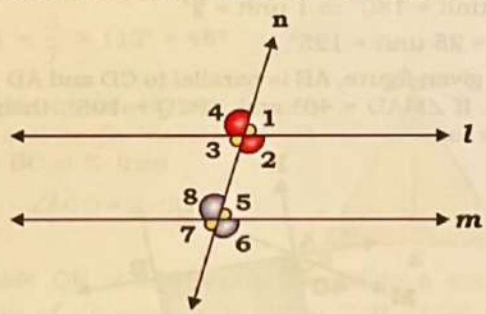
$\Rightarrow$ ∠BOC = ∠AOD

(IV) **Angles made by Transversal Lines**

Corresponding Angle | **Alternate Angle** | **Consecutive Interior Angle**

When two parallel lines are intersected by a transversal. They form pairs of corresponding angles, Alternate angle, Consecutive Interior angles.

Lines' and 'm' are intersected by the transversal 'n'. Then



Corresponding Angle

∠1 = ∠5, ∠4 = ∠8, ∠3 = ∠7 and ∠2 = ∠6

Alternate Interior Angle

∠3 = ∠5 and ∠2 = ∠8

Alternate Exterior Angle

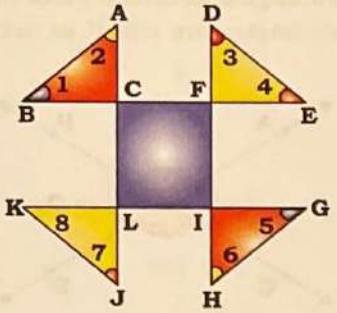
1 = ∠7 and ∠4 = ∠6

Consecutive Interior Angles

∠2 + ∠5 = 180° & ∠3 + ∠8 = 180°

Conversely, if a transversal intersects two lines in such a way that a pair of alternate interior angles is equal, then the two lines are parallel.

Ex. Angles are shown in the given figure. What is the value of $\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 + \angle 7 + \angle 8$?



HINTS $\angle C = 180° - (\angle 1 + \angle 2) \Rightarrow \angle F = 180° - (\angle 3 + \angle 4)$

$\Rightarrow \angle I = 180° - (\angle 5 + \angle 6) \Rightarrow \angle L = 180° - (\angle 7 + \angle 8)$

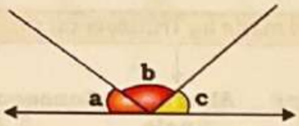
CFIL is quadrilateral.

$\therefore$ Sum of angles of CFIL is 360°.

$\Rightarrow 720° - (\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 + \angle 7 + \angle 8) = 360°$

$\Rightarrow 360° = (\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 + \angle 7 + \angle 8)$

Ex. In the given figure, if $\frac{b}{a} = 5$ and $\frac{c}{a} = \frac{6}{5}$ then what is the value of b.



HINTS b : a = (5 : 1) × 5 = 25 : 5

c : a = 6 : 5

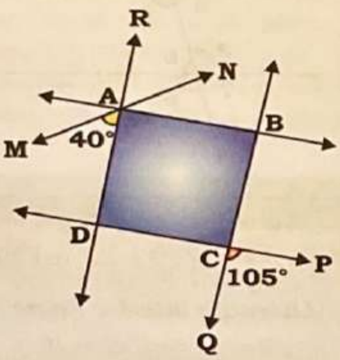
$\therefore$ a : b : c = 5 : 25 : 6

$\angle a + \angle b + \angle c = 180°$ (Straight line)

$\Rightarrow$ 36 unit = 180° $\Rightarrow$ 1 unit = 5°

$\therefore \angle b = 25$ unit = 125°

Ex. In the given figure, AB is parallel to CD and AD is parallel to BC. If $\angle MAD = 40°$ and $\angle PCQ = 105°$, then $\angle NAB$ is equals to?



HINTS

$\angle BCD = \angle PCQ = 105°$ (Vertically opposite angle)

$\angle CDA = 180° - 105° = 75°$

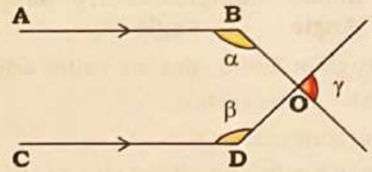
$\angle RAB = \angle CDA = 75°$ (corresponding angles)

$\angle RAN = \angle MAD = 40°$ (Vertically opposite angle)

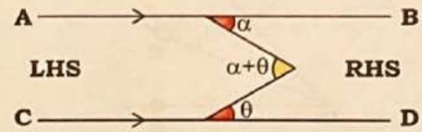
$\therefore \angle NAB = \angle RAB - \angle RAN = 75° - 40° = 35°$

Some Important Results

☞ If AB||CD then $\alpha + \beta + \gamma = 360°$



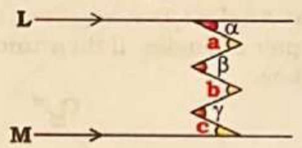
☞ If AB||CD, then sum of angle on left hand side is equal to sum of angle on right hand side.



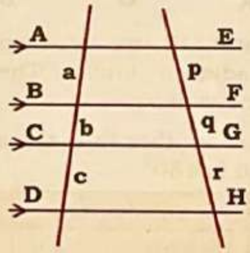
RHS = LHS

$\Rightarrow \theta + \alpha = \alpha + \theta$

☞ If L || M, $a + b + c = \alpha + \beta + \gamma$



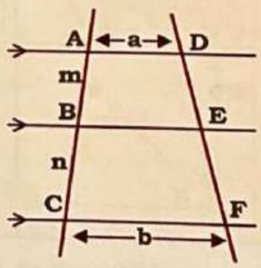
☞ When AE|| BF|| CG|| DH and these lines are cut by two transversal lines.



Then, a : b : c = p : q : r

$$\frac{a}{a+b+c} = \frac{p}{p+q+r}$$

☞ When AD|| BE||CF and these lines are cut by two transversal lines.

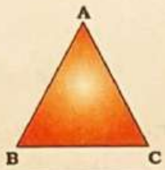


Then, $\frac{AB}{BC} = \frac{DE}{EF} = \frac{m}{n}$

$$BE = \frac{an + bm}{m + n}$$

TRIANGLE

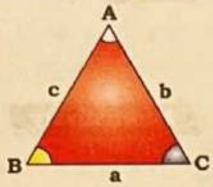
A triangle is a polygon with three sides and three angles. It is the fundamental shape in geometry formed by connecting three non-collinear points.



Fundamental Properties of Triangle

☞ Sum of all three angles of a triangle is always 180°

$\angle A + \angle B + \angle C = 180°$



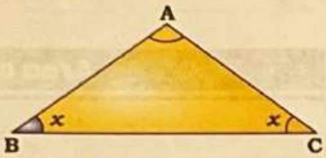
Ex. One of the angles of a triangle is 108° and the other two angles are equal. What is the measure of each of these equal angles?

HINTS We know,

$\angle A + \angle B + \angle C = 180°$

$\Rightarrow 108° + x + x = 180°$

$\Rightarrow 2x = 72° \Rightarrow x = 36°$



☞ (i) Angles opposite to the equal sides of a triangle are equal.

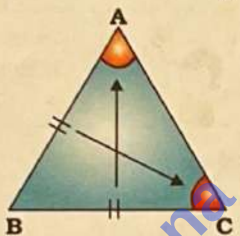
(ii) Sides opposite to the equal angles of a triangle are equal.

In ΔABC, If AB = BC, then

$\angle A = \angle C$

In ΔABC, If $\angle A = \angle C$, then

AB = BC

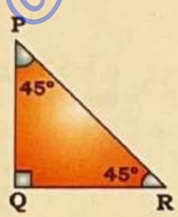


Ex. In ΔPQR, $\angle Q = 90°$, PQ = 8 cm and $\angle PRQ = 45°$. Find the length of QR.

HINTS

Using property 2(ii),

QR = PQ = 8 cm



☞ (i) The angle opposite to the greater side is always greater than the angle opposite to the smaller side.

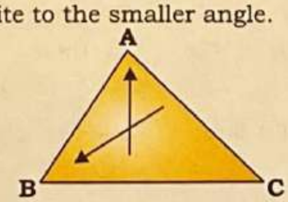
(ii) The side opposite to the greater angle is always greater than the side opposite to the smaller angle.

In ΔABC, If BC > AC, then

$\angle A > \angle B$

In ΔABC, If $\angle A > \angle C$, then

BC > AB



Ex. The ratio of the angles, ∠P, ∠Q and ∠R of a ΔPQR is 2 : 4 : 9, then which of the following is true?

(a) PQ > QR > RP

(b) PQ > RP > QR

(c) QR > RP > PQ

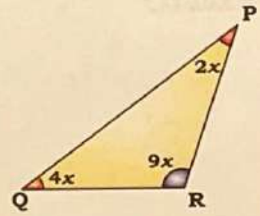
(d) PR > PQ > QR

HINTS

Here, In ΔPQR

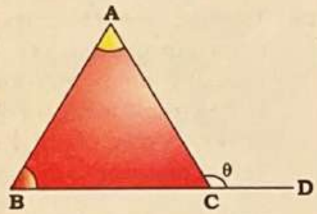
$9x > 4x > 2x$

So, PQ > PR > QR



☞ If a side of triangle is produced, then the exterior angle so formed is equal to the sum of the two interior opposite angles.

$\angle ACD = \angle CAB + \angle ABC$



Ex. The side BC of ΔABC is produced to D. If $\angle ACD = 112°$ and $\angle B = \frac{3}{4}\angle A$, then find the measure of ∠B.

HINTS We know,

Exterior angle ($\angle ACD$)

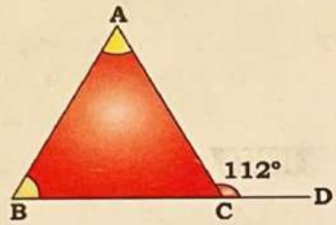
$= \angle A + \angle B$

$\because \angle B = \frac{3}{4}\angle A$

$\Rightarrow \angle A = \frac{4}{3}\angle B$ ....(i)

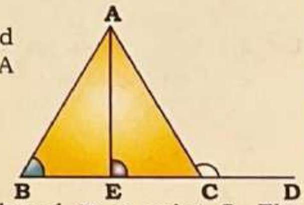
$\therefore 112° = \frac{4}{3}\angle B + \angle B$

$\Rightarrow \angle B = \frac{3}{7} \times 112° = 48°$



☞ In ΔABC, the side BC is produced to D and angle bisector of ∠A meets BC at E, then

$\angle ABC + \angle ACD = 2\angle AEC$.



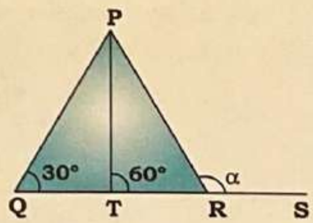
Ex. The side QR of ΔPQR is produced to a point S. The bisector of ∠P meets side QR at T. If $\angle PQR = 30°$ and $\angle PTR = 60°$, find ∠PRS.

HINTS

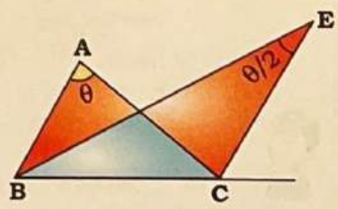
$\angle PQR + \angle PRS = 2 \times \angle PTR$

$\Rightarrow 30° + \alpha = 2 \times 60°$

$\Rightarrow \alpha = 120° - 30° = 90°$



☞ In a triangle, the angle formed between internal bisector of one base angle and external bisector of the other base angle is half of the remaining vertex angle.



According to this property, $\angle BEC = \frac{\angle A}{2}$.

Ex. In ΔABC, $\angle B = 72°$ and $\angle C = 44°$. Side BC is produced to D. Then bisectors of $\angle B$ and $\angle ACD$ meet at E. What is the measure of $\angle BEC$?

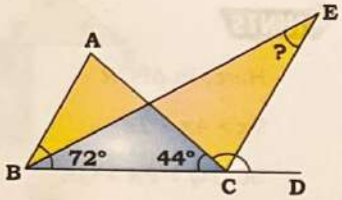
HINTS We know,

$\angle A + \angle B + \angle C = 180°$

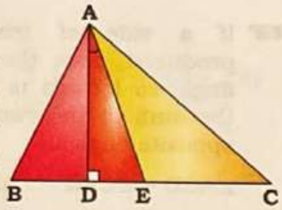
$\Rightarrow \angle A = 180° - 72° - 44°$

$\Rightarrow \angle A = 64°$

$\therefore BEC = \frac{\angle A}{2} = \frac{64°}{2} = 32°$



☞ The angle between perpendicular drawn from a vertex to opposite side and angle bisector of that vertex angle is half of difference between other two remaining vertex angles.



If $AD \perp BC$ and AE is angle bisector of $\angle A$, then

$\angle DAE = \left|\frac{\angle B - \angle C}{2}\right|$

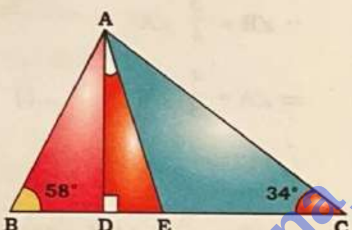
Ex. In ΔABC, AD is perpendicular to BC and AE is the bisector of $\angle BAC$. If $\angle ABC = 58°$ and $\angle ACB = 34°$, then find the measure of $\angle DAE$.

HINTS

$\angle DAE = \frac{\angle B - \angle C}{2}$

$= \frac{58° - 34°}{2} = \frac{24°}{2}$

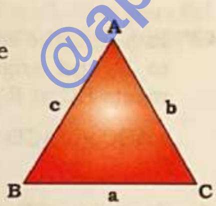
$= 12°$



☞ (i) Sum of any two sides of a triangle is always greater than the third side.

$a + b > c$, $b + c > a$

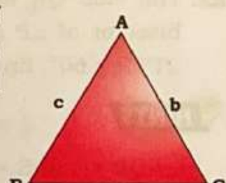
$c + a > b$



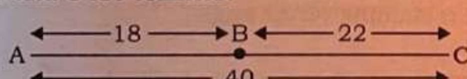
(ii) The difference of the length of any two sides of a triangle is always smaller than the third side.

$|a - b| < c$, $|b - c| < a$

$|c - a| < b$



1. If one side of a triangle is greater than the sum of the other two sides then a triangle can't be formed.
2. When the length of one side is equal to sum of the length of other two sides then the points are collinear & triangle is not formed (just a straight line back and forth).i.e., if a + b = c then point A, B and C are collinear.

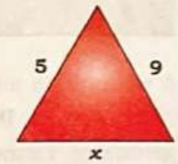


Ex. Three sides of a triangle are 5 cm, 9 cm, and x cm. The minimum integral value of x is.

HINTS

Value of x lies between, $4 < x < 14$

Thus, Minimum integral value of x is 5.



Ex. If the sides of a triangle are 7, 12 and x, and x is an integer, then find the number of possible values of x.

HINTS $12 + 7 > x > 12 - 7$

$\Rightarrow 19 > x > 5$

$x = (19 - 5) - 1 = 13$

Alternatively

No. of possible values of x = 2 × smallest side – 1

= 2 × 7 – 1 = 13

"Number of Possible value of x
= 2 × smallest side – 1"

Area of Triangle

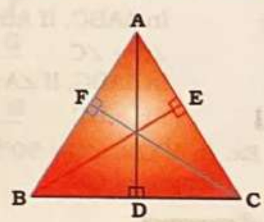
(A) Area of $\Delta ABC = \frac{1}{2}$ × base × height

To calculate the area of ΔABC, we take any of the sides as base and the corresponding perpendicular from the vertex to the base as the height.

In ΔABC,
$AD \perp BC$, $BE \perp AC$ and
$CF \perp AB$.

$ar(\Delta ABC) = \frac{1}{2} \times BC \times AD$

$= \frac{1}{2} \times AC \times BE = \frac{1}{2} \times AB \times CF$

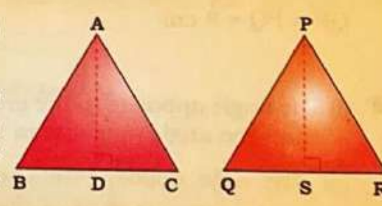


(i) If the height of the two triangles is equal, the ratio of their areas is proportional to the ratio of their bases.

- If AD = PS, then

Ar (ΔABC) : Ar (ΔPQR)
= BC : QR

(ii) If the bases of the two triangles is equal, the ratio of their areas is proportional to the ratio of their height.



- If BC = QR, then

Ar (ΔABC) : Ar (ΔPQR) = AD : PS

(B) Area of scalene triangle

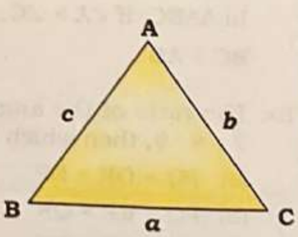
Area of ΔABC

$= \sqrt{s(s-a)(s-b)(s-c)}$

(Heron's formula)

Where, Semi Perimeter (s)

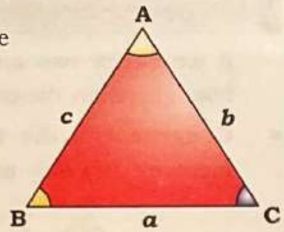
$= \frac{a+b+c}{2}$



(C) When two sides and the angle between these two sides are given

Area of $\Delta ABC = \frac{1}{2}$ ab sinC

$= \frac{1}{2}$bc sinA $= \frac{1}{2}$ac sinB



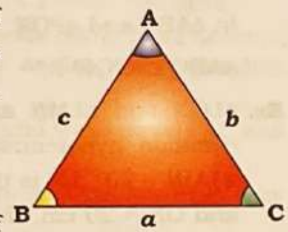
Sine & Cosine Rule

(a) Sine Rule

The ratio of side and sine of opposite angle of a triangle is equal to twice of circum radius.

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R;$$

Where, R is the circumradius of triangle.



(b) Cosine Rule

If two sides and the angle between those sides are given, then we can find the opposite side by Cosine Rule.

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}, \quad \cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Ex. In ΔABC, $\angle B = 30°$ and $\angle C = 45°$. If BC = 50cm, then find the length of AB.

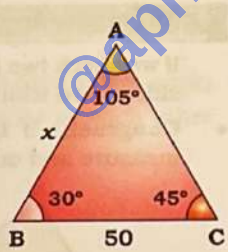
HINTS

$$\Rightarrow \frac{50}{\sin 105°} = \frac{x}{\sin 45°}$$

$$\Rightarrow \frac{50}{\frac{(\sqrt{3}+1)}{2\sqrt{2}}} = \frac{x}{\frac{1}{\sqrt{2}}}$$

$$\left(\because \sin 105° = \frac{(\sqrt{3}+1)}{2\sqrt{2}}\right)$$

$$\Rightarrow x = \frac{100}{(\sqrt{3}+1)} \times \frac{\sqrt{3}-1}{\sqrt{3}-1} = 50\left(\sqrt{3}-1\right)$$



Ex. In ΔABC, AB = 12cm and AC = 10cm, and $\angle BAC = 60°$. What is the length of side BC?

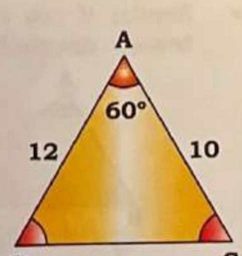
HINTS $\cos A = \frac{AB^2 + AC^2 - BC^2}{2 \times AB \times AC}$

$$\Rightarrow \cos 60° = \frac{(12)^2 + (10)^2 - BC^2}{2 \times 12 \times 10}$$

$$\Rightarrow \frac{1}{2} = \frac{144 + 100 - BC^2}{2 \times 120}$$

$$\Rightarrow 120 = 244 - BC^2$$

$$\Rightarrow BC^2 = 124 \Rightarrow BC = 2\sqrt{31} \text{ cm}$$



Congruency & Similarity in Triangles

Congruency in Triangles

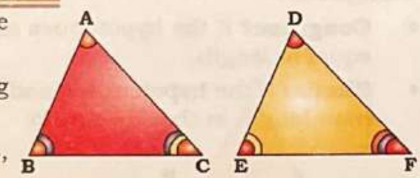
- Two triangles are Congruent, if

(i) Their corresponding angles are equal

$\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$

(ii) Their corresponding sides are equal.

AB = DE, BC = EF, CA = FD



Similarity in Triangles

- Two triangles are similar, if

(i) Their corresponding angles are equal

$\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$

(ii) Their corresponding sides are in the same ratio (or proportion).

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$$

All congruent figures are similar, but not all similar figures are congruent.

Similarity in Regular Figures

Any two figures could be tested to check for similarity and congruence. In the case of regular figures, this is easiest – any two regular figures with the same number of sides will be similar to each other.

For example if we take two regular hexagons, or two circles, or two equilateral triangles, or two squares, or two regular pentagons, each pair of figures will be similar without any further checking required.

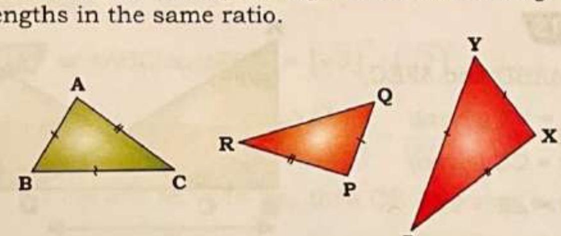
The figure may not always look similar-one should test to make sure.

Conditions for Congruency and Similarity of Triangles

SSS (Side-Side-Side) Test

If we check the three sides of two triangles, then the triangles are

- **Congruent** if three pairs of sides of the two triangles are equal in length.
- **Similar** if the corresponding sides of two triangles have lengths in the same ratio.



In ΔABC and ΔPQR: AB = PQ, BC = QR, AC = PR

$\therefore \Delta ABC \cong \Delta PQR$

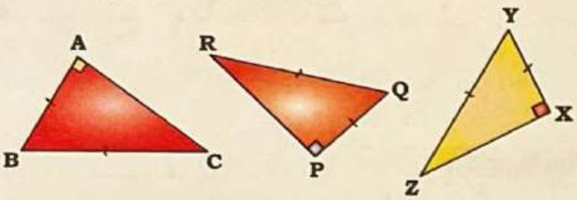
In ΔPQR and ΔXYZ: $\frac{PQ}{XY} = \frac{QR}{YZ} = \frac{PR}{XZ}$

$\therefore \Delta PQR \sim \Delta XYZ$

Hypotenuse Side Test

If we check the sides of two right-angled triangles, then the triangles are

- **Congruent** if the hypotenues and one shorter sides are equal in length.
- **Similar** if the hypotenuses and one pair of shorter sides have length in the same ratio.



In ΔABC and ΔPQR: $AB = PQ$, $BC = RQ$

$\therefore \Delta ABC \cong \Delta PQR$

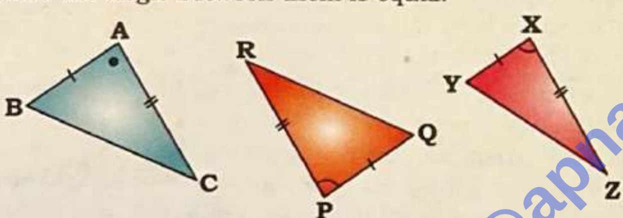
In ΔPQR and ΔXYZ: $\frac{PQ}{XY} = \frac{QR}{YZ}$

$\therefore \Delta PQR \sim \Delta XYZ$

SAS (Side-Angle-Side) Test

If we check two side and its included angle of two triangles then the triangle are

- **Congruent** if the two sides are equal in length and the angle between equal sides is also equal.
- **Similar** if the two sides have lengths in the same ratio and the angle between them is equal.



In ΔABC and ΔPQR: $AB = PQ$, $AC = PR$, $\angle A = \angle P$

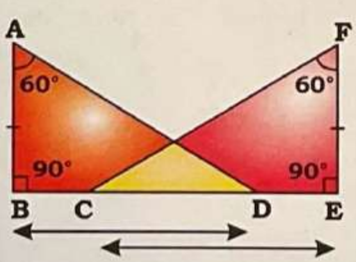
$\therefore \Delta ABC \cong \Delta PQR$

In ΔPQR and ΔXYZ: $\frac{PQ}{XY} = \frac{QR}{YZ}$, $\angle P = \angle X$

$\therefore \Delta PQR \sim \Delta XYZ$

Ex. In ΔABD and ΔFEC, $\angle BAD = 60°$, $l(BD) = l(EC)$, $\angle ABD = \angle FEC = 90°$, and $l(AB) = l(FE)$. Find the ratio of $\angle BAD$ and $\angle FCE$.

HINTS



In ΔABD and ΔFEC,

$AB = FE$ (given)

$BD = CE$ (given)

$\angle B = \angle E = 90°$

$\Delta ABD \cong \Delta FEC$ (from SAS)

$\Rightarrow \angle F = 60°$

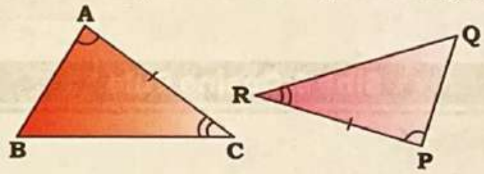
$\therefore \angle FCE = 180° - 90° - 60° = 30°$

$\therefore \frac{\angle BAD}{\angle FCE} = \frac{60°}{30°} = \frac{2}{1} = 2 : 1$

ASA (Angle-Side-Angle) Test

If we check two angles and the included side of two triangles then the triangle are

- **Congruent** if the two pairs of angle have the same measure and only one side are equal in length.

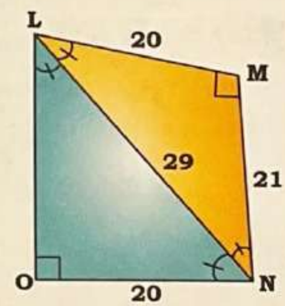


In ΔABC and ΔPQR: $BC = QR$, $\angle A = \angle P$, $\angle C = \angle R$

$\Delta ABC \cong \Delta PQR$

Ex. ΔLON and ΔLMN are two right-angled triangles with common hypotenuse LN such that $\angle LON = 90°$ and $\angle LMN = 90°$. LN is the bisector of $\angle OLM$. If $LN = 29$ cm and $ON = 20$ cm, then what is the perimeter (in cm) of ΔLMN?

HINTS



In ΔLON and ΔLMN,

$\angle LON = \angle LMN = 90°$

$LN = LN$ (common side)

$\angle OLN = \angle MLN$ (angle bisector)

$\therefore \Delta LON \cong \Delta LMN$ (By ASA)

In ΔLMN,

$MN^2 = 29^2 - 20^2 \Rightarrow MN = 21$

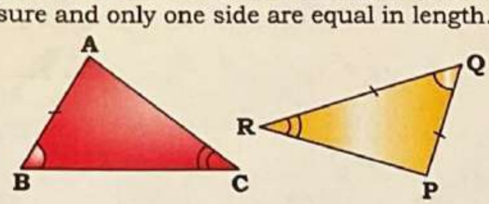
$\therefore$ Perimeter of $\Delta LMN = 29 + 20 + 21$

$= 70$ cm

AAS (Angle-Angle-Side) Test

If we check two angles and a corresponding non included side of two triangles then the triangle are

- **Congruent** if the two pairs of angles have the same measure and only one side are equal in length.



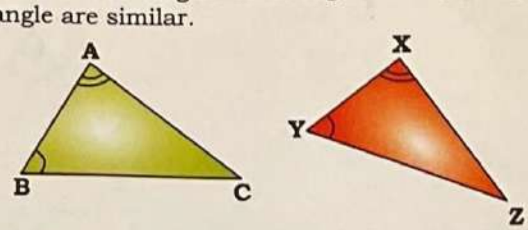
In ΔABC and ΔPQR: $AB = PQ$, $\angle B = \angle Q$, $\angle C = \angle R$

$\Delta ABC \cong \Delta PQR$

AA Test

If we check the angles of two triangles then the triangles are

- Similar if two angles of triangle are equal then both triangle are similar.

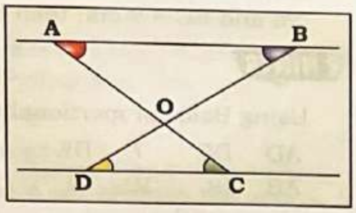


In ΔABC and ΔPQR: $\angle A = \angle X$, $\angle B = \angle Y$

$\Delta ABC \sim \Delta XYZ$

Spotting Similarity & Congruency

- Identifying similarity (and congruency) is crucial in geometry, especially for visualizing problems.
- A key to recognizing similarity is spotting equal angles.



Case 01: Parallel Lines

- When you see two parallel lines intersected by transversals, immediately look for similar triangles.
- This is because parallel lines create many pairs of equal angles (e.g., alternate interior angles, corresponding angles).

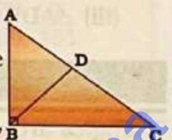
Given parallel lines AB and CD, and transversals AC and BD intersecting at O:

- Triangles ΔAOB and ΔCOD are similar because they have two pairs of equal angles:
  - ∠OAB = ∠OCD (Alternate interior angles)
  - ∠OBA = ∠ODC (Alternate interior angles)
  - ∠AOB = ∠COD (Vertically opposite angles)

Case 02: Altitude to the Hypotenuse in a Right Triangle:

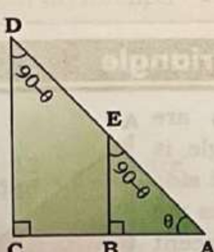
When you draw an altitude from the right angle vertex to the hypotenuse of a right-angled triangle, it creates three similar triangles:

- The original right triangle.
- The two smaller triangles formed by the altitude.

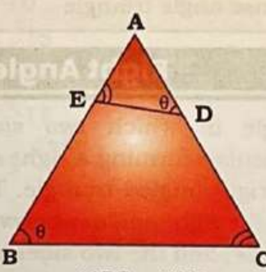


- Example (using ΔABC right-angled at B, with BD as altitude to AC):
- ΔABC ~ ΔADB (They share ∠A, and both have a right angle).
- ΔABC ~ ΔBDC (They share ∠C, and both have a right angle).
- Therefore, all three triangles are similar to each other: ΔABC ~ ΔADB ~ ΔBDC.

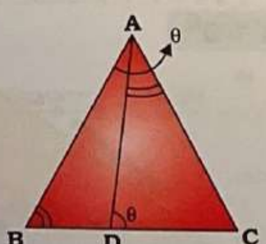
Some Similar Figure

- 

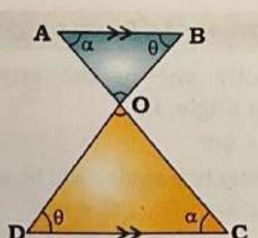
ΔABE ~ ΔACD



ΔADE ~ ΔABC

- 

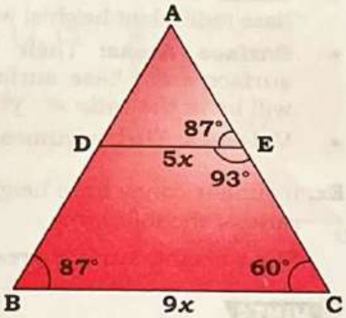
∠ADC = ∠BAC = θ
∠DAC = ∠ABC (Given)
ΔABC ~ ΔADC



ΔAOB ~ ΔCOD

Ex. In ΔABC, ∠B = 87° and ∠C = 60°. Points D and E are on the sides AB and AC, respectively, such that ∠DEC = 93° and DAE is

HINTS



∠DEA = ∠ABC

∠A = Common

⇒ ΔAED ~ ΔABC

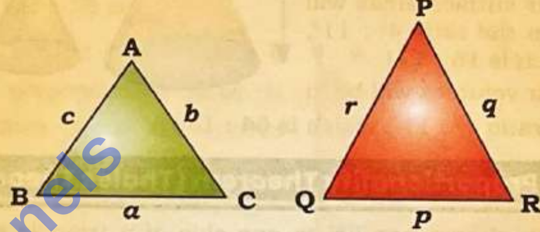
$\Rightarrow \frac{AE}{AB} = \frac{ED}{BC} = \frac{AD}{AC}$

$\Rightarrow \frac{AE}{14.4} = \frac{5}{9}$

⇒ AE = 8 cm

Properties in Similar triangles

If ΔABC and ΔPQR are similar, then



(i) $\frac{a}{p} = \frac{b}{q} = \frac{c}{r}$

(ii) Ratio of corresponding sides
= Ratio of perimeter
= Ratio of semi-perimeter(s)
= Ratio of corresponding medians
= Ratio of inradius
= Ratio of circumradius

(iii) Ratio of area
= Ratio of square of corresponding sides
= Ratio of square of perimeter
= Ratio of square of semi-perimeter
= Ratio of square of corresponding medians
= Ratio of square of inradius
= Ratio of square of circumradius

Ex. If the ratio of corresponding sides of two similar triangles is $\sqrt{3} : \sqrt{2}$, then what is the ratio of the area of the two triangles?

HINTS $ar(\Delta ABC) : ar(\Delta PQR) = (\sqrt{3})^2 : (\sqrt{2})^2 = 3 : 2$

Ex. Let ΔABC ~ ΔQPR and $\frac{ar(\Delta ABC)}{ar(\Delta QPR)} = \frac{64}{169}$. If AB = 10 cm, BC = 7 cm and AC = 16 cm, then QR (in cm) is equal to:

HINTS Here, ΔABC ~ ΔQPR & $\frac{ar(\Delta ABC)}{ar(\Delta QPR)} = \frac{64}{169}$

$\Rightarrow \frac{AC}{QR} = \sqrt{\frac{64}{169}} \Rightarrow \frac{16}{QR} = \frac{8}{13}$

$\Rightarrow QR = \frac{13 \times 16}{8} = 26$ cm

Similarity of Cones

If two cones are similar and their heights (a linear dimension) are in the ratio $x : y$.

- **Linear Dimensions:** Their other linear dimensions (like base radii, slant heights) will also be in the same ratio $x : y$.
- **Surface Areas:** Their various surface areas (total surface area, base surface area, curved surface area) will be in the ratio $x^2 : y^2$.
- **Volumes:** Their volumes will be in the ratio $x^3 : y^3$.

Ex. If similar cones have heights in the ratio 4 : 11. Find the ratio of the following:

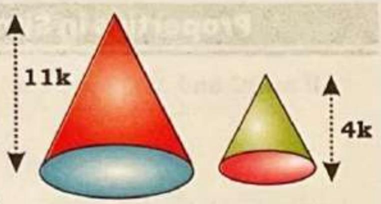
Slant height, surface area and volume

HINTS

Their base radii and slant heights will also be in the ratio 4 : 11.

Their surface areas will be in the ratio $4^2 : 11^2$, which is 16 : 121.

Their volumes will be in the ratio $4^3 : 11^3$, which is 64 : 1331.



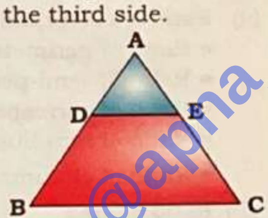
Basic Proportionality Theorem (Thales Theorem)

- A line drawn parallel to one side of a triangle divides other two sides in the same ratio.
- If a line divides any two sides of a triangle in the same ratio, the line must be parallel to the third side.

In ΔABC, If $DE \parallel BC$, then

$$\frac{AD}{DB} = \frac{AE}{EC}$$

'OR' If $\frac{AD}{DB} = \frac{AE}{EC}$, then $DE \parallel BC$



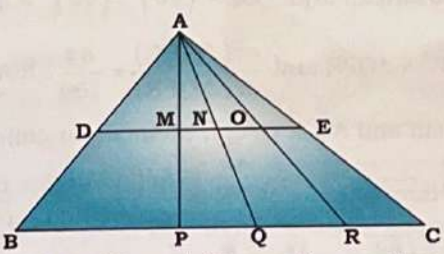
- Some of the results derived from this theorem, we will use, are as follows:

(i) $\frac{AD}{BD} = \frac{AE}{EC}$ (ii) $\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$

(iii) $\Delta ADE \sim \Delta ABC$

(iv) $\frac{Ar(\Delta ADE)}{Ar(\Delta ABC)} = \left(\frac{AD}{AB}\right)^2 = \left(\frac{AE}{AC}\right)^2 = \left(\frac{DE}{BC}\right)^2$

- A line drawn parallel to one side of a triangle divides the median, the angle bisector and the altitude of triangle in the sain ratio it divides the other two sides of the triangle.



In ΔABC, AP, AQ and AR are the median, the angle bi-sector and altitude respectively and $DE \parallel BC$, then

$$\frac{AD}{DB} = \frac{AE}{EC} = \frac{AM}{MP} = \frac{AN}{NQ} = \frac{AO}{OR}$$

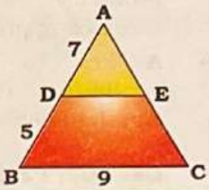
Ex. In ΔABC, D and E are the points on sides AB and AC, respectively such that $\angle ADE = \angle B$. If AD = 7 cm BD = 5 cm and BC = 9 cm, then DE (in cm) is equal to____

HINTS

Using Basic proportionality theorem,

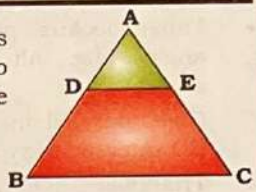
$$\frac{AD}{AB} = \frac{DE}{BC} \Rightarrow \frac{7}{12} = \frac{DE}{9}$$

$$\Rightarrow DE = \frac{63}{12} = 5.25 \text{ cm}$$



Mid-Point Theorem

The line segment joining the mid-points of any two sides of a triangle is parallel to the third side and is half of the magnitude of third side and vice-versa.



- If D and E are mid-points of AB and AC, respectively, then $DE \parallel BC$ and $DE = \frac{BC}{2}$
- $DE \parallel BC$ and $DE = \frac{BC}{2}$, then D and E are the mid-points of AB and AC respectively. In this case

(i) $\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC} = \frac{1}{2}$ (ii) $\frac{AD}{DB} = \frac{AE}{EC} = 1$

(iii) $\Delta ADE \sim \Delta ABC$ (iv) $\frac{Ar(\Delta ADE)}{Ar(\Delta ABC)} = \frac{1}{4}$

Types of Triangle

Triangles are classified on the basis of angles and sides–

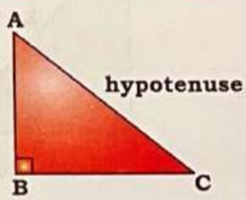
Types of Triangle

On the basis of Angle
- Right angle triangle
- Acute angle triangle
- Obtuse angle triangle

On the basis of Sides
- Scalene triangle
- Isosceles triangle
- Equilateral triangle

Right Angle Triangle

A triangle in which two sides are perpendicular, forming a right angle, is called a right-angled triangle. The side opposite the right angle is known as the hypotenuse, and the two sides adjacent to the right angle are called legs.



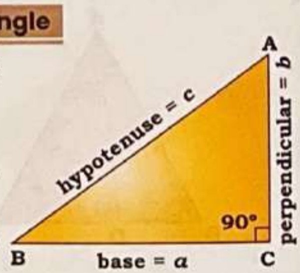
Properties of a Right angle Triangle

(i) Exactly one of the angle is right angle, i.e.

$\angle B = 90°$

- Exactly two angles will be acute. i.e. $\angle A < 90°$, $\angle C < 90°$
- One angle is equal to the sum of other two angle, i.e.

$\angle B = \angle A + \angle C = 90°$



(ii) (a) **Pythagoras Theorem:** In a right triangle, the square of the hypotenuse is equal to the sum of the square of the other two sides.

If a, b and c be three sides of a right-angled triangle, then according to the Pythagoras Theorem,

$$a^2 + b^2 = c^2$$

For Example: $3^2 + 4^2 = 5^2$

(b) **Pythagorean Triplets:** A set of three integers a, b, c which satisfy Pythagoras Theorem ($a^2 + b^2 = c^2$) or are the sides of a right angle triangle is called Pythagorean triplets.

- The smallest Pythagorean triplet is (3, 4, 5).
- If (a, b, c) be a Pythagorean triplets, then (ak, bk, ck) or $\left(\frac{a}{k}, \frac{b}{k}, \frac{c}{k}\right)$ will also be the Pythagorean triplet.

Following Pythagorean triplets are frequently used in the examinations.

3 4 5 $\xrightarrow{\times 1/2}$ 1.5, 2, 2.5
3 4 5 $\xrightarrow{\times 3}$ 9, 12, 15
3 4 5 $\xrightarrow{\times 2}$ 6, 8, 10
3 4 5 $\xrightarrow{\times 4}$ 12, 16, 20
3 4 5 $\xrightarrow{\times 5}$ 15, 20, 25

5, 12, 13, $\xrightarrow{\times 2}$ 10, 24, 26

7, 24, 25, $\xrightarrow{\times 1/2}$ 3.5 12, 12.5
7, 24, 25, $\xrightarrow{\times 2}$ 14, 48, 50
7, 24, 25, $\xrightarrow{\times 3}$ 21, 72, 75

1, 1, $\sqrt{2}$
9, 40, 41
11, 60, 61
12, 35, 37
16, 63, 65
13, 84, 85
20, 21, 29
39, 80, 89
36, 77, 85
65, 72, 97
20, 99, 101

$(2n, n^2 - 1, n^2 + 1)$ $\left(\frac{n^2+1}{2}, \frac{n^2-1}{2}, n\right)$

Tricks for Making Triplets

(a) For even number →

$6 \rightarrow \frac{6}{2} = 3 \xrightarrow{\text{Square}} 9 \begin{cases} \xrightarrow{-1} 8 \\ \xrightarrow{+1} 10 \end{cases}$

$8 \rightarrow \frac{8}{2} = 4 \xrightarrow{\text{Square}} 16 \begin{cases} \xrightarrow{-1} 15 \\ \xrightarrow{+1} 17 \end{cases}$

$12 \rightarrow \frac{12}{2} = 6 \xrightarrow{\text{Square}} 36 \begin{cases} \xrightarrow{-1} 35 \\ \xrightarrow{+1} 37 \end{cases}$

(b) For odd number →

$3 \xrightarrow{\text{Square}} 9 \rightarrow \frac{9}{2} = 4.5 \begin{cases} \xrightarrow{-0.5} 4 \\ \xrightarrow{+0.5} 5 \end{cases}$

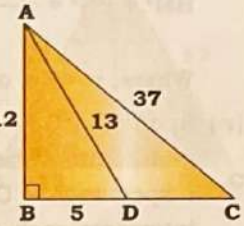
$5 \xrightarrow{\text{Square}} 25 \rightarrow \frac{25}{2} = 12.5 \begin{cases} \xrightarrow{-0.5} 12 \\ \xrightarrow{+0.5} 13 \end{cases}$

$7 \xrightarrow{\text{Square}} 49 \rightarrow \frac{49}{2} = 24.5 \begin{cases} \xrightarrow{-0.5} 24 \\ \xrightarrow{+0.5} 25 \end{cases}$

"(20, 21, 29) and (20, 99, 101) are triplets."

Ex. ΔABC is right angled at B and D is a point of BC such that BD = 5 cm, AD = 13 cm and AC = 37 cm, then find the length of BC in cm.

HINTS



AB = 12 cm (By triplet of 5, 12, 13)

$(12)^2 + (BC)^2 = (37)^2$

$\Rightarrow (BC)^2 = (37)^2 - (12)^2$

$\Rightarrow BC^2 = 25 \times 49$

$\Rightarrow BC = 5 \times 7 = 35$

Alternatively

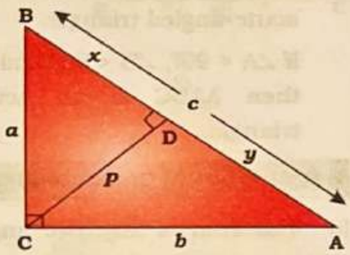
(5, 12, 13) and (12, 35, 37) are triplet.

With the help of triplet, AB = 12cm satisfy the given condition.

$\therefore$ BC = 35 cm

(iii) In a right angle ΔABC, if a perpendicular (CD) is drawn from the vertex which is right angle (∠C) to the hypotenuse (AB), then,

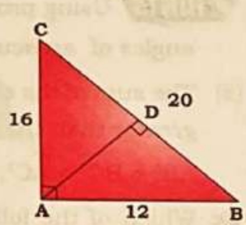
ΔABC ~ ΔBCD ~ ΔACD



- $a^2 = x \times c$
- $p^2 = x \times y$
- $b^2 = y \times c$
- $a \times b = p \times c$
- $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$

Ex. In ΔABC, ∠A = 90°, AD⊥BC at D. If AB = 12 cm and AC = 16 cm, then what is the length (in cm) of BD?

HINTS



$AB^2 = BD \times BC$

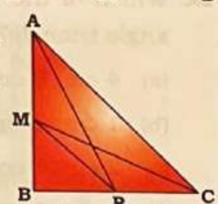
$\Rightarrow 12^2 = BD \times 20$

$\Rightarrow 144 = BD \times 20$

$\Rightarrow$ BD = 7.2cm

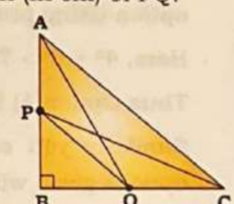
(iv) If P and M are the two points on the sides BC and AB respectively of ΔABC, right-angled at B, then

$AP^2 + MC^2 = AC^2 + MP^2$



Ex. Points P and Q are on the sides AB and BC respectively of a ΔABC, right angled at B. If AQ = 11 cm, PC = 8 cm, and AC = 13 cm, then find the length (in cm) of PQ.

HINTS



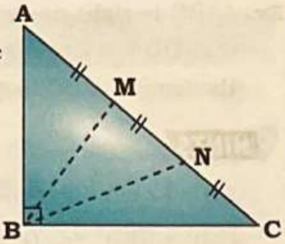
$AQ^2 + PC^2 = AC^2 + PQ^2$

$\Rightarrow 121 + 64 = 169 + PQ^2$

$\Rightarrow 16 = PQ^2 \Rightarrow PQ = 4$

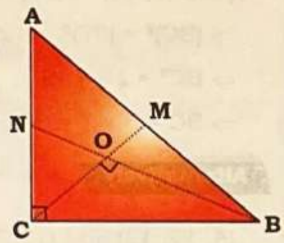
(v) ΔABC is a right angle triangle, right angle at B. If M, N are points on AC such that AM = MN = NC then

$$BM^2 + BN^2 = \frac{(n-1)(2n-1)AC^2}{6n}$$



Where, n = No. of equal parts

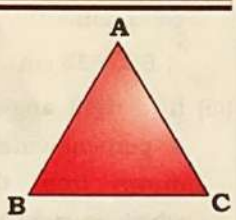
(vi) In ΔABC, $\angle C = 90°$, M and N are mid-points of sides AB and AC respectively. CM and BN intersects each other at O and $\angle BOC$ is 90° then $BN = \frac{BC\sqrt{6}}{2}$



Acute Angle Triangle

If each of the angles of a triangle is acute, then the triangle is called an acute-angled triangle.

If $\angle A < 90°$, $\angle B < 90°$ and $\angle C < 90°$, then ΔABC is an acute-angled triangle.



Properties of an acute angle triangle

(i) The sum of any two angles of the triangle is always greater than the third angle.

$\angle A + \angle B > \angle C$, $\angle B + \angle C > \angle A$, $\angle C + \angle A > \angle B$

Ex. Which one of the following cannot be the ratio of angles in acute angle triangle?

(a) 2 : 5 : 8 (b) 4 : 1 : 4

(c) 2 : 3 : 4 (d) 1 : 1 : 1

HINTS Using property (i), Option (a) can't be the ratio of angles of an acute angle triangle.

(ii) The sum of the squares of any two sides of the triangle is greater than the square of the third side.

$AB^2 + BC^2 > AC^2$, $BC^2 + AC^2 > AB^2$, $AC^2 + AB^2 > BC^2$

Ex. Which of the following can be the 3 sides of an acute angle triangle?

(a) 4 cm, 8 cm, 7 cm

(b) 1 cm, 2 cm, 3 cm

(c) 2 cm, 4 cm, 3 cm

(d) 9 cm, 12 cm, 15 cm

HINTS In such type of question you have to check each option using property (ii).

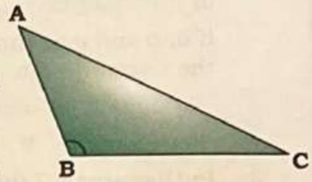
Here, $4^2 + 8^2 > 7^2$, $8^2 + 7^2 > 4^2$ & $7^2 + 4^2 > 8^2$

Thus Option(A) is right answer.

Similarly you can check other options but all other options given will not satisfy the property.

Obtuse Angle Triangle

If one of the angle of a triangle is obtuse, then the triangle is called an obtuse-angled triangle.



ΔABC is obtuse angle triangle.

Properties of an obtuse-angled triangle

(i) Exactly one of the angle of the triangle is obtuse and other two angles are acute.

$\angle B > 90°$ and $0° < \angle A, \angle C < 90°$

(ii) The sum of the two acute angles of the triangle is less than the obtuse angle.

$(\angle A + \angle C) < \angle B$

(iii) The sum of the square of two smaller sides is less than the square of the third side (the largest side).

- $AB^2 + BC^2 < AC^2$

Ex. In an obtuse angle ΔPQR, $\angle Q$ is obtuse angle if side PQ = 11cm, QR = 15cm then find minimum possible integer length of side PR?

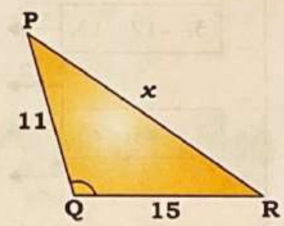
HINTS

$11^2 + 15^2 < x^2$

$\Rightarrow 121 + 225 < x^2$

$\Rightarrow 346 < x^2$

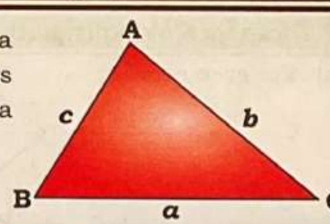
$\Rightarrow x_{min} = 18.60 \sim 19$ cm



Scalene Triangle

If all of the three sides of a triangle are of different lengths then the triangles is called a scalene triangle.

$AB \neq BC \neq CA \Rightarrow a \neq b \neq c$



Properties of a scalene triangle

(i) No two sides are equal in length.

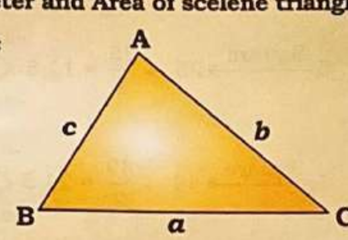
$AB \neq BC \neq CA$ or $a \neq b \neq c$

(ii) No two angles are equal, i.e., $\angle A \neq \angle B \neq \angle C$

(iii) Triangle may be one of the acute angled, right angled or obtuse angled.

(iv) **Perimeter, Semi-perimeter and Area of scelene triangle**

- Perimeter = $a + b + c$
- Semi-perimeter(s) $= \frac{a+b+c}{2}$
- Area of ΔABC $= \sqrt{s(s-a)(s-b)(s-c)}$



Ex. The sides of the scalene triangle ABC are in the ratio 3 : 5 : 6 and the semi-perimeter is 42 cm, what is the difference of the largest and the smallest sides of the triangle?

HINTS Semi-perimeter = $\frac{3x+5x+6x}{2}$

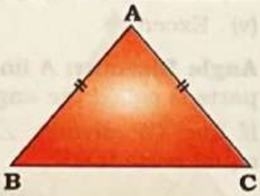
$\Rightarrow 42 = 7x \Rightarrow x = 6$

Required Difference = $6x - 3x = 3 \times 6 = 18$ cm

Isosceles Triangle

If two sides of a triangle are equal in length, then the triangle is called an isosceles triangle.

If AB = AC ≠ BC, then ΔABC is an isosceles triangle.



Properties of an isosceles triangle

(i) The length of the two sides are equal, but the length of third side can be smaller or larger than the equal sides.

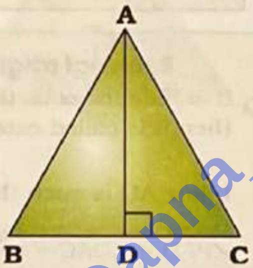
AB = AC & BC > AB or BC < AB

(ii) Two angles opposite to the equal sides are equal.

∠B = ∠C (opposite to the side AC and AB)

(iii) In an isosceles ΔABC, the median, the angle bisector, perpendicular bisector and the altitude of unequal side is the same and it divides ΔABC in two congruent (equal) right-angled triangle.

In ΔABC, AD is the median, the angle bisector, perpendicular bisector and the altitude.

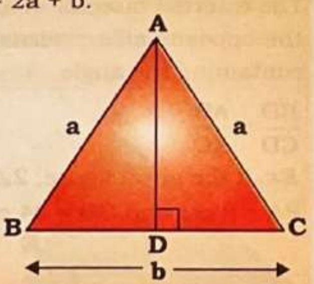


AD ⊥ BC and BD = DC, then

ΔABD ≅ ΔACD

(iv) **Perimeter, semi-perimeter and Area**

- Perimeter = AB + BC + CA = 2a + b.
- Semi perimeter $= \frac{AB+BC+CA}{2} = a + \frac{b}{2}$
- $AD = \frac{\sqrt{4a^2 - b^2}}{2}$
- Area of $\Delta ABC = \frac{b}{4}\sqrt{4a^2 - b^2}$



In isosceles triangle ABC, ΔABD and ΔACD are the two congruent right-angled triangle. Use Pythagorean triplets in these right angled triangle.

Ex. The perimeter of an isosceles triangle is 50 cm. If the base is 18 cm, then find the length of equal sides.

HINTS Perimeter, 2a + b = 50

$\Rightarrow 2a + 18 = 50$

$\Rightarrow a = 16$

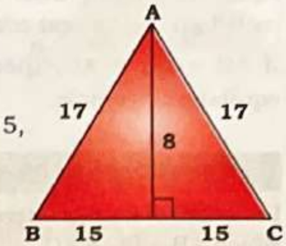
Ex. The length of perpendicular dropped on the base of an isosceles triangle is 8 cm. If its perimeter is 64 cm, what is the area of this triangle?

HINTS

Check 30 + 34 = 64

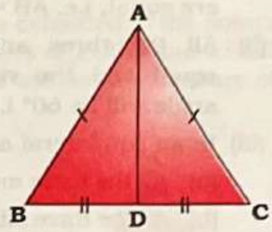
[Apply concept of triplet, (8, 15, 17) to solve quickly]

Area $= \frac{1}{2} \times 30 \times 8 = 120$ cm²



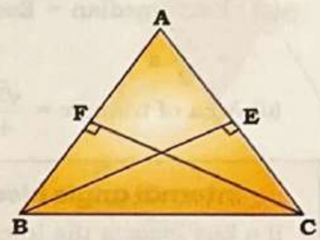
(v) If a line (which joins the common vertex of two equal sides of a triangle) bisects the base then that line is perpendicular to the base and vice versa.

If AB = AC and BD = DC, then ∠ADB = ∠ADC = 90°

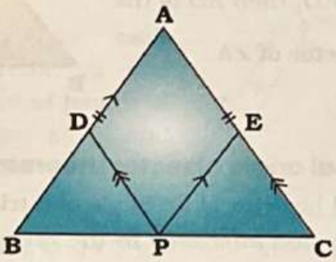


(vi) • If AB = AC and BE ⊥ AC and CF ⊥ AB, then BE = CF

• If AB = AC, E and F are the mid-points, then BE = CF



(vii) ABC is an isosceles triangle (AB = AC). If P is a point on the side BC and DP || AC and EP || AB then DP + PE = AB = AC

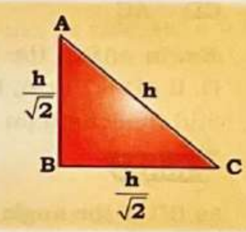


Isosceles Right Angle Triangle

- Ratio of sides

$\Rightarrow AB : BC : AC = 1 : 1 : \sqrt{2}$

- Area $= \frac{\text{Hypotenuse}^2}{4}$
- Perimeter $= h(\sqrt{2}+1)$



Ex. If the hypotenuse of the isosceles right angle triangle equals the side of a square whose area is 256 cm². Find the area of triangle.

HINTS Side of square = $\sqrt{256} = 16$.

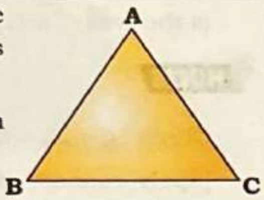
∴ side of square = $h = 16$

Area $= \frac{h^2}{4} = \frac{16^2}{4} = \frac{256}{4} = 64$ cm².

Equilateral Triangle

If all the three sides of a triangle are equal in length, then the triangle is called an equilateral triangle.

If AB = BC = AC, then DABC is an equilateral triangle.

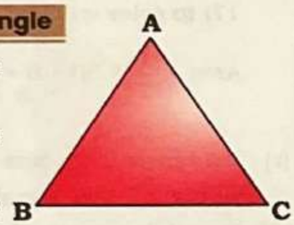


Properties of an equilateral triangle

Let ΔABC is an equilateral triangle in which AB = BC = AC

(i) The length of all the three sides are equal, i.e. AB = BC = AC

(ii) All the three angles will be equal and the value of each angle will be 60° i.e. $\angle A = \angle B = \angle C$



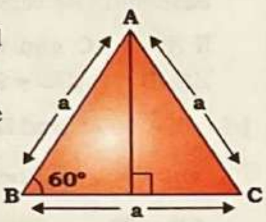
(iii) In an equilateral ΔABC,

(a) All the three medians are equal

(b) All the three altitudes are equal

(c) Each median = Each altitude $= \frac{\sqrt{3}}{2}a$

(d) Area of triangle $= \frac{\sqrt{3}}{4}a^2$



(iv) In equilateral ΔABC, incentre, circumcentre, orthocentre and centroid are at the same point.

(v) If altitudes of a triangle are equal, the triangle is an equilateral triangle.

Centres of Triangle

We will study 5 types of centres of a triangle

(i) Incentre (ii) Circumcenter

(iii) Orthocentre (iv) Centroid

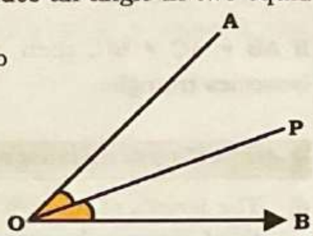
(v) Excentre

Angle Bisector: A line which divides an angle in two equal parts is called the angle bisector.

If line OP divides $\angle AOB$ in two equal parts then

$\angle AOP = \angle BOP = \frac{\angle AOB}{2}$, then

OP is an angle bisector.



Angle bisector of an angle of a triangle

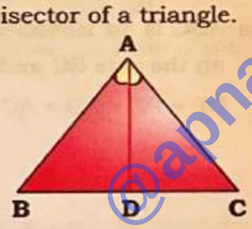
Interior or Internal | Exterior or External

Internal angle bisector of a triangle

If a line bisects the Internal angle of a triangle, then it is called internal angle bisector of a triangle.

If line AD is such that

$\angle BAD = \angle CAD$, then AD is the interior bisector of $\angle A$.



Internal angle bisector theorem

The Internal bisector of an angle of a triangle divides the opposite side internally in the ratio of the sides containing the angle, i.e.

$$\frac{BD}{CD} = \frac{AB}{AC}$$

Ex. In ΔABC, the bisector of $\angle B$ meets AC at point D. If AB = 12 cm, BC = 18 cm and AC = 15 cm, then find the length (in cm) of AD.

HINTS

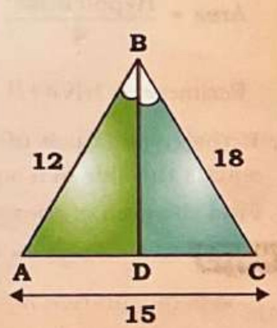
As BD is the angle bisector of $\angle ABC$

$\Rightarrow \frac{AB}{BC} = \frac{AD}{DC} \Rightarrow \frac{12}{18} = \frac{AD}{DC}$

$\Rightarrow AD : DC = 2 : 3$

Also, AC = 15 cm

$\therefore AD = \frac{15}{(2+3)} \times 2 = 6$ cm



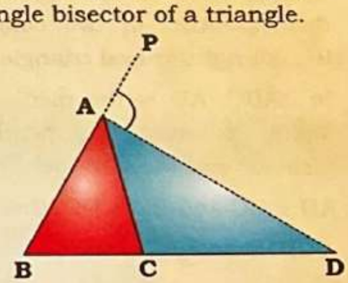
External angle bisector of a triangle

If a line bisects the exterior angle of a triangle, then it is called external angle bisector of a triangle.

If line AD is such that

$\angle PAD = \angle DAC = \left(\frac{\pi - A}{2}\right)$,

then AD is the external bisector of $\angle A$.

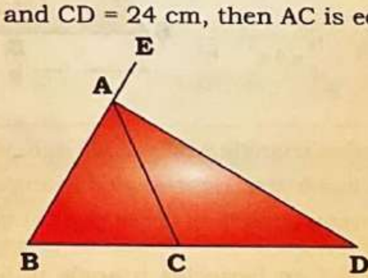


External angle bisector theorem

The external bisector of an angle of a triangle divides the opposite side externally in the ratio of the sides containing the angle

$$\frac{BD}{CD} = \frac{AB}{AC}$$

Ex. If the given figure, $2\angle EAD = \angle EAC$, BC = 40 cm, BA = 8 cm and CD = 24 cm, then AC is equal to



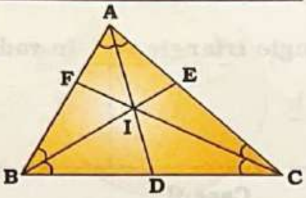
HINTS

Since AD is angle bisector

$\frac{AB}{AC} = \frac{BD}{CD} \Rightarrow \frac{8}{AC} = \frac{64}{24} \Rightarrow AC = 3\text{cm}$

Incentre

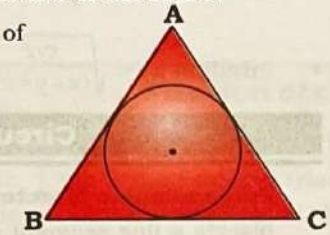
The point of intersection of the internal bisector of the angles of a triangle is called the incentre.



- If AD, BE and CF are the internal angle bisectors of the ∠A, ∠B and ∠C respectively, then all the three bisectors of the angles pass through a point, which is called the incentre of the ΔABC.
- The incentre is the centre of the incircle of a triangle.
- It is generly denoted by 'I'.

I → Incentre of ΔABC

I → Centre of the circle.



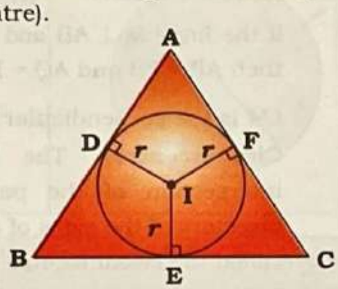
Properties of incentre

(i) All the three internal bisectors of the angles of a triangle pass through a point (incentre).

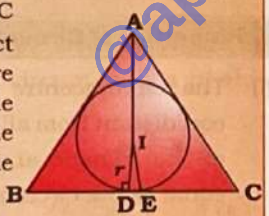
(ii) The incentre of the triangle is equidistance from all the three sides of the triangle and the distance is equal to the inradius.

If circle touches the side AB, BC and AC at points D, E and F respectively and I be the incentre, then

ID = IE = IF (inradius)



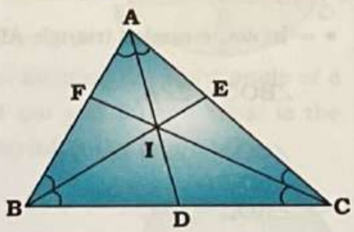
The bisector of ∠A of ΔABC may or may not intersect side BC at point E where the incircle touches the side BC of the triangle and the same is true for other angle bisectors.



The bisector of ∠A meets BC at D and the side BC touches the circle at E.

(iii)

- $\angle BIC = 90° + \frac{\angle A}{2}$
- $\angle AIC = 90° + \frac{\angle B}{2}$
- $\angle AIB = 90° + \frac{\angle C}{2}$



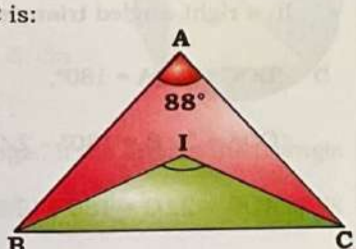
Ex. In ΔABC, ∠A = 88°. If I is the incentre of the triangle, then the measure of ∠BIC is:

HINTS

$\angle BIC = 90° + \frac{\angle A}{2}$

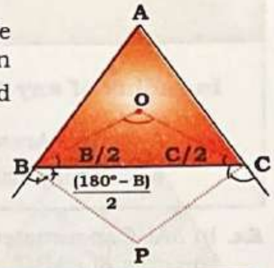
$= 90° + \frac{88°}{2}$

$= 90° + 44° = 134°$



(iv) The angle between the external bisectors of two angles of a triangle is equal to the difference between right angle and half of the third angle.

$\angle BPC = 90° - \frac{\angle A}{2}$



If in ΔABC, the internal bisectors of ∠B and ∠C meet at O and the external bisectors of ∠B and ∠C meet at P, then ∠BOC + ∠BPC = 180°.

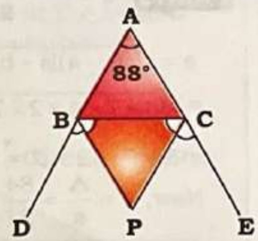
Ex. The sides of ΔABC, AB and AC are extended to the points D and E, respectively. The bisectors of ∠CBD and ∠BCE meet at P. If ∠A = 88°, then what will be the measure of ∠P?

HINTS

$\angle BPC = 90° - \frac{1}{2}\angle A$

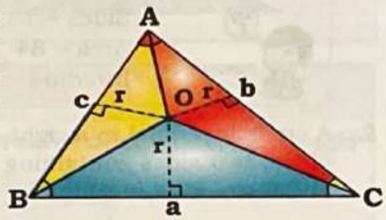
$\angle BPC = 90° - \frac{1}{2} \times 88°$

$\angle BPC = 46°$

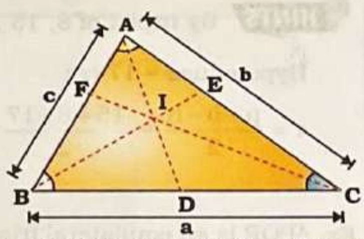


(v) The ratio of area of triangle formed by incentre and two vertex are in ratio in their corresponding sides.

Ar ΔBOC : Ar ΔAOC : Ar ΔAOB = a : b : c



(vi) Incentre divides each angle bisector in the ratio of sum of length of two adjacent sides and opposite side.



Here, AD, BE and CF are angle bisectors of ∠A, ∠B and ∠C respectively.

AI : ID = b + c : a

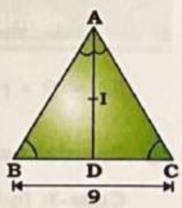
BI : IE = a + c : b

CI : IF = a + b : c

Ex. The perimeter of ΔABC is 24 cm and its side, BC = 9 cm. AD is the bisector of ∠BAC, while I is the incentre AI : ID is equal to:

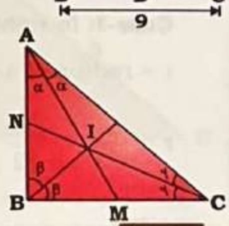
HINTS

$\frac{AI}{ID} = \frac{AB + AC}{BC} = \frac{15}{9}$



(vii) ABC is a right angle triangle at B. If BN and BM are given and I is the incentre of triangle then

$BI^2 = BN \times BM$.



In-radius

In-radius of any triangle	In-radius of right angle triangle	In-radius of equilateral triangle
$r = \frac{\Delta}{s} = \frac{\text{Area}}{\text{Semi perimeter}}$	$r = \frac{p+b-h}{2}$	$r = \frac{a}{2\sqrt{3}} = \frac{\text{Side}}{2\sqrt{3}}$

Ex. In ΔABC, perimeter is 24 cm and in-radius is 7 cm. Find the area of ΔABC.

HINTS $s = 24$ cm, $r = 7$cm

$$r = \frac{ar\Delta ABC}{\text{Semi perimeter}} \Rightarrow 7 = \frac{ar\Delta ABC}{12}$$

$\Rightarrow$ ar $\Delta ABC = 84$ cm²

Ex. Find in-radius of a triangle whose sides are 13 cm, 14cm and 15 cm.

HINTS $s = \frac{13+14+15}{2} = 21$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{21 \times 8 \times 7 \times 6}$$

$$= \sqrt{3 \times 7 \times 4 \times 2 \times 7 \times 2 \times 3}$$

$$= 3 \times 7 \times 2 \times 2 = 84 \text{ cm}^2$$

Now, $r = \frac{\Delta}{s} = \frac{84}{21} = 4$ cm

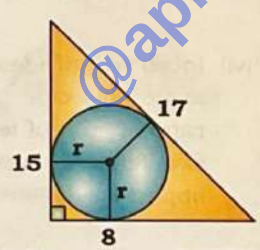
Sides = 13,14,15
Area = 84
Inradius = 4

Ex. A circle inscribed in a right-angled triangle. The lengths of the two sides containing the right angles are 15 cm and 8 cm. Find in radius?

HINTS By triplet of 8, 15 , 17

Hypotenuse = 17 cm

$$r = \frac{p+b-h}{2} = \frac{15+8-17}{2} = 3 \text{ cm}$$



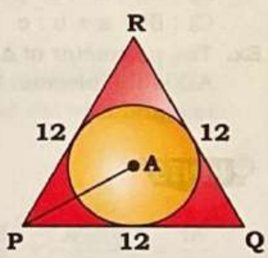
Ex. ΔPQR is an equilateral triangle and the incentre of ΔPQR is point A. If the side of the triangle is 12 cm, then what is the length of PA?

HINTS

$$r = \frac{a}{2\sqrt{3}} = \frac{12 \times \sqrt{3}}{2\sqrt{3} \times \sqrt{3}}$$

$$= 2\sqrt{3} \text{ cm}$$

$$PA = 2 \times r = 2 \times 2\sqrt{3} = 4\sqrt{3} \text{ cm}$$

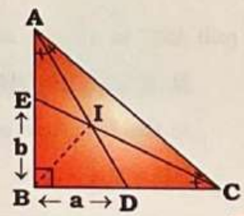


Special cases:

Case-I: In right angle triangle ABC,

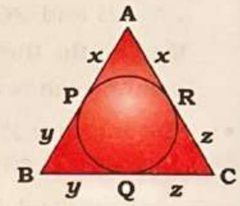
r = radius of ΔABC

$$r = \frac{BI}{\sqrt{2}} = \frac{\sqrt{ab}}{\sqrt{2}}$$



Case-II:

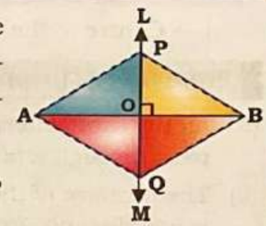
- Semi-perimeter (s) = $x + y + z$
- Area of $\Delta ABC = \sqrt{(x+y+z)xyz}$
- Inradius (r) = $\sqrt{\frac{xyz}{(x+y+z)}}$



Circumcentre

Perpendicular bisector: If a line bisects a line segment perpendicularly, the line is called a perpendicular bisector.

If the line LM $\perp$ AB and AO = OB, then AP = PB and AQ = BQ



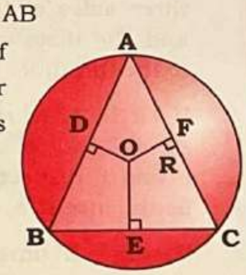
LM is the perpendicular bisector of AB

Circumcentre: The point of intersection of the perpendicular bisectors of the sides of a triangle is called the circumcentre.

OD $\perp$ AB and AD = BD

OE $\perp$ BC and BE = EC

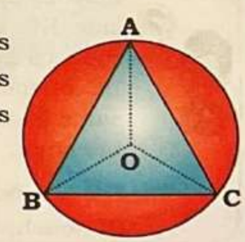
OF $\perp$ AC and CF = FA



Properties of Circumcentre

(i) The circumcentre of a triangle is equidistant from all the three vertices of the triangle and the distance is equal to the circumradius, i.e.

OA = OB = OC = R (circumradius)



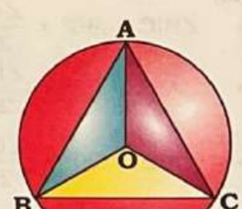
(ii) If O be the circumcentre, then

- In acute-angled triangle ABC,

$\angle BOC = 2\angle A$,

$\angle COA = 2\angle B$ &

$\angle AOB = 2\angle C$

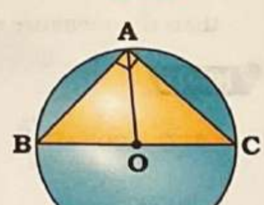


- In a right-angled triangle ABC,

$\angle BOC = 2\angle A = 180°$,

$\angle COA = 2\angle B = 180° - 2\angle C$,

$\angle AOB = 2\angle C = 180° - 2\angle B$

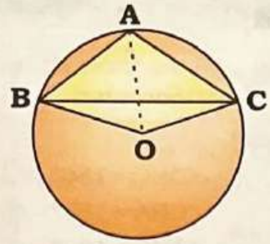


- In an obtuse-angled triangle ABC,

  $\angle BOC = 2(180^\circ - \angle A)$,

  $\angle COA = 2\angle B$ &

  $\angle BOA = 2\angle C$



Ex. From the circumcentre L of ΔXYZ, perpendicular LM is drawn on side YZ. If $\angle YXZ = 60^\circ$, then the measure of $\angle YLM$ is:

HINTS $\angle YXZ = 60^\circ$

LM $\perp$ YZ (Given)

$\Rightarrow \angle YLZ = 2 \times \angle YXZ = 120^\circ$

In ΔYLM & ΔZLM

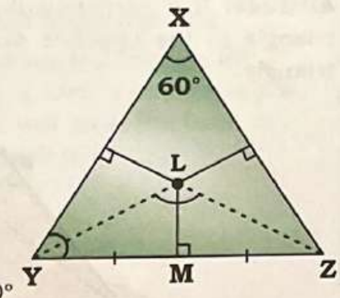
YM = MZ

$\angle YML = \angle LMZ = 90^\circ$

LY = LZ (Circum radius)

$\Rightarrow \Delta YLM \cong \Delta ZLM$ (By RHS)

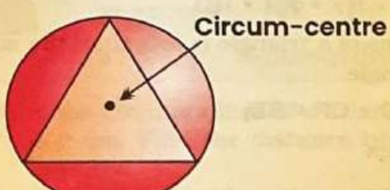
$\therefore \angle YLM = \angle ZLM = \frac{120^\circ}{2} = 60^\circ$



Position of circum-centre

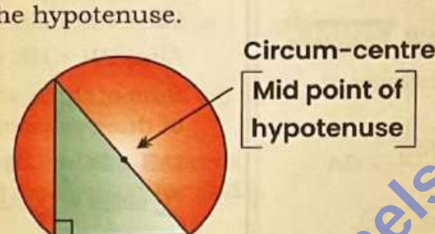
Acute angle triangle

In an acute angle triangle, the circum-centre is always inside the triangle.



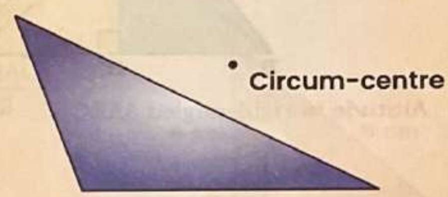
Right angle triangle

In a right-angled triangle, the circum-centre is at the mid-point of the hypotenuse.



Obtuse angle triangle

In an obtuse-angled triangle, the circum-centre is always outside the triangle and will be at the front of the angle which is obtuse.



Circum-radius

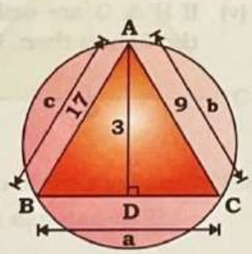
Circum-radius of any triangle	Circum-radius of right angle triangle	Circum-radius of equilateral triangle
$R = \frac{\text{Product of all three sides}}{4 \times \text{Area of the triangle}} = \frac{abc}{4\Delta}$	$R = \frac{\text{Hypotenuse}}{2} = \frac{h}{2}$	$R = \frac{\text{Side}}{\sqrt{3}} = \frac{a}{\sqrt{3}}$

Ex. In a ΔABC, AB = 17 cm, AC = 9 cm, AD is perpendicular on BC & AD = 3 cm. Find the circum radius of this triangle.

HINTS

$$R = \frac{abc}{4\Delta} = \frac{a \times 9 \times 17}{4 \times \frac{1}{2} \times a \times 3}$$

$= 25.5$ cm

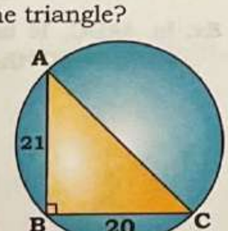


Ex. The lengths of the two sides forming the right angle of a right-angled triangle are 21 cm and 20 cm. What is the radius of the circle circumscribing the triangle?

HINTS

AC = 29 (By triplet of 20, 21, 29)

$$R = \frac{\text{Hypotenous}}{2} = \frac{29}{2} = 14.5 \text{ cm}$$



Ex. ΔABC is an equilateral triangle. If the area of the triangle is $36\sqrt{3}$ then what is the radius of circle circumscribing the ΔABC?

HINTS $\frac{\sqrt{3}}{4}a^2 = 36\sqrt{3}$

$\Rightarrow a^2 = 144$

$\Rightarrow a = 12$

$$\therefore R = \frac{a}{\sqrt{3}} = \frac{12 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}}$$

$= 4\sqrt{3}$ cm

Relationship between semi-perimeter, inradius & circum-radius of right angle triangle

- Area of Δ = r.s

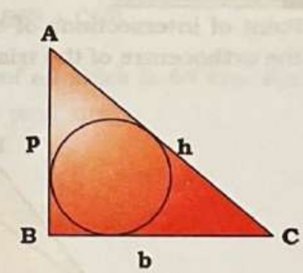
  Put, r = s – 2R

  $\Delta = s(s - 2R)$

- Δ = r.s

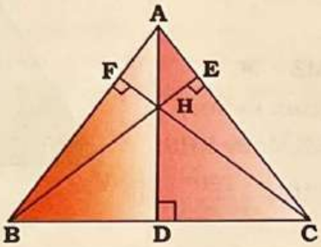
  Put, s = r + 2R

  $\Delta = r(r + 2R) = r^2 + 2Rr$



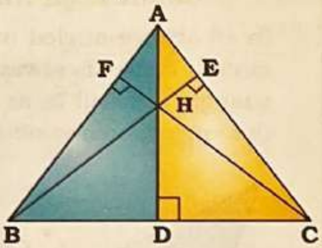
Orthocentre

Altitude: If a perpendicular is drawn from a vertex of a triangle to the opposite side is called the altitude of the triangle.

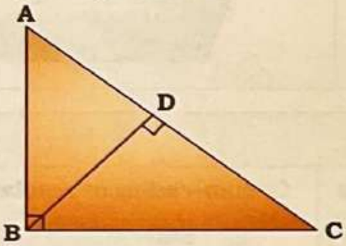


If AD ⊥ BC, therefore AD is the altitude.

(i) Altitudes in acute angle ΔABC

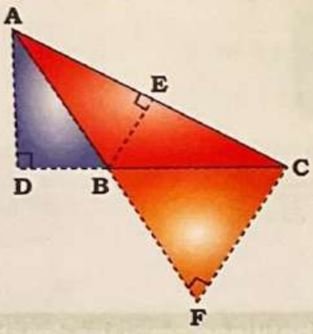


(ii) Altitude in right-angled ΔABC



Two altitudes of a right-angled triangle ABC are side AB and BC and the third altitude is BD.

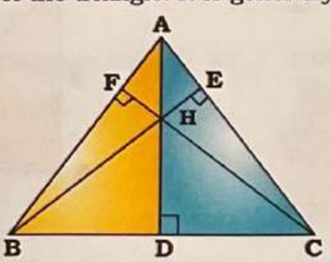
(iii) Altitude in obtuse-angled ΔABC



Three altitudes of ΔABC are AD, BE and CF.

Orthocentre

Point of intersections of the altitudes of a triangle is called the orthocentre of the triangle. It is generally denoted by H.



Properties of Orthocentre

(i) All the three altitudes of a triangle pass through a point (orthocentre)

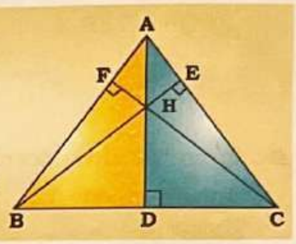
(ii)

$\angle BHC = 180^\circ - \angle A$

$\angle CHA = 180^\circ - \angle B$

$\angle AHB = 180^\circ - \angle C$

It is true for all types of triangle.



Ex. In an obtuse-angled triangle ABC, ∠A is the obtuse angle and O is the orthocentre. If ∠BOC = 54°, then ∠BAC is:

HINTS

$\angle BAC = 180^\circ - 54^\circ = 126^\circ$

(iii) ΔBFH ~ ΔCEH

BH × HE = CH × HF

Also, BH × HE = CH × HF = AH × HD

(iv) Sum of three altitudes of a triangle is less than the sum of three sides of triangle.

(AB + BC + CA) > (AD + CF + BE)

$\sum$ sides > $\sum$ Altitudes

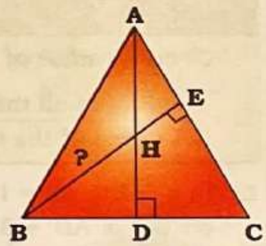
Ex. In DABC, AD and BE are the altitude of the triangle meets at H such that AH = 12 cm, HD = 9 cm, and HE = 4 cm. Find BH?

HINTS

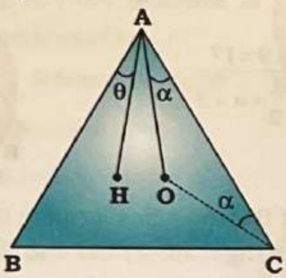
AH × HD = BH × HE

⇒ 12 × 9 = BH × 4

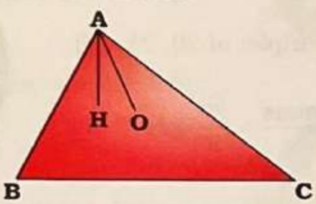
⇒ BH = 27 cm



(v) If H & O are orthocentre & circumcentre (as shown in the figure) then, θ = α



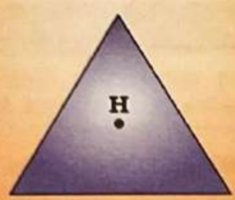
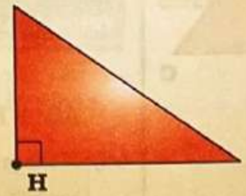
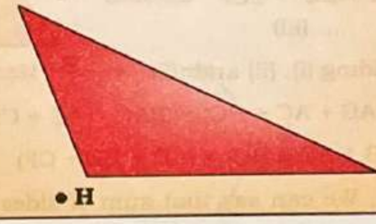
Ex. In ΔABC, H is the orthocenter, O is circumcenter. If ∠BAH = 30° then find ∠OAC?



HINTS Given, ∠BAH = 30°

∠OAC = ∠BAH = 30°

Position of orthocentre

Acute angle triangle	Right angle triangle	Obtuse angle triangle
In an acute-angled triangle the orthocentre is always inside the triangle.	In right-angled triangle the orthocentre is on the vertex of the triangle at which triangle is the right-angled.	In an obtuse-angled triangle the orthocentre is always outside of the triangle and will be at the back of the angle which is obtuse.
		

- Distance between orthocentre and circum-centre of a right angle triangle $= \frac{\text{Hypotenuse}}{2} = \frac{h}{2}$
- Distance between circum-centre and incentre in any triangle $= \sqrt{R^2 - 2Rr}$

Where, R = circum-radius & r = inradius

Ex. If the circumradius of a triangle is 6 cm and inradius is 2 cm. Find the distance between circum-centre and incentre.

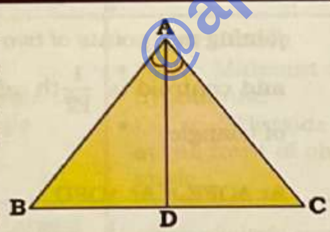
HINTS R = 6, r = 2

$\therefore d = \sqrt{R^2 - 2Rr}$

$= \sqrt{36 - 2\times6\times2} = \sqrt{12} = 2\sqrt{3}$

Centroid

Median: If a line segment from a vertex of a triangle bisects the opposite side, the line segment is called the median of the triangle.



If line segment AD bisects BC, then AD is the median.

Centroid: The point of intersection of the medians of the triangle is called the centroid. It is denoted by G.

Properties of Centroid

(i) All the three median of a triangle pass through a point (Centroid).

(ii) Centroid of all types of triangles are always inside the triangle.

(iii) Apollonius Theorem

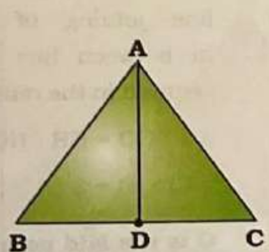
If AD is median of ΔABC, then

$AB^2 + AC^2 = 2(AD^2 + CD^2)$

'OR'

Length of median (AD)

$= \frac{1}{2}\sqrt{2AC^2 + 2AB^2 - BC^2}$



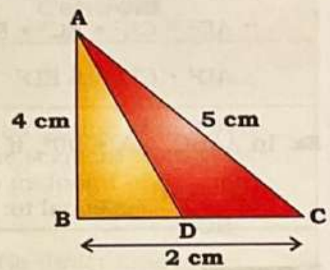
Ex. If the sides of ΔABC are 5 cm, 4 cm and 2 cm, then find the length of the median inserted from its smallest angled vertex?

HINTS We know that smallest angle is the angle opposite to smallest side.

$AD = \frac{\sqrt{2AB^2 + 2AC^2 - BC^2}}{2}$

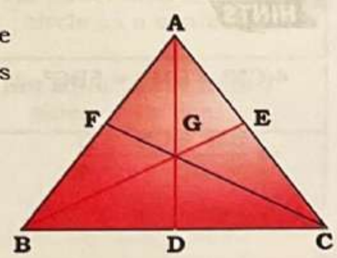
$= \frac{\sqrt{32 + 50 - 4}}{2}$

$= \frac{\sqrt{78}}{2} = \sqrt{\frac{39}{2}}$ cm



(iv) If AD, BE and CF are medians of ΔABC and G is the centroid, then

$\frac{AG}{GD} = \frac{BG}{GE} = \frac{CG}{GF} = \frac{2}{1}$



(v) $\frac{AG^2 + BG^2 + CG^2}{AB^2 + BC^2 + CA^2} = \frac{1}{3}$

Ex. G is the centroid of ΔABC. If the length of sides of ΔABC are 8 cm, 10 cm, and 12 cm, then find the $AG^2 + BG^2 + CG^2$?

HINTS

$AG^2 + BG^2 + CG^2 = \frac{1}{3}[64 + 100 + 144] = \frac{308}{3}$

(vi) $3(AB^2 + BC^2 + CA^2) = 4(AD^2 + BE^2 + CF^2)$

Ex. In ΔABC, the sum of square of all sides is 64 cm. Find the sum of square of all three medians.

HINTS

$3(a^2 + b^2 + c^2) = 4(m_a^2 + m_b^2 + m_c^2)$

$\Rightarrow 3 \times 64 = 4 \times R$

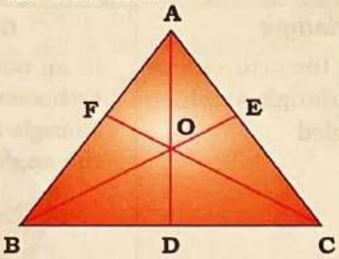
$\Rightarrow R = 48$

(vii) The sum of any two sides of a triangle is greater than twice the median drawn to the third side.

$AB + AC > 2AD$ ....(i)

$AB + BC > 2BE$ ....(ii)

$AC + BC > 2CF$ ......(iii)



Adding (i), (ii) and (iii)

$2(AB + AC + BC) > 2(AD + AE + CF)$

$(AB + AC + BC) > (AD + BE + CF)$

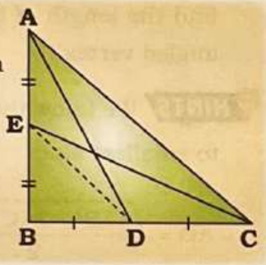
So, We can say that sum of sides (perimeter) is always greater than sum of all median.

(viii) If ΔABC is Right angle triangle and AD & CE are medians, then

- $AC^2 = 4ED^2$
- $4(AD^2 + CE^2) = 5AC^2$

$\because AD^2 + CE^2 = AC^2 + ED^2$

$\therefore AD^2 + CE^2 = 5ED^2$

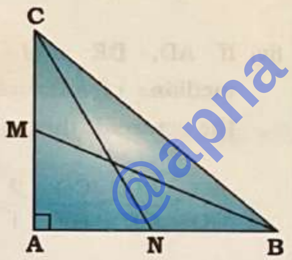


Ex. In ΔABC, $\angle A = 90°$, if BmM and CN are two medians, $\frac{BM^2 + CN^2}{BC^2}$ is equal to:

HINTS

$4(CN^2 + BM^2) = 5BC^2$

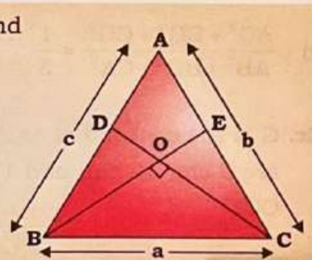
$\Rightarrow \frac{BM^2 + CN^2}{BC^2} = \frac{5}{4}$



(ix) If BE & CD are medians and $BE \perp CD$, then

$AB^2 + AC^2 = 5BC^2$

$\Rightarrow c^2 + b^2 = 5a^2$



Ex. If In ΔABC, BE and CF are two medians perpendicular to each other and if AB = 19 cm and AC = 22 cm then the length of BC is:

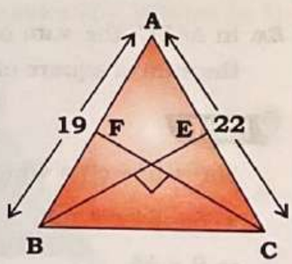
HINTS $AB^2 + AC^2 = 5BC^2$

$\Rightarrow 19^2 + 22^2 = 5BC2$

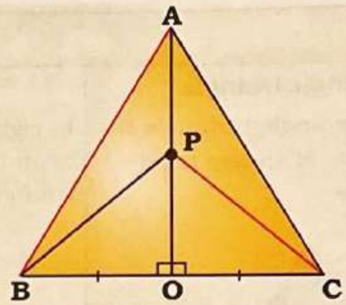
$\Rightarrow 361 + 484 = 5BC^2$

$\Rightarrow 845 = 5BC^2 \Rightarrow BC^2 = 169$

$\Rightarrow BC = 13$

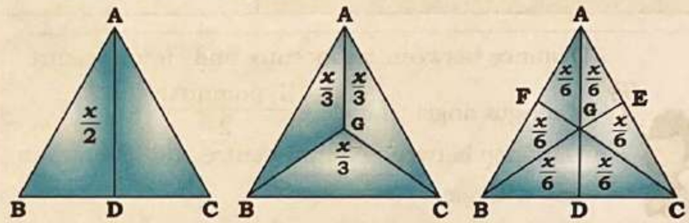


(x)



$AB^2 + PC^2 = AC^2 + BP^2$

(xi) If the area of $\Delta ABC = x$, AD, BE and CF are three medians and G is centroid, then

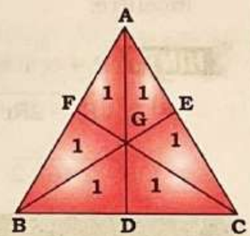


Ex. In a ΔABC, AD, BE and CF are medians intersect at G. If area of ΔABC is 204 cm². Find the area of quadrilateral GDCE.

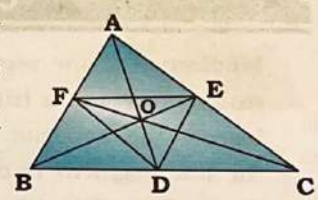
HINTS

Here, 6 unit = 204 cm²

$\therefore$ 2 unit = 68 cm²



(xii) Area of triangle formed by joining mid points of two sides and centroid is $\frac{1}{12}$th of area of triangle.



$Ar\ \Delta OFE = Ar\ \Delta OFD$

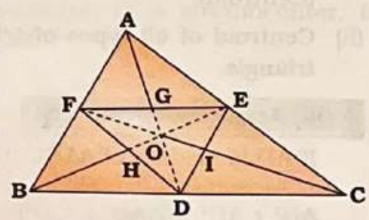
$= Ar\ \Delta OED = \frac{1}{12}\ Ar\ \Delta ABC$

$\Delta OFE \sim \Delta OCB$

(According to Mid-Point theorem FE || BC so that

$\angle EFO = \angle OCB$, $\angle OEF = \angle OBC$)

(xiii) The line segment joining the mid points of two sides divides the line joining of vertex in between line to the centroid in the ratio 3 : 1.



$AG : GO = BH : HO$

$= CI : IO = 3 : 1$

G is the Mid point of AD and FE

(xiv) If three medians of a triangle are given then

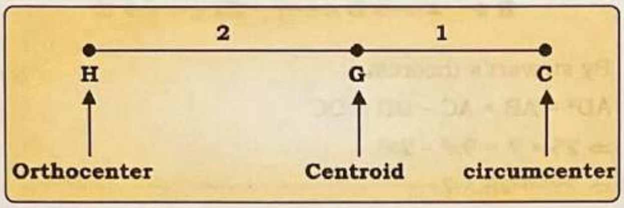
Area of $\Delta = \frac{4}{3}$ (Area of triangle considering medians as side)

$= \frac{4}{3}\sqrt{S_m(S_m - m_1)(S_m - m_2)(S_m - m_3)}$

Where, $S_m = \frac{m_1 + m_2 + m_3}{2}$

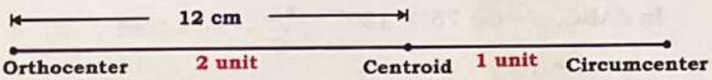
Relation between Orthocenter, centroid and circumcenter

In any triangle, orthocenter, centroid and circumcenter are co-linear and centroid divides the line in 2:1 as shown in figure



Ex. In a triangle the distance between the centroid and orthocentre is 12 cm. Find the distance between orthocentre and circumcenter.

HINTS



2 unit → 12 cm

$\times\frac{3}{2}$ $\times\frac{3}{2}$

3 unit → $\boxed{18\text{ cm}}$

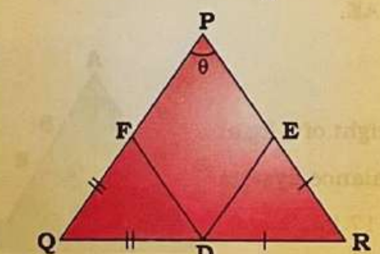
$\therefore$ Distance between orthocentre & circumcenter = 18 cm

Summary

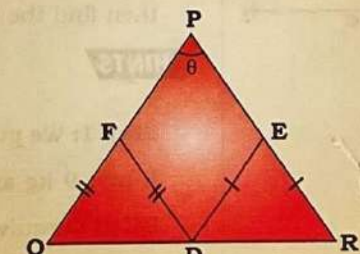
Feature	Incenter	Circumcenter	Orthocenter	Centroid
Symbol	I	O	H	G
Definition	Intersection point of internal angle bisectors.	Intersection of perpendicular bi-sectors of sides.	Intersection of altitudes	Intersection of medians (line segment from vertex to midpoint of opposite side)
Associated circle	Centre of the inscribed circle (incircle)	Centre of circumscribed circle (circumcircle)	No direct associated circle as a center	No direct associated circle as a center
Equidistant from	Sides of the triangle	Vertices of the triangle	Not equidistant from sides or vertices	Not equidistant from sides or vertices
Location	Always inside the triangle	• Acute: Inside • Right: Midpoint of hypotenuse • Obtuse: Outside at the front of obtuse angle	• Acute: Inside • Right: At the right angle vertex • Obtuse: Outside at the back of obtuse angle	Always inside the triangle
Significance	Center of largest circle that fits inside	Center of circle passing through all vertices	Related to triangle's heights	Center of mass/gravity of the triangle

Some important results

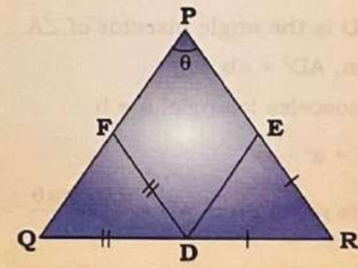
In ΔPQR, D, E, F are points on sides QR, PR and PQ respectively. If QF = QD, RE = DR and $\angle RPQ = \theta$ then $\angle EDF = 90° - \frac{\theta}{2}$



In ΔPQR, D, E, F are points on sides QR, PR and PQ respectively. If QF = FD, DE = ER and $\angle RPQ = \theta$ then $\angle EDF = \theta$



In ΔPQR, D, E, F are points on sides QR, PR and PQ respectively. If QD = DF, DR = RE and $\Delta RPQ = \theta$ then $\angle EDF = 180° - 2\theta$



Ex. In a ΔABC, points P, Q and R are taken on AB, BC and CA, respectively, such that BQ = PQ and QC = QR. If $\angle BAC = 75°$, what is the measure of $\angle PQR$ (in degrees)?

HINTS

In ΔABC, $\alpha + \theta + 75° = 180°$

$\Rightarrow \alpha + \theta = 105°$

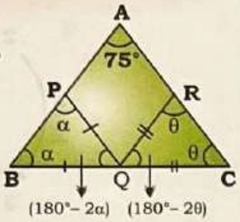
$\because$ BQC is a straight line.

$180° - 2\alpha + \angle PQR + 180° - 2\theta = 180°$

$\Rightarrow \angle PQR = 180° - 360 + 2(\alpha + \theta)$

$= -180° + 2 \times 105°$

$= -180° + 210° = 30°$



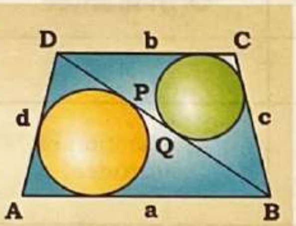
Alternatively

$\angle PQR = 180° - 2 \times \angle BAC$

$= 180° - 2 \times 75° = 180° - 150° = 30°$

☞ In trapezium ABCD, two unequal circles are inscribed such that each circle touches one pair of non-parallel sides. The diagonal DB intersects the circles at point P & Q.

$$PQ = \frac{(a+b)-(c+d)}{2}$$



☞ If n equal circles fit Perfectly along base BC of ΔABC, each touching adjacent circles and the Triangle's sides. then their radius is.

$$r = \frac{\text{Area of } \Delta ABC}{s + (n-1) \times h}$$

where, s = semi-perimeter

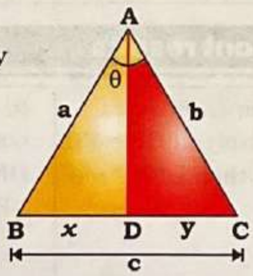
Some important theorems

(A) Stewart theorem:

It is used to find the length of any cevian.

Where $c = x + y$

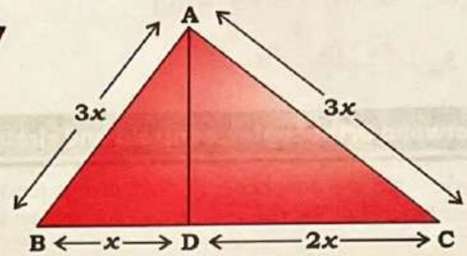
$a^2x + b^2y = AD^2c + xyc \Rightarrow a^2x + b^2y = c(AD^2 + xy)$



- If AD is the angle bisector of $\angle A$
  Then, $AD^2 = ab - xy$
- In isosceles triangle, a = b
  $AD^2 = a^2 - xy$
- If θ is given then $AD = \dfrac{2ab\cos\theta}{a+b}$

Ex. ΔABC is an equilateral triangle. D is a point on side BC such that BD : BC = 1 : 3. If $AD = 5\sqrt{7}$ cm, then the side of the triangle is:

HINTS



By stewart's theorem,

$AD^2 = AB \times AC - BD \times DC$

$\Rightarrow 25 \times 7 = 9x^2 - 2x^2$

$\Rightarrow 7x^2 = 25 \times 7$

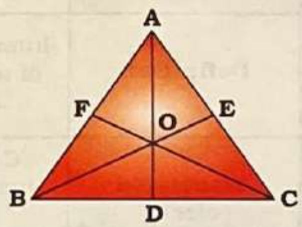
$\Rightarrow x = 5$

$\therefore$ Side $= 3x = 3 \times 5 = 15$ cm

(B) Ceva Theorem

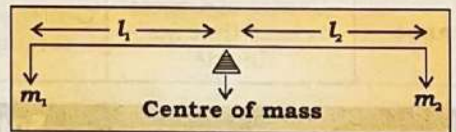
In ΔABC, AD, BE and CF are the cevians which pass through a common point O.

- $\dfrac{AF}{FB} \times \dfrac{BD}{DC} \times \dfrac{CE}{EA} = 1$
- $\dfrac{OD}{AD} + \dfrac{OE}{BE} + \dfrac{OF}{CF} = 1$
- $\dfrac{AO}{AD} + \dfrac{BO}{BE} + \dfrac{CO}{CF} = 2$



(C) Mass Point Theorem

- It is used to solve problems involving triangles and intersecting cevians by applying centers of mass principles.
- The base idea is that of a see-saw with weights at each end. The see-saw will balance if the product of mass and its distance to the fulcrum is same for each mass.

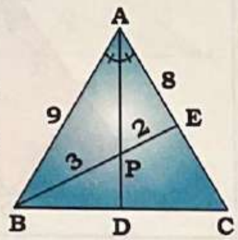


$$m_1 l_1 = m_2 l_2 \Rightarrow \frac{m_1}{m_2} = \frac{l_2}{l_1}$$

Ex. In ΔABC, AD is angle bisector of $\angle A$. AD and BE lie on BC and AC respectively such that AB = 9 cm and AC = 8 cm. AD and BE intersects at point P and BP : PE = 3 : 2, then find the value of AE.

HINTS

Step 1: We put the weight of 8 kg at B and 9 kg at C to balance system BC at D equivalent to 17 kg.



Step 2: To balance AC, we put 3 kg at A and getting resultant at E equivalent to 12 kg.

The ratio of the distances from the centre of mass would be inverse of the weights at the end points.

Therefore, AE : EC = 9 : 3 = 3 : 1

$\therefore$ (3 + 1) = 4 unit $\xrightarrow{\times 2}$ 8 cm

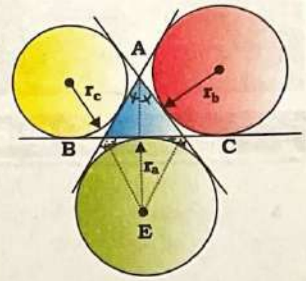
1 unit $\xrightarrow{\times 2}$ 2 cm

$\therefore$ AE = 3 unit $\xrightarrow{\times 2}$ 6 cm

AE = 6 cm

Excenter

The intersection point of internal angle bisector of one angle and bisectors of other two opposite exterior angles.



(a) $\angle BEC = 90° - \frac{\angle A}{2}$

(b) Ex-radii

$r_a = \frac{\Delta}{s-a}; r_b = \frac{\Delta}{s-b}; r_c = \frac{\Delta}{s-c}$

(c) $r_a = \frac{rs}{s-a} = \sqrt{\frac{s(s-b)(s-c)}{s-a}}$ where, $s = \frac{a+b+c}{2}$

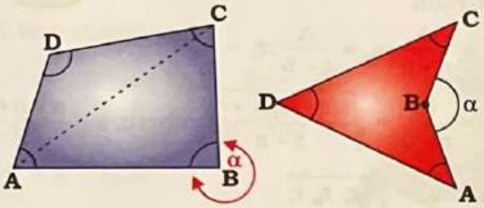
(d) $\Delta(\text{area}) = \sqrt{r\, r_a\, r_b\, r_c}$

(e) $r^2_a + r^2_b + r^2_c = (4R + r)^2 - 2s^2$

QUADRILATERAL

Quadrilateral

It is a plane figure bounded by four straight lines. The line segment which joins the opposite vertices of a quadrilateral is called diagonal of the quadrilateral.



AC, BD = diagonals

Here, both the above figures represent quadrilateral ABCD.

Properties of Quadrilateral

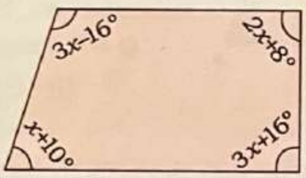
- Sum of interior angles of a quadrilateral = 360°

  i.e., $\angle A + \angle B + \angle C + \angle D = 360°$

  $\angle B_{internal} = 360° - (\angle A + \angle C + \angle D)$

  $\angle B_{external} = \alpha = \angle A + \angle C + \angle D$

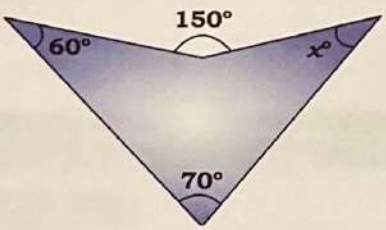
Ex. Find the value of 'x' in the given figure.



HINTS $3x - 16° + 2x + 8° + x + 10° + 3x + 16° = 360°$

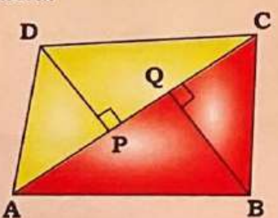
$\Rightarrow x = 38°$

Ex. Find the value of 'x' in the given figure.



HINTS $60° + 70° + x = 150° \Rightarrow x = 20°$

- ***Area of quadrilateral***



$\text{Area} = \frac{1}{2} \times AC \times DP + \frac{1}{2} \times AC \times BQ = \frac{1}{2} \times AC \times (DP + BQ)$

$= \frac{1}{2} \times \text{Diagonal} \times (\text{Sum of perpendicular dropped on diagonal})$

Ex. ABCD is a quadrilateral. The length of diagonal AC is 24 cm. The sum of perpendicular drawn from vertex B and D on the diagonal AC is 10 cm. What is the area of the quadrilateral?

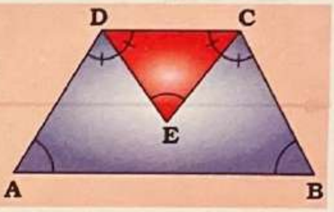
HINTS AC = 24 cm, DP + BQ = 10 cm

$\text{Area of quadrilateral ABCD} = \frac{1}{2} \times AC \times (DP + BQ)$

$= \frac{1}{2} \times 24 \times 10 = 120\text{cm}^2$

- If DE and CE are the angle bisectors of $\angle D$ and $\angle C$ respectively, then:

  $\angle CED = \frac{1}{2}(\angle A + \angle B)$



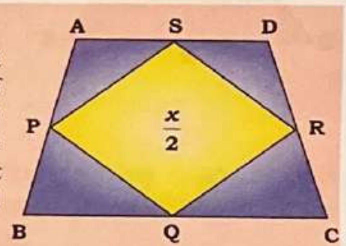
Ex. In a quadrilateral ABCD, the bisectors of $\angle C$ and $\angle D$ meet at point E. If $\angle CED = 57°$ and $\angle A = 47°$, then the measure of $\angle B$ is:

HINTS Let, $\angle B = \theta$

$\because \angle CED = \frac{1}{2}(\angle A + \angle B) \Rightarrow 57° = \frac{47° + \theta}{2}$

$\Rightarrow 114° - 47° = \theta \Rightarrow \theta = 67°$

- If ABCD be any type of quadrilateral and P, Q, R and S be the mid-points of sides AB, BC, CD and AD respectively, then: If the area of quadrilateral is x then area of quadrilateral PQRS = $\frac{\text{Area of} \square \text{ABCD}}{2}$

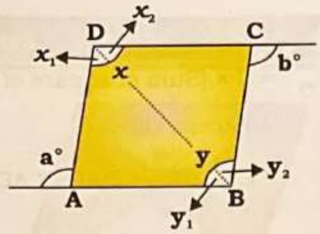


Ex. What is the area of the quadrilateral PQRS, which is formed by joining the mid-points of the adjacent sides of a quadrilateral ABCD as shown in the figure, if it is provided that $\Delta APS = 8\text{ cm}^2$, $\Delta BPQ = 12\text{ cm}^2$, $\Delta QCR = 9\text{ cm}^2$ and $\Delta RDS = 15\text{ cm}^2$.

HINTS $ar(\square PQRS) = 12 + 8 + 15 + 9 = 44\text{ cm}^2$

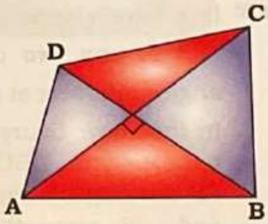
Original figure	Figure formed by joining the mid-points of sides
Square	Square
Rectangle	Rhombus
Rhombus	Rectangle
Parallelogram	Parallelogram
Quadrilateral	Parallelogram
Trapezium	Parallelogram
Kite	Rectangle

- If ABCD is a quadrilateral then $x + y = a + b$



- If diagonals of the quadrilateral intersect each other at 90°, then:

$AB^2 + CD^2 = BC^2 + AD^2$



Types of Quadrilateral

Square | Rectangle | Parallelogram | Rhombus | Trapezium | Kite

Square

A quadrilateral in which all the sides and all the angles are equal is known as **square**.

Where, a = side of square

AC = BD = d = length of diagonal

Properties of Square

- All sides are equal and parallel.
- All angles are right angles.
- Diagonals are equal and bisect each other at right angle.
- Diagonal 'BD' = side $\sqrt{2}$ = $a\sqrt{2}$
- Perimeter = 4 × side = 4a
- Area = $(\text{Side})^2 = (a)^2 = \dfrac{(\text{diagonal})^2}{2} = \dfrac{d^2}{2}$
- Radius 'r' of incircle = $\dfrac{a}{2}$
- Radius 'R' of circumcircle = $\dfrac{d}{2} = \dfrac{a}{\sqrt{2}}$
- $\dfrac{R}{r} = \dfrac{\sqrt{2}}{1}$
- $\dfrac{\text{Area of circumcircle}}{\text{Area of incircle}} = \left(\dfrac{\sqrt{2}}{1}\right)^2 = \dfrac{2}{1}$

$\dfrac{\text{Area of circumcircle}}{\text{Area of square}} = \dfrac{11}{7}$

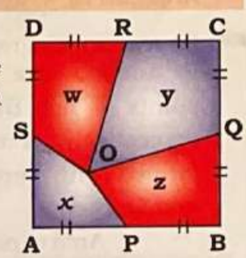
BD = d

☞ In a square ABCD, O is any point inside the square. P,Q,R,S are the midpoints of AB, BC, CD and DA respectively.

Area $(x + y)$ = Area $(w + z)$

$= \dfrac{1}{2}$ × Area of square ABCD

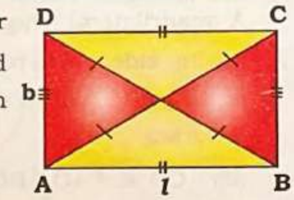
Above result is also true for rectangle.



Rectangle

A quadrilateral whose all four angles are right angles (90°) and its opposite sides are equal in length.

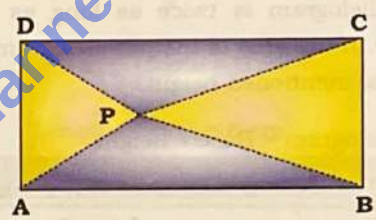
where, l = length, b = breadth

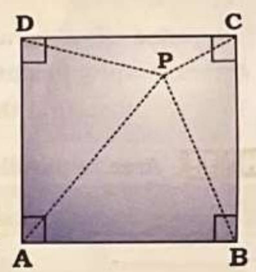


Properties of Rectangle

- Opposite sides are parallel and equal.
- Diagonals are equal and bisect each other but not at right angle.
- Perimeter = $2(l + b)$
- Area = $l \times b$
- Diagonal = $\sqrt{l^2 + b^2}$
- Of all the rectangles of any given perimeter, the square will have the maximum area.
- Bisectors of the four angles enclose a square.
- When the rectangle is inscribed in a circle, it will have the maximum area when it is a square.

☞ If P is any point inside the rectangle ABCD or square ABCD, then:





$PA^2 + PC^2 = PB^2 + PD^2$

Ex. There is a point P in a rectangle ABCD, such that PA = 4 cm, PD = 5 cm, PB = 8 cm. Find PC.

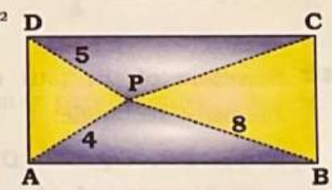
HINTS $PA^2 + PC^2 = PB^2 + PD^2$

$\Rightarrow 16 + PC^2 = 64 + 25$

$\Rightarrow PC^2 = 89 - 16$

$\Rightarrow PC^2 = 73$

$\Rightarrow PC = \sqrt{73}$



Ex. If P is a point inside a square ABCD such that PA = 15 cm, PB = 7 cm and PC = 20 cm, then the value of PD is :

HINTS $PA^2 + PC^2 = PB^2 + PD^2$

$\Rightarrow 225 + 400 = 49 + PD^2$

$\Rightarrow 625 - 49 = PD^2$

$\Rightarrow PD^2 = 576 \Rightarrow PD = 24$ cm

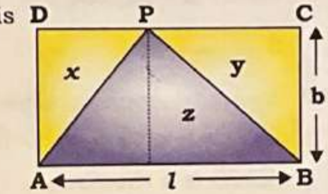
☞ ABCD is a rectangle and P is any point on side CD, then:

Area of ΔAPB

$= \dfrac{\text{Area of}\,\square\text{ABCD}}{2}$

∴ Area 'z' = Area '$x + y$'

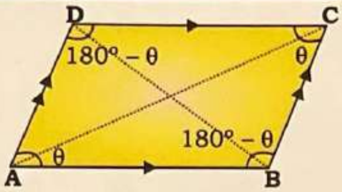
Above result is also true for square, rhombus and parallelogram.



Parallelogram

A quadrilateral whose opposite sides are parallel and equal is called parallelogram.

AB || CD and AD || BC



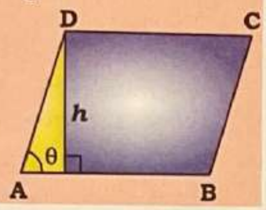
Properties of Parallelogram

- Opposite sides are parallel and equal.
- Diagonals may or may not be equal.
- Diagonals may or may not bisect each other at right angle.
- Sum of any two adjacent angles = 180°
- Area of parallelogram = Base × Height

Area = AB × h

$\because$ h = ADsinθ

$\therefore$ Area = AB × h = AB × ADsinθ



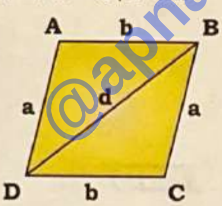
Ex. The base of a parallelogram is twice as long as its corresponding height. If the area of the parallelogram is 144 cm², then find the mentioned height.

HINTS Area of parallelogram = Base × height

$\Rightarrow 144 = 2x \times x \Rightarrow 2x^2 = 144 \Rightarrow x^2 = 72 \Rightarrow x = 6\sqrt{2}$ cm

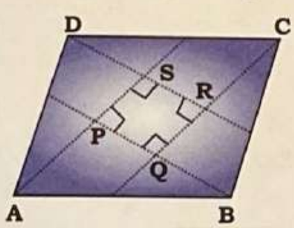
- Area $= 2\times\sqrt{s(s-a)(s-b)(s-d)}$

where, $s = \dfrac{a+b+d}{2}$



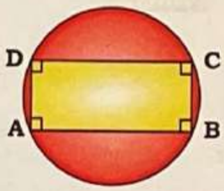
☞ Bisectors of the four angles enclose a rectangle.

If AS, BP, CQ and DR are angle bisectors of ∠A, ∠B, ∠C and ∠D respectively, then PQRS is a rectangle.



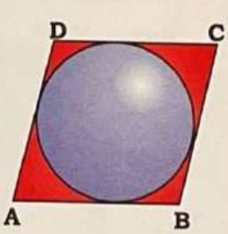
☞ A parallelogram inscribed in a circle is always a rectangle.

ABCD is a rectangle.



☞ A parallelogram circumscribed about a circle is always a rhombus.

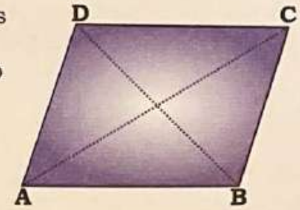
ABCD is a rhombus.



☞ Sum of square of the diagonals

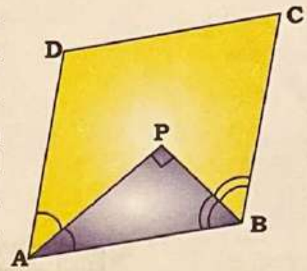
= 2 × (Sum of square of the two adjacent sides)

$AC^2 + BD^2 = 2(AB^2 + AD^2)$



☞ In a parallelogram, the bisectors of any two consecutive angles intersect at right angle.

In the above figure, there is a parallelogram ABCD. PA and PB are angle bisectors of ∠A and ∠B respectively. Here, ∠APB = 90°.



☞ If PQRS is a parallelogram,

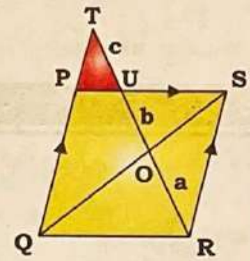
RO = a,

OU = b,

UT = c

then:

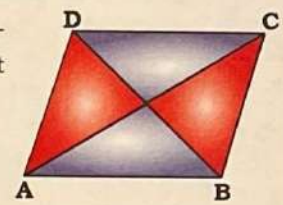
$a^2 = b(b + c)$



Area based concepts of Parallelogram

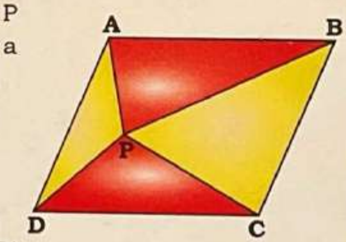
☞ Each diagonal divides the parallelogram into two congruent triangles of equal area.

- Area of ΔABC = Area of ΔADC
- Area of ΔABD = Area of ΔBDC



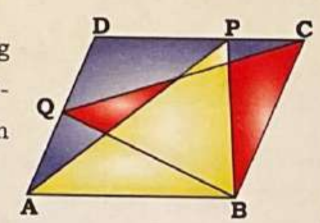
☞ In the figure given below, P is a point in the interior of a parallelogram ABCD, then:

- Area(ΔAPB) + Area(ΔPCD) $= \dfrac{1}{2}$ Area (□ABCD)
- Area(ΔAPD) + Area(ΔPBC) = Area(ΔAPB) + Area(ΔPCD)



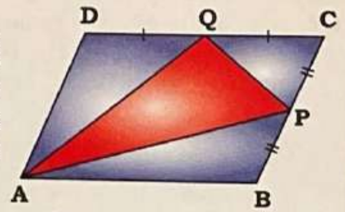
☞ If P and Q are two points lying on the sides DC and AD respectively of a parallelogram ABCD, then:

Area(ΔAPB) = Area(ΔBQC)



☞ In a parallelogram ABCD, P and Q are the mid points of side BC and CD respectively. A triangle is formed with vertices A, P and Q, then:

$$\frac{\text{Area of } \Delta APQ}{\text{Area of parallelogram ABCD}} = \frac{3}{8}$$

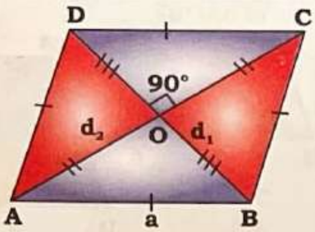


Rhombus

A **rhombus** is a parallelogram in which all the sides are equal.

a = sides of rhombus

d_1 , d_2 = digonals of rhombus



Properties of Rhombus

- Opposite sides are parallel.
- Opposite angles are equal.
- Diagonals bisect each other at right angle, but they are not necessarily equal.

  $d_1^2 + d_2^2 = 4a^2$
- Diagonals bisect the vertex angles.
- Sum of any two adjacent angles is 180°.

  $\angle A + \angle B = 180°$

Ex. The diagonals of a rhombus are 24 cm and 10 cm. The perimeter of the rhombus (in cm) is :

HINTS

$d_1^2 + d_2^2 = 4a^2$

$\Rightarrow 24^2 + 10^2 = 4a^2$

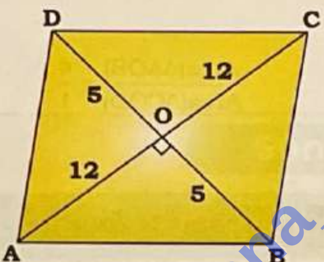
$\Rightarrow 576 + 100 = 4a^2$

$\Rightarrow a^2 = 169$

$\Rightarrow a = 13$ cm

$\therefore$ Perimeter = 4 × a

= 4 × 13 = 52 cm



Ex. If PQRS is a rhombus and $\angle SPQ = 50°$, then $\angle RSQ$ is :

HINTS

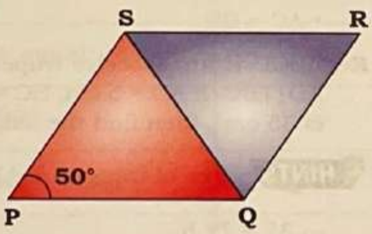
Here, $\angle SPQ = 50°$

$\angle PSR = 180° - \angle SPQ$

$= 180° - 50° = 130°$

$\therefore \angle RSQ = \dfrac{\angle PSR}{2}$

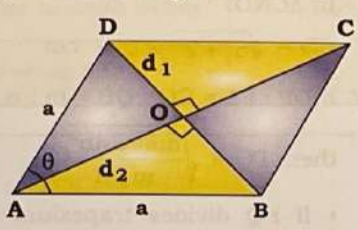
$= \dfrac{130°}{2} = 65°$



☞ Area of rhombus = $\frac{1}{2}$ × product of diagonals

$d_1 = 2a \sin\dfrac{\theta}{2}$

$d_2 = 2a \cos\dfrac{\theta}{2}$



$\because d_1 d_2 = 2a \sin\dfrac{\theta}{2} \times 2a \cos\dfrac{\theta}{2} = 2a^2 \sin\theta$

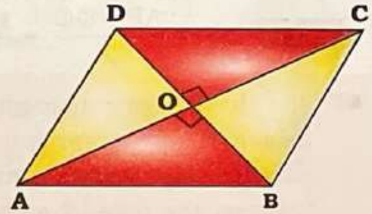
$\therefore$ Area of rhombus ABCD = $\dfrac{1}{2} \times d_1 d_2 = \dfrac{1}{2} \times 2a^2 \sin\theta = a^2 \sin\theta$

Ex. If the perimeter of a rhombus is 150 cm and length of one diagonal is 50 cm, then find the length of second diagonal and the area of rhombus.

HINTS AC = 50 cm

AB = BC = CD = DA

$= \dfrac{150}{4} = 37.5$ cm



$\therefore OB = \sqrt{AB^2 - OA^2}$

$= \sqrt{(37.5)^2 - (25)^2} = \sqrt{\dfrac{125 \times 625}{100}} = \dfrac{25}{2}\sqrt{5}$

$\therefore$ Second diagonal, BD = 2 × OB = $2 \times \dfrac{25}{2}\sqrt{5} = 25\sqrt{5}$ cm

$\therefore$ Area = $\dfrac{1}{2} \times d_1 \times d_2 = \dfrac{1}{2} \times 50 \times 25\sqrt{5} = 625\sqrt{5}$ cm²

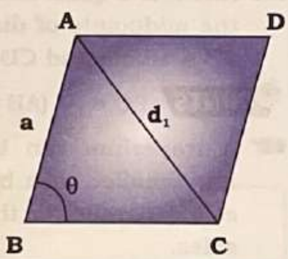
Ex. ABCD is a rhombus. AC is its smallest diagonal and $\angle ABC = 60°$. Find the length of the side of rhombus when AC = 6 cm.

HINTS d_1 = AC = 6 cm, θ = 60°

$\because d_1 = 2a.\sin\dfrac{\theta}{2} \Rightarrow 6 = 2a.\sin30°$

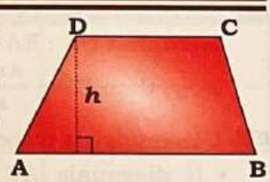
$\Rightarrow 3 = a\left(\dfrac{1}{2}\right) \Rightarrow a = 6$ cm

$\therefore$ Length of side = 6 cm



Trapezium

A quadrilateral whose two opposite sides are parallel and remaining two opposite sides are not parallel, is called **trapezium**.



Properties of Trapezium

- Area of trapezium = $\frac{1}{2}$ × (sum of parallel sides) × height

  $= \dfrac{1}{2} \times (AB + CD) \times h$

Ex. The length of parallel sides of a trapezium is 20 cm and 25 cm, respectively and the perpendicular distance between these two sides is 14 cm. What is the area (in cm²) of the trapezium?

HINTS Area = $\dfrac{1}{2} \times (20 + 25) \times 14 = 45 \times 7 = 315$ cm²

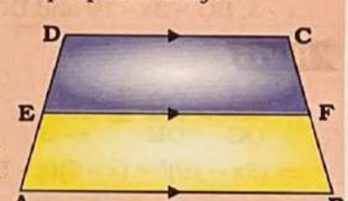
☞ Any line parallel to the parallel sides of a trapezium divides the non-parallel sides proportionally.

If AB || DC || EF, then:

$\dfrac{AE}{ED} = \dfrac{BF}{FC}$

- If E & F are the mid points of sides AD and BC respectively, then:

  $EF = \dfrac{1}{2}(AB + DC)$

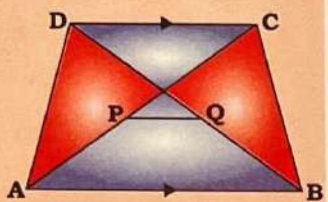


Ex. In trapezium ABCD, AB||DC. E is the mid-point of side AD and F is the mid point of side BC. If DC = 8 cm and AB = 18 cm, then EF is :

HINTS $EF = \dfrac{AB + DC}{2} = \dfrac{18+8}{2} = 13$ cm

☞ The line segment joining the mid-points of the diagonals of a trapezium is parallel to each of the parallel sides and is equal to half of the difference between the parallel sides of the trapezium.

In trapezium ABCD, AB||DC and P and Q are the mid-points of diagonals AC and BD respectively, then:

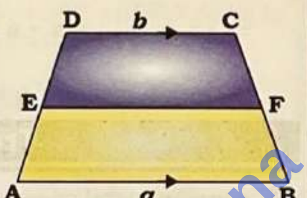


- PQ||AB||DC
- $PQ = \dfrac{1}{2}(AB - DC)$

Ex. ABCD is a quadrilateral in which AB || DC. P and Q are the midpoints of diagonals AC and BD, respectively. If AB = 18 cm and CD = 6 cm, then PQ = ?

HINTS $PQ = \dfrac{1}{2}(AB - CD) = \dfrac{1}{2} \times 12 = 6$ cm

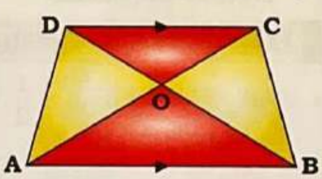
☞ A trapezium can be divided into smaller ones by drawing a line parallel to the parallel sides.



If the line EF divides the non-parallel sides in the ratio x : y, such that DE : EA = CF : FB = $x : y$, then $EF = \dfrac{ax + by}{x + y}$

☞ Diagonals of a trapezium divide each other proportionally.

- If diagonals AC and BD intersect at O, then: $\dfrac{AO}{CO} = \dfrac{BO}{DO} = \dfrac{AB}{CD}$.



Conversely, if the diagonals of a quadrilateral divide each other proportionally, then it is a trapezium.

- Ar(ΔAOB) × Ar(ΔDOC) = Ar(ΔAOD) × Ar(ΔBOC)
- Area of ΔAOD = Area of ΔBOC
- $AC^2 + BD^2 = AD^2 + BC^2 + 2(AB \times CD)$

Ex. ABCD is a trapezium where AD||BC. The diagonals AC and BD intersect each other at a point O. If AO = 3, CO = $x - 3$, BO = $3x - 19$ and DO = $x - 5$, then the value of 'x' is :

HINTS

$\because \dfrac{AO}{OC} = \dfrac{DO}{OB} \Rightarrow \dfrac{3}{x-3} = \dfrac{x-5}{3x-19}$

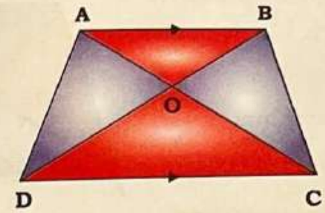
$\Rightarrow (3x - 19)^3 = (x - 3)(x - 5)$

$\Rightarrow 9x - 57 = x^2 - 5x - 3x + 15$

$\Rightarrow x^2 - 17x + 72 = 0$

$\Rightarrow x = 8, 9$

Ex. In the given figure, AB || DC. If area of ΔAOD and ΔDOC is 36 cm² and 48 cm², respectively, then what is the area of ΔAOB?

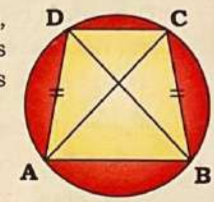


HINTS Here, Ar(ΔAOD) = Ar(ΔBOC) = 36 cm²

∵ Ar(ΔAOB) × Ar(ΔDOC) = Ar(ΔAOD) × Ar(ΔBOC)

$\Rightarrow x \times 48 = 36 \times 36 \Rightarrow x = 27$ cm²

☞ If a trapezium is inscribed in a circle, then it is a isosceles trapezium and its non parallel sides are equal and its diagonals are also equal.



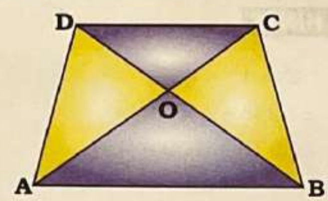
AD = BC

AC = BD

∠DAB + ∠ADC = ∠CBA + ∠BCD = 180°

☞ If in a trapezium ABCD, AB||DC and AB = 2DC, then the ratio of the area of ΔAOB and ΔCOD is 4 : 1.

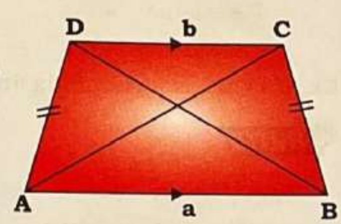
$\dfrac{\text{Area}(\Delta AOB)}{\text{Area}(\Delta COD)} = \dfrac{4}{1}$



Isosceles Trapezium

A trapezium whose non-parallel sides are equal, is called **isosceles trapezium**.

- AD = BC
- ∠DAB = ∠CBA
- AC = BD



Ex. ABCD is an isosceles trapezium in which AB = CD and AD||BC. If AD = 5 cm, BC = 9 cm and area of trapezium is 35 cm², then find the length of side CD.

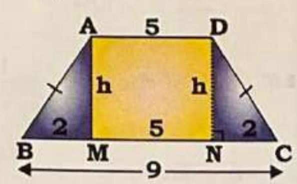
HINTS Area of trapezium ABCD = $\dfrac{1}{2} \times 14 \times h$

$\Rightarrow 35 = 7 \times h$

$\Rightarrow h = 5$ cm

In ΔCND:

$CD = \sqrt{5^2 + 2^2} = \sqrt{29}$ cm



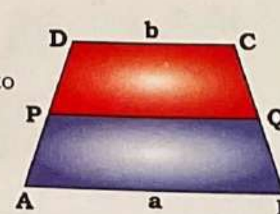
☞ If DP : PA = CQ : QB = m : n,

then: $PQ = \sqrt{\dfrac{ma^2 + nb^2}{m+n}}$

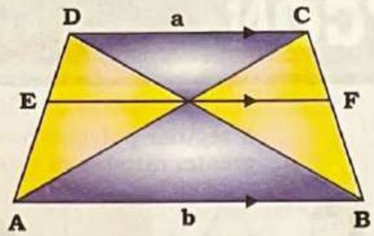
- If PQ divides trapezium into two equal area, then:

m : n = 1 : 1

and, $PQ = \sqrt{\dfrac{a^2 + b^2}{2}}$

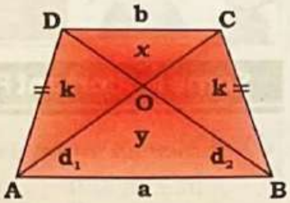


☞ The length of a parallel line segment passing through the intersection of diagonals of the isosceles trapezium is:



$$EF = \frac{2ab}{a+b}$$

☞ If ABCD is an isosceles trapezium and y, k, x and k is respective area of ΔAOB, ΔBOC, ΔCOD and ΔDOA, then:

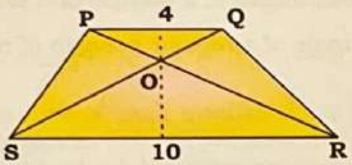


$k = \sqrt{xy}$

$x : y = b^2 : a^2$

Ex. In a trapezium PQRS, PQ is parallel to SR and diagonals PR and QS intersect at O. If PQ = 4 cm, SR = 10 cm, then what is the ratio of area of ΔPOQ to area of ΔSOR ?

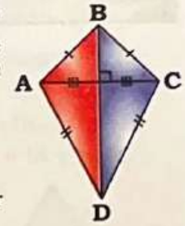
HINTS



$$\frac{ar\Delta POQ}{ar\Delta SOR} = \frac{(4)^2}{(10)^2} = \frac{4}{25}$$

Kite

A **kite** is a quadrilateral with two distinct pairs of adjacent sides that are equal in length and the diagonals are perpendicular to each other.



- AB = BC and AD = CD
- The diagonals intersect at right angle.
- The longer diagonal bisects the shorter diagonal.
- Area = $\frac{1}{2}$ × product of diagonals = $\frac{1}{2}$ × AC × BD
- Two opposite angles are equal and remaining two opposite angles are not equal.
- The angle between the two unequal sides is equal to the angle between the other two unequal sides.

  $\angle BAD = \angle BCD$

Ex. In a kite ABCD, longer diagonal BD is drawn. If $\angle ABD = 54°$ and $\angle ADB = 50°$, then what is the measure of $\angle DCB$?

HINTS

In ΔBAD:

$\angle BAD = 180° - 54° - 50° = 76°$

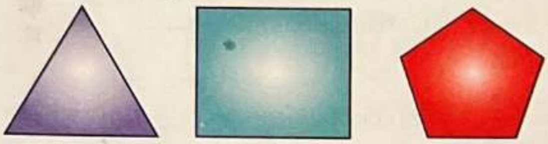
$\therefore \angle DCB = \angle BAD = 76°$

Summary

Property	Square	Rectangle	Parallelogram	Rhombus
4 sides	✓	✓	✓	✓
Opposite sides equal	✓	✓	✓	✓
All sides equal	✓	×	×	✓
Opposite angles equal	✓	✓	✓	✓
All angles 90°	✓	✓	×	×
Diagonals bisect each other	✓	✓	✓	✓
Diagonals equal	✓	✓	×	×
Diagonals perpendicular to each other	✓	×	×	✓
Diagonals bisect vertex angle	✓	×	×	✓

POLYGON

A polygon is a 'n' sided closed figure formed by line segments. The line segments are called its edges/sides. The points where two sides meet are the vertices or corner of a polygon.



Types of Polygon

- *On the basis of sides*
  - Regular polygon
  - Irregular polygon
- *On the basis of angles*
  - Convex polygon
  - Concave polygon

Regular Polygon

A regular polygon has equal lengths on all sides and identical angles at each vertex.

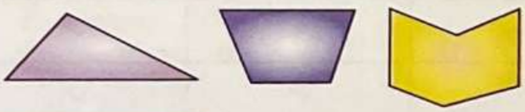
Ex. Equilateral triangle-3 sided regular polygon

Square- 4 sided regular polygon

Regular hexagon- 6 sided regular polygon

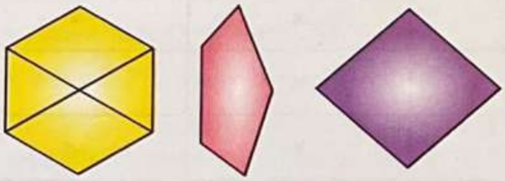
Irregular Polygon

An irregular polygon has sides that are not equal and angles that differ from one another.



Convex polygon

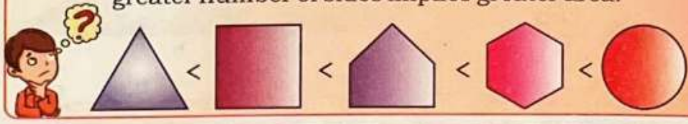
A convex polygon is a polygon with all interior angle less than 180°. In convex polygon all diagonal are in the interior of the polygon.



Concave polygon

A concave polygon is a polygon with atleast one interior angle greater than 180°. In concave polygon not all diagonal are in the interior of the polygon.

If regular polygon have the same perimeter, then greater number of sides implies greater area.



Some Important Formulae of Regular Polygon

- Sum of all internal angles of a polygon of n sides $= (n - 2)\ 180°$
- Each interior angle of a regular polygon of n sides $= \frac{(n-2)\times180°}{n}$
- Sum of all exterior angles of a polygon of n sides = 360°
- Each exterior angle of a regular polygon of n sides $= \frac{360°}{n}$
- No. of side in a regular polygon (n) $= \frac{360°}{\text{Exterior angle}}$
- Sum of each interior angle and exterior angle of a regular polygon is 180°.
- Ratio of the measure of an interior angle of a polygon of n-sides to the measure of its exterior angle is given by $= \left(\frac{n}{2}-1\right):1$
- ***Diagonal of polygon:-*** If we join any 2 non-adjacent vertices of a polygon, the line segment formed is called a diagonal.

  No. of diagonals in a polygon of n sides $= \frac{n(n-3)}{2}$, $n > 3$
- Number of sides + No. of diagonal $= \frac{n(n-1)}{2}$

Ex. Sum of all interior angles of a regular octagon is:

HINTS Sum of all Interior angles of a polygon $= (n-2) \times 180°$

Here, n = 8

Sum of all interior angles of regular octagon

$= (8-2) \times 180° = 6 \times 180° = 1080°$

Ex. The sum of all the interior angles of a polygon is 1440°. The number of sides of this polygon is-

HINTS Sum of interior angles of polygon $= (n-2) \times 180°$

$\Rightarrow (n-2) \times 180° = 1440° \Rightarrow (n-2) = 8$

$\Rightarrow n = 10$

Ex. The measure of each exterior angle of a regular octagon is:

HINTS Sum of exterior angles of a regular octagon = 360°

Each exterior angle of a regular octagon $= \frac{360°}{8} = 45°$

Ex. Each interior angle of a regular polygon is 144°. The number of sides of the polygon is :

HINTS Each interior angle of regular polygon = $\frac{(n-2)\times 180°}{n}$

$\Rightarrow 144° = \frac{(n-2)\times 180°}{n}$

$\Rightarrow 144°n = 180°n - 360° \Rightarrow 36°n = 360° \Rightarrow n = 10$

Thus, Number of sides = 10.

Alternatively

Interior angle + Exterior angle = 180°

$\Rightarrow$ 144° + Exterior angle = 180° $\Rightarrow$ Exterior angle = 36°

No. of side in a regular polygon (n) = $\frac{360°}{\text{Exterior angle}}$

$= \frac{360°}{36°} = 10$

Ex. In a polygon, the interior and exterior angles are in the ratio 4 : 1. The number of sides of the polygon is :

HINTS Using property, $\frac{n}{2} - 1 = 4 \Rightarrow n = 10$

Ex. The measure of each interior angle of a cyclic polygon with 'n' sides is 144°. So, what will be the measure of the angle formed by each of its arms at the geometric center of the polygon?

HINTS Each interior angle of a regular polygon

$= \frac{(n-2)}{n} \times 180° \Rightarrow \frac{(n-2)}{n} \times 180° = 144° \Rightarrow n = 10$

The angle at the geometric center of the polygon is 360°.
As the polygon is regular, the central angle is equally divided into 10 parts.

Thus, The angle formed by each of its arms at the geometric center = $\frac{360°}{10} = 36°$

Ex. The number of diagonals in a polygon of 17 sides is:

HINTS $\therefore$ Number of diagonals = $\frac{17(17-3)}{2} = 119$

Ex. A polygon has 35 diagonals. The number of sides in the polygon is:

HINTS No. of diagonal = 35

$\Rightarrow \frac{n(n-3)}{2} = 35$

On solving we get, n = 10.

Ex. If one of the interior angles of a regular polygon is $\frac{15}{16}$ times of one of the interior angles of a regular decagon, then find the number of diagonals of the polygon.

HINTS Each interior angle of a polygon = $\frac{(n-2)\times 180°}{n}$

Each interior angle of decagon = $\frac{(10-2)\times 180°}{10} = 144°$

ATQ, $\frac{(n-2)\times 180°}{n} = \frac{15}{16}\times 144° \Rightarrow n = 8$

$\therefore$ Number of diagonal = $\frac{n(n-3)}{2} = \frac{8(8-3)}{2} = 20$

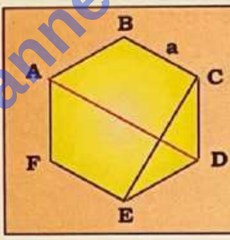
- Perimeter of regular polygon = n × a;
  Where n is the number of sides, a is the length of the side.
- Area of a regular polygon of n sides where length of each side is 'a' = $n\frac{a^2}{4}\cot\left(\frac{180°}{n}\right)$
- Inradius = $\frac{a}{2}\cot\left(\frac{\pi}{n}\right)$
- Circumradius = $\frac{a}{2}\text{cosec}\left(\frac{\pi}{n}\right)$
- Number of non overlaping triangles formed by joining the vertices of polygon = (n – 2)
  Where, n = no. of sides

Ex. In 11-sided polygon, how many non overlaping triangles can be formed by joining the vertices.

HINTS Number of non overlaping triangles formed by joining the vertices of polygon = (n – 2) = 11 – 2 = 9

Regular Hexagon

A regular hexagon is a closed shape polygon which has six equal sides and six equal angles.

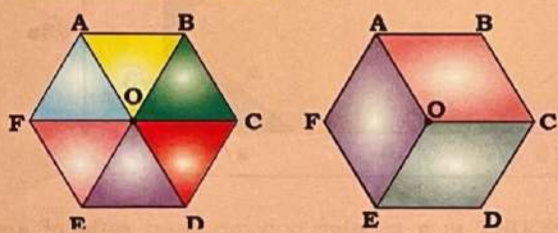


Larger Diagonal (d_1) = 3
AD = BE = CF = 2a

Smaller Diagonal (d_2) = 6
AC = BD = CE = DF = EA = BF = $\sqrt{3}$a

Properties of regular hexagon

- It has 6 vertices.
- Sum of interior angle = 720°
- Interior angle = 120°, Exterior angle = 60°
- Number of diagonals in regular hexagon = 9
- Perimeter of regular hexagon = 6 × Each Side
- Hexagon is made up of 6 equilateral triangle or 3 rhombus of equal area.

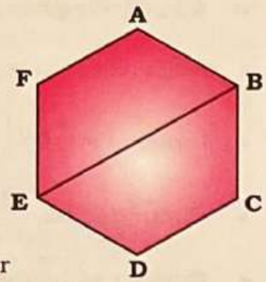


Area of regular hexagon

- When Side is given = $\frac{3\sqrt{3}}{2}(\text{Side})^2$
- When Larger diagonal is given = $\frac{3\sqrt{3}}{8}(d_1)^2$
- When Smaller diagonal is given = $\frac{\sqrt{3}}{2}(d_2)^2$

Ex. ABCDEF is a regular hexagon and BE = 14 cm, find
(i) Perimeter of hexagon
(ii) Area of hexagon

HINTS



BE = 14 cm

$AF = \frac{14}{2} = 7$ cm

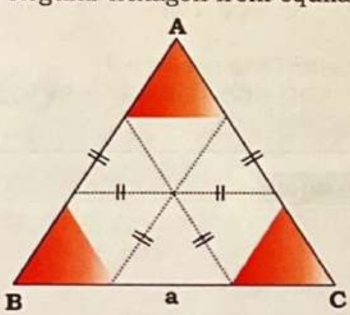
$\therefore$ Perimeter = 6 × 7 = 42 cm

(ii) Area of hexagon when larger diagonal is given

$= \frac{3\sqrt{3}}{8}(d_1)^2 = \frac{3\sqrt{3}}{8}(14)^2 = \frac{147\sqrt{3}}{2}$

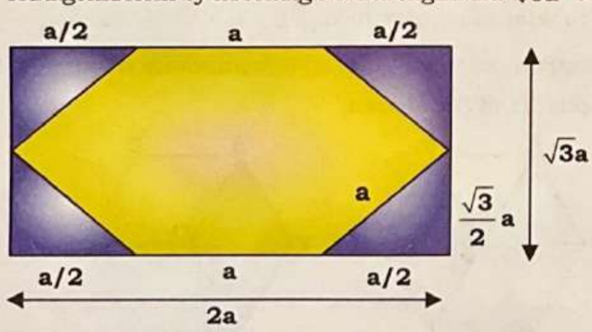
- Inradius of regular hexagon = $\frac{\sqrt{3}}{2}$ × Side
- Circumradius of regular hexagon is equal to the length of its side.
- Inradius : circumradius = $\sqrt{3} : 2$
- Area of circumcircle : area of hexagon : area of incircle
  $= \pi a^2 : \frac{3\sqrt{3}}{2}a^2 : \pi\frac{3a^2}{4} = 4\pi : 6\sqrt{3} : 3\pi$

☞ Regular hexagon from equilateral Δ

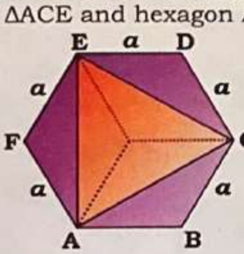


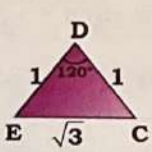
Ar(hexagon) : Ar(ΔABC) = 6 : 9 = 2 : 3

☞ Hexagonal form by a rectangle of 2a length and $\sqrt{3}$a breadth:



☞ If ABCDEF is a regular polygon then ratio of area of ΔACE and hexagon ABCDEF is 1 : 2.

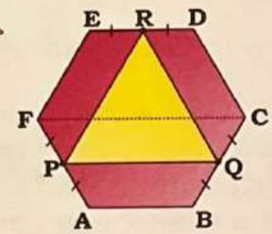


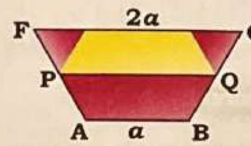


ΔEAC = Equilateral triangle of side $\sqrt{3}$a

$\therefore \frac{\text{Area of } \Delta EAC}{\text{Area of hexagon ABCDEF}} = \frac{\frac{\sqrt{3}}{4}\times(\sqrt{3}a)^2}{\frac{3\sqrt{3}}{2}\times a^2} = \frac{1}{2}$

☞





P, Q and R are the midpoint of side AF, BC and ED respectively.

$PQ = \frac{2a+a}{2} = \frac{3a}{2}$

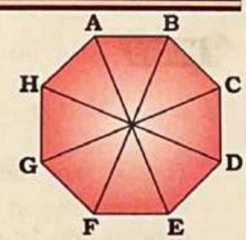
ΔPQR = Equilateral triangle of side $\frac{3a}{2}$

$\therefore \frac{\text{Area of } \Delta PQR}{\text{Area of hexagon ABCDEF}} = \frac{\frac{\sqrt{3}}{4}\times\frac{9}{4}a^2}{\frac{3\sqrt{3}}{2}\times a^2} = \frac{3}{8}$

Regular Octagon

A regular octagon is a closed eight-sides polygon where all side have equal length and all interior angles are equal.

- A Well known example of regular octagon is stop sign on roads.



Properties of Regular Octagon

- Sum of the interior angles is 1080°.
- Sum of the exterior angles is 360°.
- The interior angle at each vertex is 135° and exterior angle at each vertex is 45°
- Perimeter of regular octagon = 8 × Side
- No. of diagonal = $\frac{8(8-3)}{2} = \frac{8\times5}{2} = 20$
- Length of diagonal of octagon = $a\left(\sqrt{4+2\sqrt{2}}\right)$
- Area of regular octagon = $2a^2\left(1+\sqrt{2}\right)$
  Where a is side of regular octagon.
- Inradius of regular octagon = $\frac{a}{2\sqrt{2}-2} = \frac{a(\sqrt{2}+1)}{2}$
- Circumradius of regular octagon = $\frac{a}{2}\left(\sqrt{4+2\sqrt{2}}\right)$

Ex. Find the area of regular octagon having perimeter of 16 cm.

HINTS Perimeter of regular octagon = 8 × side

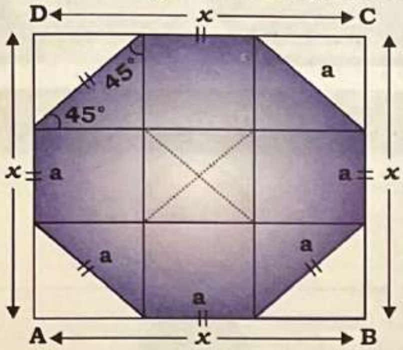
$\Rightarrow$ 16 = 8 × side

$\Rightarrow$ Side = 2 cm

Area of octagon = $2a^2\left(1+\sqrt{2}\right) = 2 \times 2^2 \times \left(\sqrt{2}+1\right)$

$= 8\left(\sqrt{2}+1\right)$

☞ Regular octagon formed by square of side x : –



(a) Side of square $(x) = a(\sqrt{2}+1)$

(b) Side of octagon (a) $= x(\sqrt{2}-1)$

Ex. A square of side 4 cm is cut down into regular octagon. Find the area of octagon.

HINTS Side of octagon $= x(\sqrt{2}-1) = 4(\sqrt{2}-1)$

Area of octagon $= 2a^2(1+\sqrt{2})$

$= 2 \times 16\ (\sqrt{2}-1)^2(\sqrt{2}+1) = 32(\sqrt{2}-1)$

$= 32 \times 0.41 = 13.25\ \text{cm}^2$

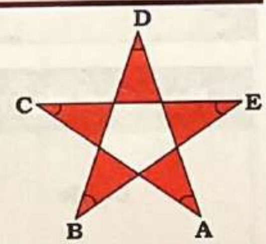
Star

Star is formed by extending sides of a regular polygon.

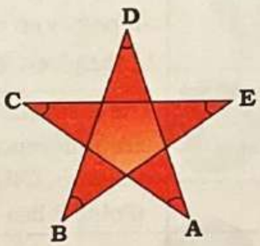
Sum of outer angle

$= (n - 4) \times 180°$,

where n be the no. of outer triangle in star.



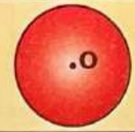
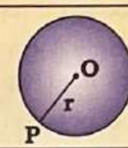
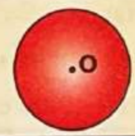
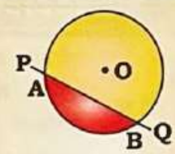
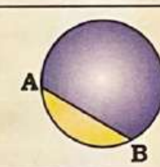
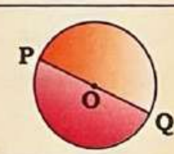
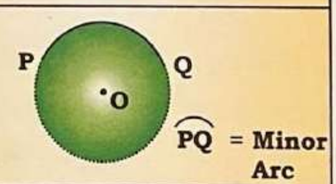
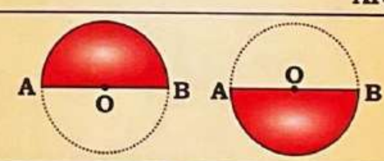
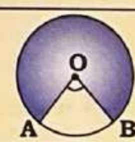
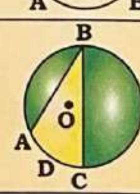
Ex. Find the sum of $\angle A + \angle B + \angle C + \angle D + \angle E$ in the given figure.

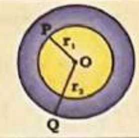
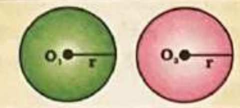
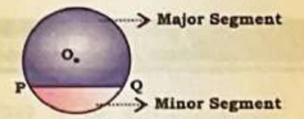
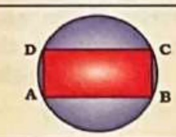
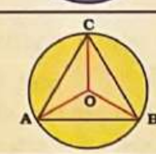
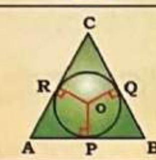
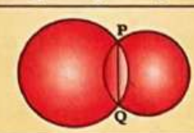


HINTS Sum of all outer angle $= (n - 4) \times 180°$

$= (5 - 4) \times 180° = 180°$

Circle

Nomenclature	Definition	Diagram
Circle	A circle is a set of points on a plane which lie at a fixed distance from a fixed point. It is a round shape figure that has no corners or edges.	
Centre	The fixed point inside the circle that is equidistant from all points on the circle's circumference, is called the centre. In the given diagram, 'O' is the centre of the circle.	
Radius	The distance between the centre and any point on the circumference of a circle is called radius(r). In the given diagram, OP is the radius of the circle. (Point P lies on the circumference.)	
Circumference	The circumference of a circle is the length of boundary of the circle which is equal to $2\pi r$.	
Secant	A line segment which intersects the circle in two distinct points, is called secant. In the given diagram, secant PQ intersects circle at two points at A and B.	
Chord	A line segment whose end points lie on the circumference of the circle. In the given diagram, AB is a chord.	
Diameter	A chord which passes through the centre of the circle is called the diameter of the circle. The length of the diameter is twice the length of the radius. In the given diagram, PQ is the diameter of the circle. (O is the centre of the circle)	
Arc	Any two points on the circumference of circle divide the perimeter of circle into two parts. The smaller part is called minor arc and the larger part is called major arc. It is denoted as $\frown$. In the given diagram, $\overset{\frown}{PQ}$ is arc.	
Semicircle	The diameter of a circle divides the circle into two equal parts. Each part is known as semicircle.	
Central Angle	An angle formed at the centre of a circle, is called the central angle. In the given diagram, $\angle AOB$ is the central angle.	
Inscribed angle	When two chords have one common end point, then the angle between these two chords at the common point is called the inscribed angle. $\angle ABC$ is the inscribed angle by the arc $\overset{\frown}{ADC}$.	

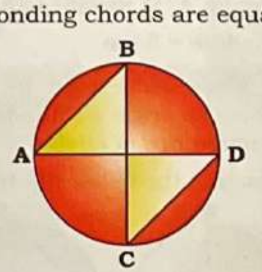
Nomenclature	Definition	Diagram
Measure of an arc	Basically it is the central angle formed by an arc. e.g., (a) Measure of angle at the centre of circle = 360° (b) Measure of angle at the centre of semicircle =180° (c) Measure of a minor arc = ∠POQ (d) Measure of a major arc = 360° – ∠POQ m(arc PRQ) = m ∠POQ m(arc PSQ) = 360° – m(arc PRQ)	
Concentric circles	Circles having the same centre at a plane are called the concentric circles. In the given diagram there are two circles with radii 'r_1' and 'r_2' having the common (or same) centre 'O'. They are concentric circles.	
Congruent circles	Circles having equal radii are called congruent circles.	
Segment of a circle	A chord divides a circle into two regions. These two regions are called the segments of a circle. (a) Major segment (b) Minor segment ***The major and minor segments of a circle are called the alternate segments of each other.***	
Cyclic Quadrilateral	A quadrilateral whose all the four vertices lie on the circumference of the circle.	
Circum-circle	A circle which passes through all the three vertices of a triangle is called circumcircle. Thus the circumcentre is always equidistant from the vertices of the triangle. Circumradius 'R'= OA = OB = OC	
Incircle	A circle which touches all the three sides of a triangle i.e., all the three sides of the triangle are tangents to the circle is called an incircle. Incircle's centre is always equidistant from the sides of a triangle. Inradius, 'r' = OP = OQ = OR	
Common Chord	If two circles intersect at two points and the points of intersection of these circles are joined by a line segment, then this line segment is called the common chord. Common chord = PQ	

Fundamental properties, theorems and important results based on arcs & chords

☞ The diameter is the longest chord in the circle.

☞ If two arcs of a circle (or of congruent circles) are congruent, then corresponding chords are equal.

$\widehat{AB} = \widehat{CD}$ ⇐
$\therefore$ AB = CD



Conversely, if two chords of a circle (or of congruent circles) are equal, then their corresponding arcs (minor, major or semi-circular) are congruent.

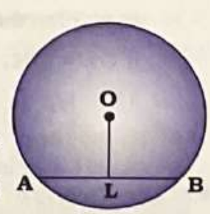
☞ A perpendicular from the centre of a circle to a chord bisects the chord.

If OL ⊥ AB, then AL = LB.

Conversely, the line joining the centre of a circle to the mid-point of a chord is perpendicular to the chord.

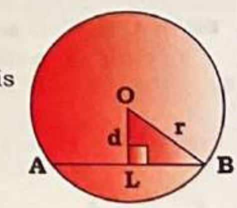
If AL = LB, then

OL ⊥ AB.

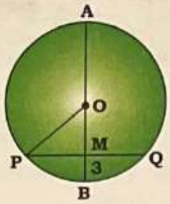


☞ If OL ⊥ AB, then AL = LB and ΔOLB is a right-angled triangle, i.e.,

$OL^2 + LB^2 = OB^2$



Ex. In a given circle, the chord PQ is of length 18 cm. AB is the perpendicular bisector of PQ at M. If MB = 3 cm, then the length of AB = ?



HINTS Let, radius of circle be 'x'.

OM = OB – MB = $x - 3$

In ΔOMP, using Pythagoras Theorem

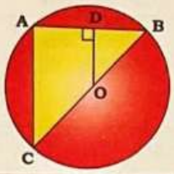
$(OP)^2 = (OM)^2 + (MP)^2$

$\Rightarrow (x)^2 = (x-3)^2 + (9)^2 \Rightarrow x = 15$ cm

$\therefore$ Diameter 'AB' = 15 × 2 = 30 cm

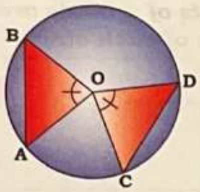
☞ If in a circle, OD is perpendicular to the chord AB, O is centre of the circle and BC is the diameter as shown below, then:

CA = 2 × OD



Angles subtended by chords

☞ Equal chords of a circle subtend equal angles at the centre and conversely, if the angles subtended by the chords at the centre (of a circle) are equal, then the chords are equal. If AB = CD, then



∠AOB = ∠COD

Conversely, if ∠AOB = ∠COD,

then AB = CD.

Ex. AB and CD are two chords of a circle such that AB = CD = 5 cm and r = 4 cm. If O is the centre of the circle, then (∠AOB – ∠COD) is:

HINTS Equal chords subtend equal angles at the centre.

$\therefore$ ∠AOB = ∠COD

$\Rightarrow$ ∠AOB – ∠COD = 0

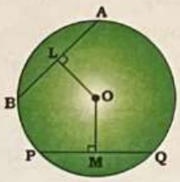
☞ Equal chords of a circle are equidistant from centre.

If AB = PQ, then OL = OM

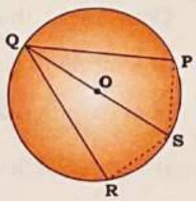
Conversely, if two chords are equidistant from the centre, then chords are equal.

If OL = OM, then

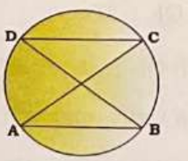
AB = PQ



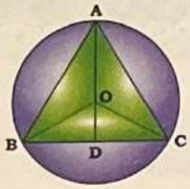
☞ If PQ and QR are the chords of a circle equidistant from the centre, then the diameter passing through Q bisects ∠PQR and ∠PSR.



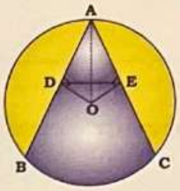
☞ If A, B, C, D are four consecutive points on a circle and AB = CD, then AC = BD.



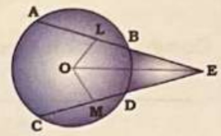
☞ If bisector AD of ∠BAC of ΔABC passes through the centre O of the circumcircle of ΔABC, then AB = AC.



☞ AB and AC are two equal chords of circle whose centre is O and if OD ⊥ AB and OE ⊥ AC, then ΔADE is an isosceles triangle.



☞ If two equal chords AB and CD of a circle with centre O, when produced to meet at a point E, as shown in the given figure, then BE = DE and AE = CE.

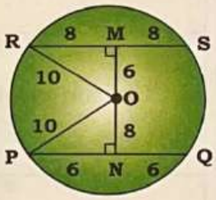


☞ If two chords of a circle bisect one another, then they must be diameters.

Ex. PQ and RS are the two parallel chords of a circle whose radius is 10 cm and the centre is O. If PQ = 12 cm and RS = 16 cm and both lie on the opposite sides of the centre, find the perpendicular distance between PQ and RS.

HINTS

Using Pythagoras theorem in ΔOMR and ΔONP:



$(OM)^2 = (OR)^2 - (MR)^2 = (10)^2 - (8)^2$

$\Rightarrow$ OM = 6

$(ON)^2 = (OP)^2 - (NP)^2 = (10)^2 - (6)^2$

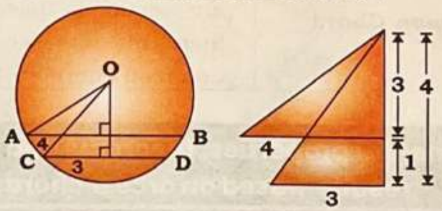
$\Rightarrow$ ON = 8

$\therefore$ Perpendicular distance between two chords

= OM + ON = 6 + 8 = 14 cm

Ex. AB = 8 cm and CD = 6 cm are two parallel chords on the same side of the centre of a circle. The distance between them is 1 cm. The radius of the circle is :

HINTS



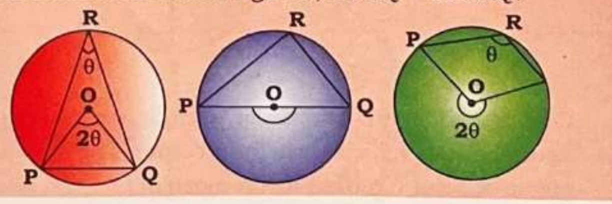
work on triplets and options.

Triplet (3, 4, 5) satisfies all conditions.

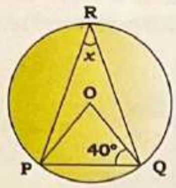
$\therefore$ Radius = 5 cm

☞ The angle subtended by a chord at the centre is double the angle subtended by it at any point on the circumference of circle.

In each of the below figures, ∠POQ = 2∠PRQ.



Ex. If O is the centre, then the value of 'x' in the given figure is :



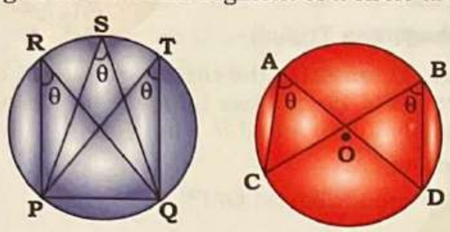
HINTS In ΔPOQ,

$\angle POQ = 180° - (40° + 40°)$

$\angle POQ = 100°$

$\therefore \quad x = \frac{100°}{2} = 50°$

☞ The angles in the same segment of a circle are equal i.e.,

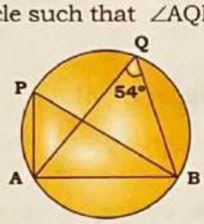


(i) (ii)

(i) $\angle PRQ = \angle PSQ = \angle PTQ$

(ii) $\angle CAD = \angle CBD$

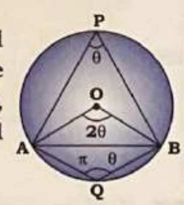
Ex. In the given figure, AB is a chord and P and Q are two points on the circle such that $\angle AQB = 54°$, then $\angle APB$ is:



HINTS We know that angles in the same segment of circle are equal.

$\therefore \angle APB = \angle AQB = 54°$

☞ If AB is a chord, O is the centre and P and Q are any points in the major and the minor segments of the circle respectively, then the angles subtended by the chord AB at P and Q are supplementary.

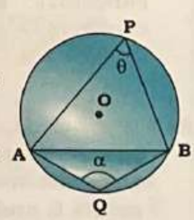


If $\angle APB = \theta$, then

$\angle AOB = 2\theta$ & $\angle AQB = \pi - \theta$.

☞ The angle subtended by a chord in the major segment is acute and that in the minor segment is obtuse.

If AB is a chord and P and Q are the points on the circumference, then:



$\angle APB < 90°$ and $\angle AQB > 90°$

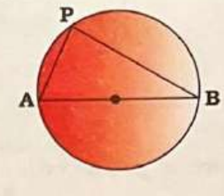
☞ **The diameter of a circle subtends:**

(a) An obtuse angle (α) at a point in the interior of the circle.

(b) An acute angle (β) at a point in the exterior of the circle.

(c) A right angle (γ) at a point on the circumference of the circle.

Ex. In the given figure, the centre of the circle lies on AB and P is any point on the circumference of circle and $\angle ABP = 35°$, then $\angle PAB$ is :

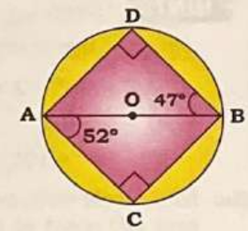


HINTS $\angle APB = 90°$ (Angle made by the diameter at the circumference of the circle)

$\angle BAP = 90° - \angle ABP = 90° - 35° = 55°$

Ex. AB is the diameter of a circle with centre O. C and D are two points on the circle on either side of AB, such that $\angle CAB = 52°$ and $\angle ABD = 47°$. What is the difference (in degree) between the measures of $\angle CAD$ and $\angle CBD$?

HINTS



$\angle BAD = 90° - 47° = 43°$

$\angle CBA = 90° - 52° = 38°$

$\angle CAD = 52° + 43° = 95°$

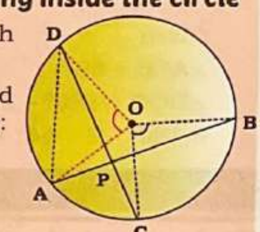
$\angle CBD = 47° + 38° = 85°$

$\therefore \angle CAD - \angle CBD = 95° - 85° = 10°$

Important results (i)

☞ **When two chords are intersecting inside the circle**

Two chords AB and CD of a circle with centre O, intersect each other at P. If $\angle AOD = x°$ and $\angle BOC = y°$, then the value of $\angle APC$ is:

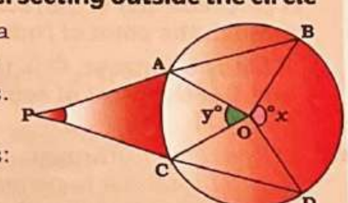


$\angle APD = \angle BPC = \frac{(x° + y°)}{2}$

$\angle APC = 180° - \frac{(x° + y°)}{2}$

☞ **When two chords are intersecting outside the circle**

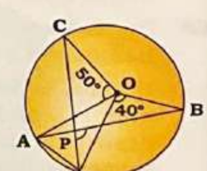
Two chords AB and CD of a circle with centre O, intersect each other at P when produces. If $\angle BOD = x°$ and $\angle AOC = y°$, then the value of $\angle BPD$ is:



$\angle BPD = \frac{(x° - y°)}{2}$

Ex. Two chords AB and CD of a circle whose centre is O meet each other at the point P and $\angle AOC = 50°$ and $\angle BOD = 40°$, then $\angle BPD$ is equal to:

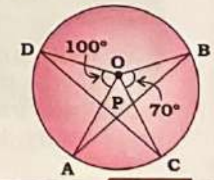
HINTS



$\angle BPD = \frac{(x° + y°)}{2} = \frac{50° + 40°}{2} = 45°$

Ex. Two chords AB and CD of a circle with centre O, intersect each other at P. If $\angle AOD = 100°$ and $\angle BOC = 70°$, then the value of $\angle APC$ is :

HINTS



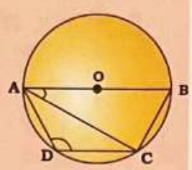
$\angle APC = 180° - \left(\frac{100° + 70°}{2}\right)$

$= 180° - 85° = 95°$

Important results (ii)

☞ AB is diameter of circle & if ∠BAC is given, then:

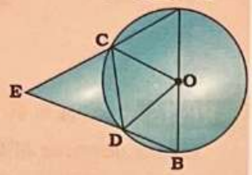
$$\angle DAC = 90^\circ - 2\,\angle BAC$$



☞ In a circle with centre O, AC and BD are two chords.

AC and BD meet at E when produced. If AB is diameter and ∠AEB = x°, then the measure of ∠DOC is:

$$\angle DOC = 180^\circ - 2\,\angle AEB$$



Ex. In a circle with centre O, AB is the diameter and CD is a chord such that ABCD is a trapezium. If ∠BAC = 40°, then ∠CAD is equal to :

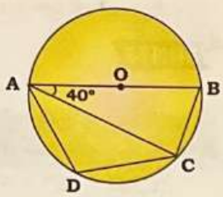
HINTS Given,

∠BAC = 40°

∠DAC = 90° – 2 × ∠BAC

= 90° – 2 × 40°

= 90° – 80° = 10°



Ex. In a circle with centre O, AC and BD are two chords. AC and BD meet at E when produced. If AB is the diameter and ∠AEB = 68°, then the measure of ∠DOC is :

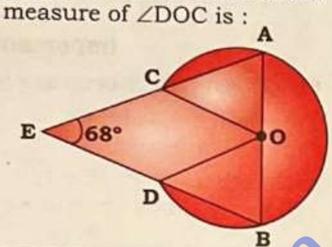
HINTS

Given,

∠AEB = 68°

∠DOC = 180° – 2∠AEB

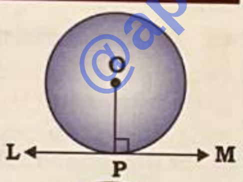
= 180° – 2 × 68° = 44°



Tangent

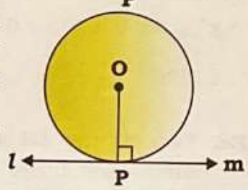
☞ The tangent at any point on a circle is perpendicular to the radius drawn through the point of contact.

If LM is a tangent, O is the centre and P is the point of contact, then OP ⊥ LM.



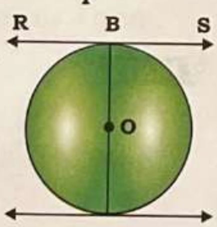
☞ A line drawn through the end-point of a radius is perpendicular to the radius and it is the tangent to the circle.

If O is the centre, P is the point of contact and OP ⊥ lm, then $\overleftrightarrow{lm}$ is the tangent.



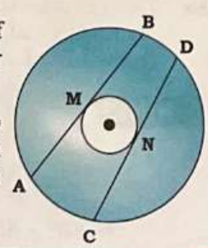
☞ The tangents at the end points of a diameter of a circle are parallel.

If AB is a diameter of a given circle and PQ and RS are the tangents drawn to the circle at points A and B respectively, then PQ || RS.



☞ In two concentric circles, all chords of the outer circle which touch the inner circle are of equal length.

If in the figure, AB and CD are two chords of the outer circle which touch the inner circle at M and N respectively, then, AB = CD.



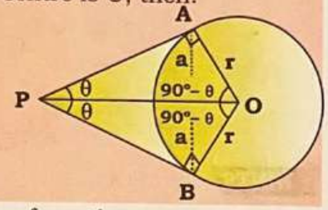
☞ Two tangents PA and PB are drawn from an external point P on a circle, whose centre is O, then:

(a) ΔPAO ≅ ΔPBO

(b) $PA = PB = \dfrac{ar}{\sqrt{r^2 - a^2}}$

(c) ∠PAO = ∠PBO = 90°

(d) ∠APO = ∠BPO



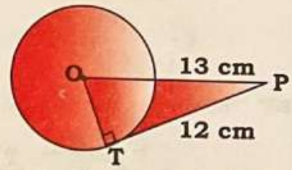
Ex. From a point P, 13 cm away from the centre, a tangent PT of length 12 cm is drawn. Find the radius of the circle.

HINTS

Given,

OT = 5 cm

(by Pythagorean Triplet)



Ex. There is a chord AB in the circle of radius 10 cm. Length of chord AB is 12 cm. Two tangents are drawn from an external point P to A and B. Find:

(i) Length of PA.

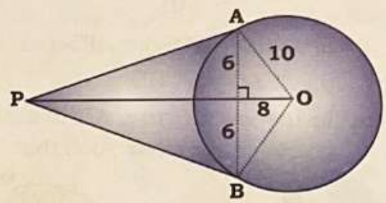
(ii) Area of quadrilateral OAPB.

HINTS

(i) PA =

$$\frac{ar}{\sqrt{r^2 - a^2}} = \frac{6\times10}{\sqrt{10^2 - 6^2}} = \frac{60}{8}$$

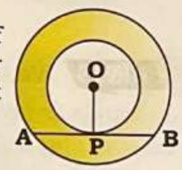
= 7.5 cm



(ii) Area of quadrilateral OAPB = 2 × Area of ΔOAP

$$= 2 \times \frac{1}{2} \times AO \times AP = 2 \times \frac{1}{2} \times 10 \times 7.5 = 75\ \text{cm}^2$$

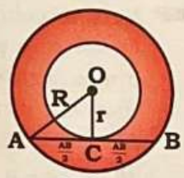
☞ In two concentric circles, a chord of larger circle which is tangent to smaller circle is bisected at the point of contact i.e., AP = BP.



☞ If circles are concentric, then:

Length of AB = 2AC

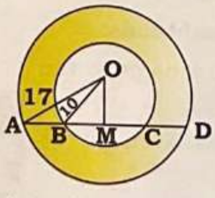
$= 2\sqrt{R^2 - r^2}$

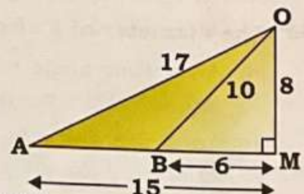


☞ If 'l' is a line intersecting the two concentric circles, whose common centre is O, at the points A, B, C and D, then AB = CD.

Ex. The radius of two concentric circles is 17 cm and 10 cm. A straight line ABCD intersects the larger circle at points A and D and intersects the smaller circle at points B and C. If BC = 12 cm, then the length of AD = ?

HINTS



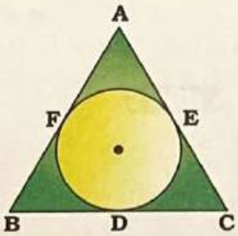


By applying the concept of triplets.

∴ Length of AD = 15 × 2 = 30 cm

☞ If ΔABC is an isosceles triangle in which AB = AC, circumscribed about a circle, then the base is bisected by the point of contact.

In ΔABC, if AB = AC then, BD = DC.



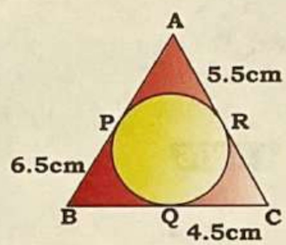
☞ If the incircle of ΔABC touches the sides BC, CA and AB at D, E and F respectively, then:

AF + BD + CE = AE + BF + CD

$= \frac{1}{2} \times$ (Perimeter of ΔABC)

Ex. A circle is inscribed inside a ΔABC. It touches sides AB, BC and AC at the points P, Q and R respectively. If BP = 6.5 cm, CQ = 4.5 cm and AR = 5.5 cm, then the perimeter (in cm) of the ΔABC is:

HINTS



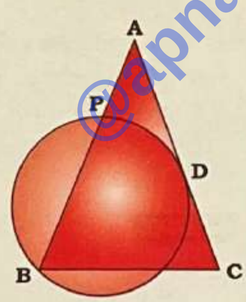
Perimeter of ΔABC

= 2(AR + BP + QC)

= 2 × 16.5 = 33 cm

Ex. ABC is an isosceles triangle with AB = AC. A circle passing through B touches AC at the middle point and intersects AB at P, then AP : AB is:

HINTS



$\because$ D is the midpoint of AC.

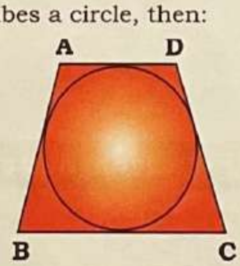
$\therefore AD = \frac{AC}{2} = \frac{AB}{2}$ $(\because AB = AC)$

$\because AP \times AB = AD^2$

$\Rightarrow AP \times AB = \frac{AB^2}{4} \Rightarrow \frac{AP}{AB} = \frac{1}{4}$

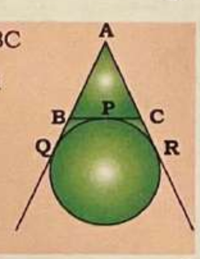
☞ If a quadrilateral ABCD circumscribes a circle, then:

$\boxed{AB + CD = BC + AD}$

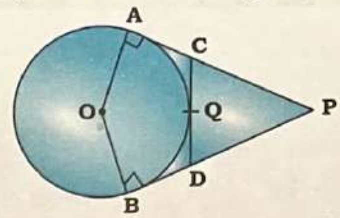


☞ If a circle is touching the side BC of ΔABC at P and touching AB and AC produced at Q and R respectively, then:

$\boxed{AQ = \frac{1}{2} \times \text{(Perimeter of } \Delta ABC)}$



Ex. In the given figure, circle with centre O have two tangents PA, PB. CD is also tangent, then 2PB is:



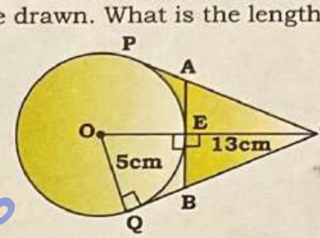
HINTS

$PB = \frac{1}{2} \times$ (Perimeter of ΔPDC)

$\Rightarrow PB = \frac{1}{2} \times (PD + DC + PC)$

$\Rightarrow 2PB = PD + DC + PC$

Ex. In the given figure, from a point T, 13 cm away from the center O of a circle of radius 5 cm, the two tangents PT and QT are drawn. What is the length of AB?



HINTS

QT = 12 cm (Pythagoras theorem)

Let, QB = x cm

ET = 13 – 5 = 8 cm

BT = $(12 - x)$ cm

In ΔBET:

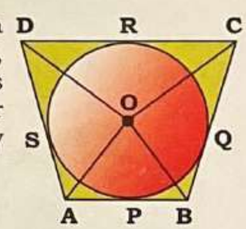
$BT^2 = BE^2 + ET^2$

$\Rightarrow (12 - x)^2 = x^2 + 8^2$

$\Rightarrow x = \frac{10}{3}$ cm

$\therefore AB = 2BE = \frac{2 \times 10}{3} = \frac{20}{3}$ cm

☞ If a circle touches the sides of a quadrilateral ABCD at P, Q, R, S respectively, then the angles subtended at the centre by the pair of opposite sides are supplementary i.e.,



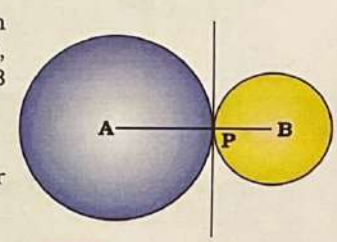
$\angle AOB + \angle COD = 180°$ and

$\angle AOD + \angle BOC = 180°$

☞ (a) If two circles touch each other externally at point P, then P lies on the line AB joining their centres.

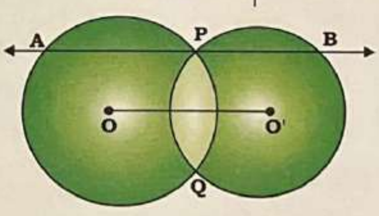
A, B and P are collinear.

Distance 'd' between their centres = AP + BP



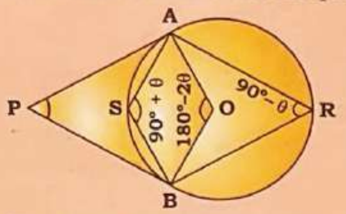
☞ (b) If two circles whose centres are O and O' intersect at P & Q and through P, a line parallel to OO' intersecting the circles at A and B is drawn, then

AB = 2 × OO'.



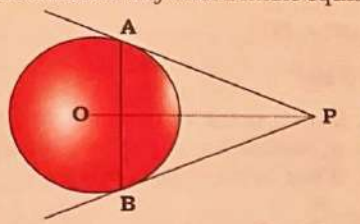
Angle subtended by two tangents from an external point

☞ PA and PB are two tangents, O is the center of the circle and R and S are the points on the circle.



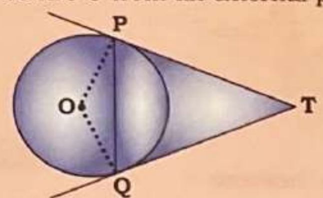
If $\angle APB = 2\theta$, then:
$\angle AOB = 180° - 2\theta$
$\angle ARB = 90° - \theta$
$\angle ASB = 90° + \theta$

☞ The tangents from a point outside the circle at the extremities of any chord make equal angles with the chord.



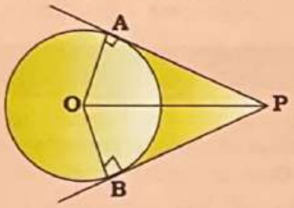
$\angle PAB = \angle PBA$

☞ If two tangents TP and TQ are drawn to a circle with centre O from an external point T, then:



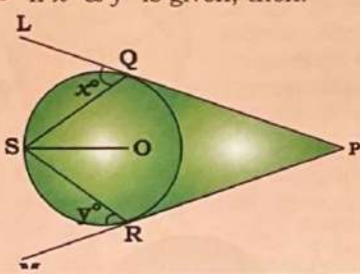
$\angle PTQ = 2\angle OPQ$

☞ The angle between the two tangents drawn from an external point of a circle is supplementary to the angle subtended by the line-segment joining the points of contact at the centre.



$\angle AOB + \angle APB = 180°$

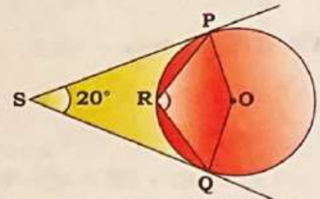
☞ If $x°$ & $y°$ is given, then:



$\angle QSR = 180° - (x° + y°)$

Ex. P and Q are two points on a circle with centre O. R is a point on the minor arc of the circle between the points P and Q. The tangents to the circle at the point P and Q meet each other at the point S. If $\angle PSQ = 20°$, then $\angle PRQ$ = ?

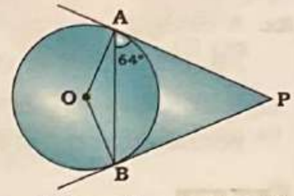
HINTS



$2\theta = 20° \Rightarrow \theta = 10°$
$\angle PRQ = 90° + \theta$
$= 90° + 10° = 100°$

Ex. Two tangents from a point P are drawn to a circle which touch the circle at points A and B, respectively. If O is the centre of the circle and $\angle PAB = 64°$, then $\angle APB$ is :

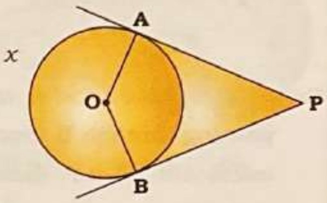
HINTS



$\angle OAB = 90° - 64° = 26°$
$\therefore \angle APB = 2\angle OAB$
$= 2 \times 26° = 52°$

Ex. The tangents at two points A and B on the circle with centre O intersects at P. If in quadrilateral PAOB, $\angle AOB : \angle APB = 5 : 1$, the measure of $\angle APB$ is:

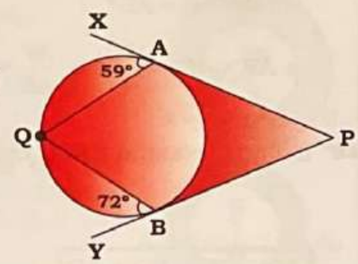
HINTS



Let, $\angle AOB = 5x$ and $\angle APB = x$
$\therefore \angle APB + \angle AOB = 180°$
$\Rightarrow 6x = 180° \Rightarrow x = 30°$
$\therefore \angle APB = 30°$

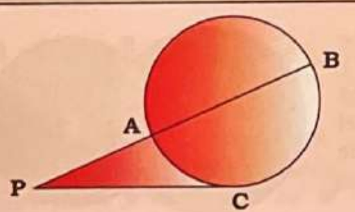
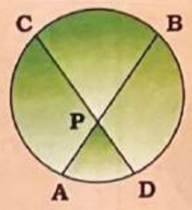
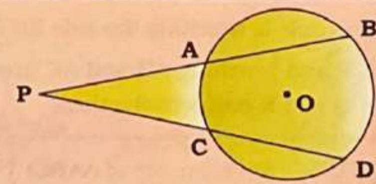
Ex. In a circle with centre O, PAX and PBY are the tangents to the circle at points A and B respectively, from an external point P. Q is any point on the circle such that $\angle QAX = 59°$ and $\angle QBY = 72°$. What is the measure of $\angle AQB$?

HINTS



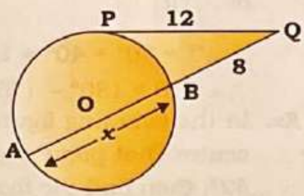
$\angle AQB = 180° - (59° + 72°)$
$= 49°$

Relationship between tangent, secant and chord

Tangent Secant Theorem	When chords are intersecting inside the circle	When chords are intersecting outside the circle
 $PA \times PB = PC^2$	 $PA \times PB = PC \times PD$	 $PA \times PB = PC \times PD$

Ex. In a circle centred at O, PQ is a tangent at P. Furthermore, AB is the chord of the circle and is extended to Q. If PQ = 12 cm and QB = 8 cm, then the length of AB is equal to:

HINTS



We know that:

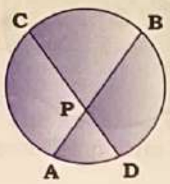
$PQ^2 = QB \times AQ$

$\Rightarrow 12 \times 12 = 8 \times (x + 8)$

$\Rightarrow 18 = x + 8$

$\Rightarrow x = 10$

Ex. In the given figure, PA = 4 cm, PB = 9 cm, PC = x cm and CD = $6x - 3$, then the value of 'x' is:



HINTS

$AP \times PB = PC \times PD$

$\Rightarrow 4 \times 9 = x \times (6x - 3 - x)$

$\Rightarrow 36 = x(5x - 3)$

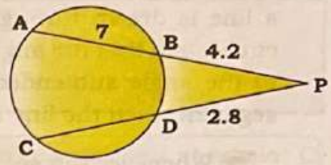
$\Rightarrow 36 = 5x^2 - 3x$

$\Rightarrow 5x^2 - 3x - 36 = 0$

$\Rightarrow x = 3$ cm

Ex. Two chords AB and CD of a circle are produced to intersect each other at a point P outside the circle. If AB = 7 cm, BP = 4.2 cm and PD = 2.8 cm, then the length of CD is:

HINTS



$\because PA \times PB = PC \times PD$

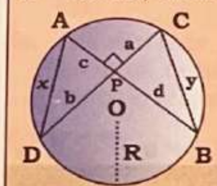
$\Rightarrow 11.2 \times 4.2 = (2.8 + CD) \times 2.8$

$\Rightarrow 2.8 + CD = \dfrac{11.2 \times 4.2}{2.8}$

$\Rightarrow CD = 16.8 - 2.8 = 14$ cm

Important results

Chords AB and CD of a circle intersect at P and are perpendicular to each other. If AD = x, BC = y, PC = a, PD = b, PA = c, PB = d and R = radius, then:



$$R = \frac{\sqrt{AP^2 + PD^2 + CP^2 + BP^2}}{2}$$

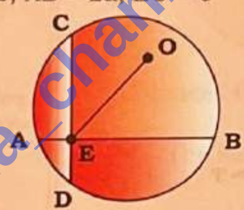
or, $a^2 + b^2 + c^2 + d^2 = 4R^2$

- If $\angle APD = 90°$, $R = \dfrac{\sqrt{BC^2 + AD^2}}{2}$

  or, $x^2 + y^2 = 4R^2$

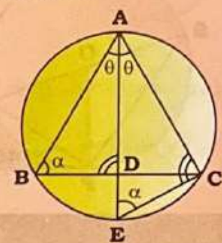
If O is any point inside the circle, then:

CD = 2b, AB = 2a, EO = c



$$r = \frac{\sqrt{a^2 + b^2 + c^2}}{2}$$

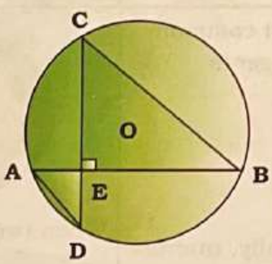
In the figure, AE is angle bisector of ∠BAC, then:



$\boxed{AB \times AC + AE \times DE = AE^2}$

Ex. Chords AB and CD of a circle intersect at E and are perpendicular to each other. Segments AE, EB and ED are of length 2cm, 6cm, and 3cm respectively. The length of the diameter of the circle (in cm) is:

HINTS



$\because AE \times BE = CE \times ED$

$\Rightarrow 2 \times 6 = CE \times 3$

$\Rightarrow CE = 4$cm

$AD = \sqrt{4+9} = \sqrt{13}$

$BC = \sqrt{16+36} = \sqrt{52}$

$r = \dfrac{\sqrt{BC^2 + AD^2}}{2}$

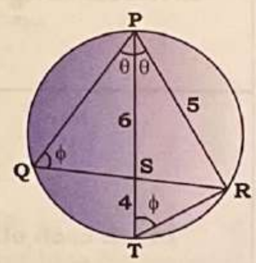
$= \dfrac{\sqrt{52+13}}{2} = \dfrac{\sqrt{65}}{2}$

$\therefore$ Diameter = 2r

$= 2 \times \dfrac{\sqrt{65}}{2} = \sqrt{65}$ cm

Ex. ΔPQR is inscribed in a circle. The bisector of ∠P cut QR at S and the circle at T. If PR = 5 cm, PS = 6 cm and ST = 4 cm, then the length of PQ is:

HINTS



$PT^2 = PQ \times PR + PT \times ST$

$\Rightarrow 10^2 = PQ \times 5 + 10 \times 4$

$\Rightarrow 100 = 5PQ + 40$

$\Rightarrow 5PQ = 60$

$\Rightarrow PQ = 12$ cm

Alternatively

In ΔPQS and ΔPTR:

∠QPS = ∠TPR (∵ PT is angle bisector)

∠PQR = ∠PTR

(Angles formed by same chord PR)

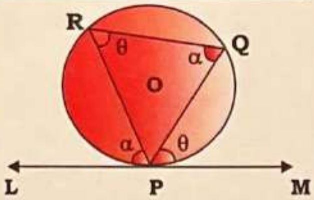
∴ ΔPQS ~ ΔPTR

$\Rightarrow \dfrac{PS}{PR} = \dfrac{PQ}{PT} \Rightarrow \dfrac{6}{5} = \dfrac{PQ}{10}$

$\Rightarrow PQ = 12$ cm

Alternate Segment Theorem

☞ If a chord is drawn through the point of contact of a tangent, then the angle which the chord makes with the tangent is equal to the angle made by that chord in the alternate segment.

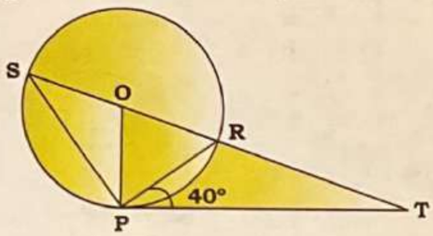


If $\angle MPQ = \theta$, then
$\angle PRQ = \theta$ and
if $\angle LPR = \alpha$, then
$\angle PQR = \alpha$

The converse of the above property also holds true. If a line is drawn through an end-point of a chord of a circle such that the angle formed with the chord is equal to the angle subtended by the chord in the alternate segment, then the line is a tangent to the circle.

Whenever you see the terms 'Chord' and 'Tangent' together in a question and you have to find angle, then you must check the applicability of alternate segment theorem.

Ex. In the given figure, O is the centre of the circle and PT is the tangent at P. If $\angle RPT = 40°$, then $\angle RTP$ is:



HINTS

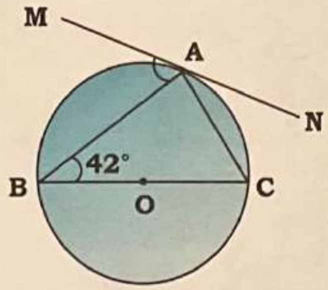
$\angle TPR = \angle PSR = 40°$ (Alternate segment theorem)

$\angle SPR = 90°$ (angle made by diameter at the circumference of circle)

$\angle SPT = 90° + 40° = 130°$

$\therefore \angle RTP = 180° - (130° + 40°) = 10°$

Ex. In the following figure, MN is a tangent to a circle with centre O at point A. If BC is the diameter and $\angle ABC = 42°$, then find the measure of $\angle MAB$.



HINTS

$\angle BAC = 90°$ (angle made by diameter at the circumference of circle)

In $\triangle ABC$:

$\angle ABC + \angle BAC + \angle BCA = 180°$

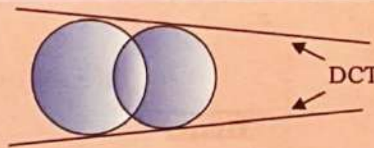
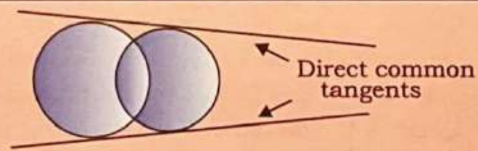
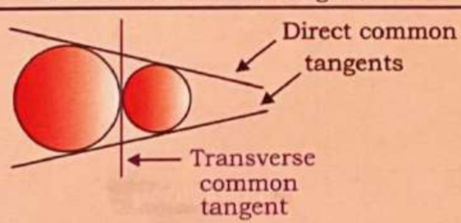
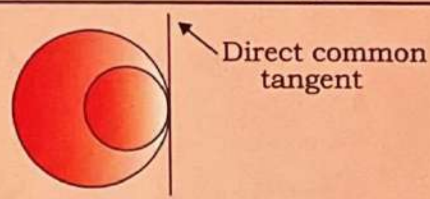
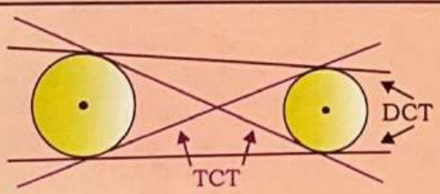
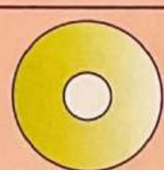
$\Rightarrow 42° + 90° + \angle BCA = 180°$

$\Rightarrow \angle BCA = 48°$

$\Rightarrow \angle BCA = \angle BAM = 48°$

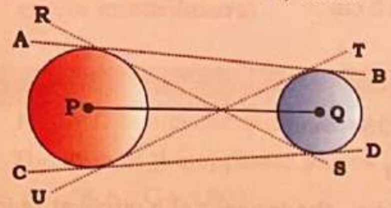
(By Alternate Segment theorem)

Maximum and Minimum possible common tangents

Circles	Maximum common tangents	Minimum common tangents
Intersect each other	 Number of common tangents = 2	 Number of common tangents = 2
Touch each other	 When two circles touch externally, number of common tangents = 3	 When two circles touch internally, number of common tangent = 1
Neither touch nor intersect	 Number of common tangents = 4	 Number of common tangents = 0

Length of common tangent

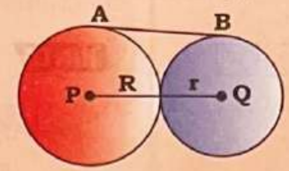
When two circles with centres 'P' and 'Q' having respective radii 'R' and 'r' are separated such that the distance between their centers is 'd', then:



- Length of Direct Common Tangent AB and CD = $\sqrt{d^2-(R-r)^2}$
- Length of Transverse Common Tangent RS and TU = $\sqrt{d^2-(R+r)^2}$

Special cases of Direct Common Tangent

When two circles of radii 'R' and 'r' touch each other externally, then:

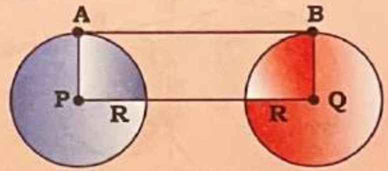


Distance between the centres = R + r

Length of Direct Common Tangent

$= \sqrt{(R+r)^2-(R-r)^2} = 2\sqrt{Rr}$

If radius of circles is same.



DCT = $\sqrt{(PQ)^2-(R-R)}$

$\Rightarrow AB = \sqrt{(PQ)^2} \Rightarrow AB = PQ$ i.e. length of direct common tangent is equal to distance between the centres of circle.

Length of direct common tangent is greater than the length of transverse common tangent.

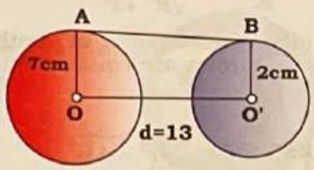
Ex. Two circles are of radii 7 cm and 2 cm. Their centres being 13 cm apart. The length of direct common tangent to the circles between the points of contact is :

HINTS AB = DCT

$DCT = \sqrt{d^2-(r_1-r_2)^2}$

$= \sqrt{169-(7-2)^2}$

$= \sqrt{169-25} = \sqrt{144} = 12$ cm



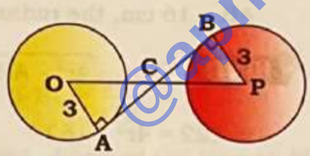
Ex. The distance between the centres of two equal circles, each of radius 3 cm, is 10 cm. The length of the transverse common tangent is:

HINTS d = 10 cm,

r = R = 3 cm

$TCT = \sqrt{d^2-(R+r)^2}$

$= \sqrt{100-36} = \sqrt{64} = 8$ cm

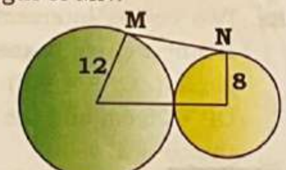


Ex. Two circles having radii 12 cm and 8 cm, respectively, touch each other externally. A common tangent is drawn to these circles which touch the circles at M and N, respectively. What is the length of MN?

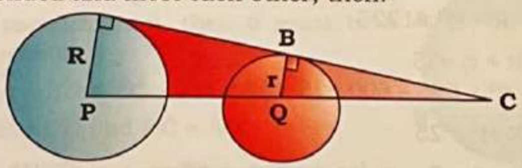
HINTS

$MN = 2\sqrt{Rr}$ cm

$= 2\sqrt{12\times8} = 8\sqrt{6}$



☞ If DCT and the line joining the centres of the circles are extended and meet each other, then:

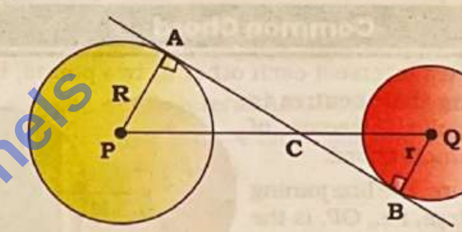


C externally divides PQ in the ratio of radius of two circles.

$\frac{CP}{CQ} = \frac{R}{r}$ (by similarity)

$CP = \left(\frac{R}{R-r}\right)\times PQ \Rightarrow CQ = \left(\frac{r}{R-r}\right)\times PQ$

☞ If TCT and the line joining the centres meet at a point C, then:

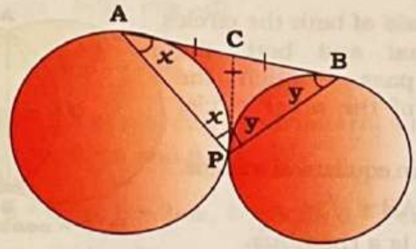


C divides PQ in the ratio of radius of two circles, i.e.,

$\frac{CP}{CQ} = \frac{R}{r}$ (by similarity) $\Rightarrow CP = \frac{R}{R+r}\times PQ \Rightarrow CQ = \frac{r}{R+r}\times PQ$

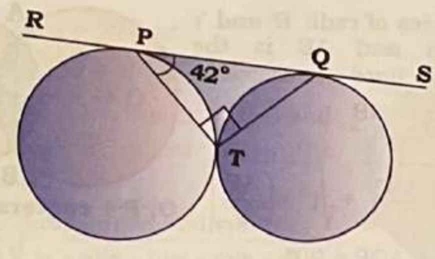
☞ If two circles externally touch each other at P and AB is the direct common tangent (DCT) of the circles, then ∠APB is always right angle.

If ∠BAP = $x°$, then ∠ABP = $(90-x)°$.



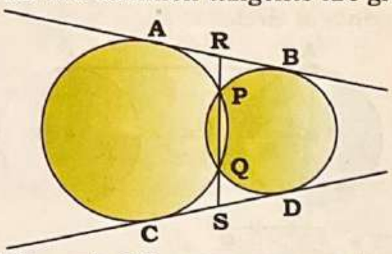
Ex. Two circles touch each other externally at T. RS is a direct common tangent to the two circles touching the circles at P and Q respectively. If ∠TPQ = 42°, then ∠PQT is:

HINTS



∠PQT = 90° − 42° = 48°

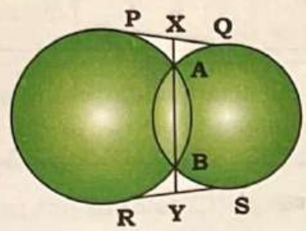
☞ If two circles intersect each other at two distinct points and two common tangents are given, then:



$RS^2 = AB^2 = PQ^2$

Ex. PQ and RS are common tangents to two circles intersecting at A and B. A and B, when produced on both the sides, meet the tangents PQ and RS at X and Y, respectively. If AB = 3 cm and XY = 5 cm, then PQ is:

HINTS



Given,

AB = 3 cm, XY = 5 cm

We know that,

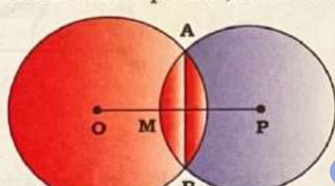
$PQ^2 = XY^2 - AB^2$

$\Rightarrow PQ = \sqrt{25-9} = 4$ cm

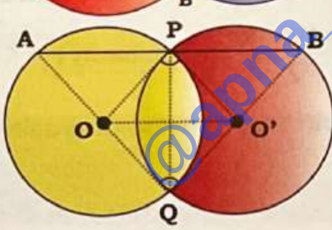
Common Chord

☞ If two circles intersect each other at two points, then the line joining their centres is perpendicular bisector of their common chord.

In the figure, the line joining their centres, i.e., OP, is the perpendicular bisector of the common chord AB.



(i) If two equal circles intersect at P and Q and a straight line through P meets the circles at A and B respectively, then QA = QB.



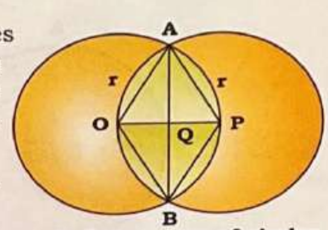
(ii) If two circles with centres O and O' intersect at P and Q, then ∠OPO' = ∠OQO'.

(a) When radii of both the circles are equal and both the circles pass through the centres of the other circle, then:

ΔOPA is an equilateral triangle.

OP = r, AB = $\sqrt{3}$r

∴ OBPA is a rhombus.



O, P = centers of circles

Ex. Two equal circles of radius 4 cm intersect each other such that each passes through the centre of the other. The length of the common chord is :

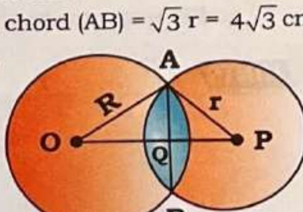
HINTS Length of the common chord (AB) = $\sqrt{3}$ r = $4\sqrt{3}$ cm

(b) Two circles of radii 'R' and 'r' intersect and AB is the common chord, then:

$AQ = BQ = \dfrac{AB}{2}$

$$OP = \sqrt{R^2 - \left(\frac{AB}{2}\right)^2} + \sqrt{r^2 - \left(\frac{AB}{2}\right)^2}$$

∠AQO = ∠AQP = 90°



O, P = centers of circles

Ex. The length of the common chord (AB) of two intersecting circles is 24 cm. If the diameters of the circles are 30 cm and 26 cm respectively, then the distance between the centres is :

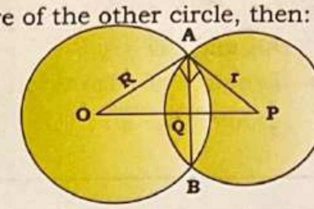
HINTS $AQ = \dfrac{24}{2} = 12$ cm

$OQ = \sqrt{225-144} = \sqrt{81} = 9$cm

$PQ = \sqrt{169-144} = \sqrt{25} = 5$cm

OP = 9 + 5 = 14 cm

(c) When ∠OAP = 90°, i.e., the tangent of a circle at point A or B pass through the centre of the other circle, then:



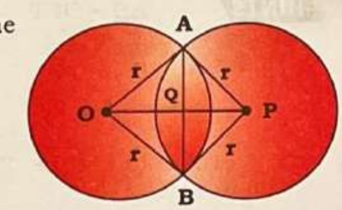
☞ $OP = \sqrt{R^2 + r^2}$

☞ $OQ = \dfrac{R^2}{\sqrt{R^2 + r^2}}$

☞ $QP = \dfrac{r^2}{\sqrt{R^2 + r^2}}$

☞ $AB = \dfrac{2Rr}{\sqrt{R^2 + r^2}}$

(d) When radii of both the circles are equal, then:



☞ $OP = \sqrt{4r^2 - AB^2}$

☞ $OQ = PQ = \dfrac{OP}{2}$

Ex. Two equal circles whose centres are O and P intersect each other at the points A and B. OP = 12 cm and AB = 16 cm, the radius of the circle is:

HINTS $OP = \sqrt{4r^2 - AB^2}$

$\Rightarrow 12^2 = 4r^2 - 16^2$

$\Rightarrow 144 + 256 = 4r^2$

$\Rightarrow 4r^2 = 400$

$\Rightarrow r = 10$ cm

Ex. Two circles intersect at A and B. The tangent at A to the first circle passes through the center of the second circle (∠OAP = 90°). If the distance between centers is OP = 25 cm and the chord AB = 24cm , find the radii of the circles.

HINTS $(R + r)^2 = 625 + 600 = 1225$

$\Rightarrow (R + r)^2 = 1225$

$\Rightarrow (R + r) = 35$ …(i)

$(R - r)^2 = 625 - 600 = 25$

$\Rightarrow (R-r)^2 = 25$

$\Rightarrow (R- r) = 5$ …(ii)

from eq^n (i) and (ii)

$R = \dfrac{35+5}{5} = 20$cm,

$r = \dfrac{35-5}{2} = 15$cm